



ELECT MAINT Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT/2025-26/SWYDESP/406410

Date:

03/09/2025(10:34:15)

Subject:Review of LMI for AIS maintenance practices
LMIOGNOPSELEC042 Rev:1

LMI for 'Guidelines on AIS maintenance practices' LMI/OGN/OPS/ELEC/042 Rev:0 Jan 2023 is due for periodic review.

Corporate OS has reviewed OGN-OPS-ELECT-042 and has issued revision 01 in April-2025. On the basis of OGN revision, LMI/OGN/OPS/ELEC/042 Rev:0 has been reviewed and suggested changes are incorporated in LMI/OGN/OPS/ELEC/042 Rev-1.

LMI/OGN/OPS/ELEC/042 Rev 1 is put up for review and approval please.

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Note No:5

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NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

Guidelines on AIS maintenance practices

LMI/OGN/OPS/ELEC/042 Rev:1

JULY - 2025

APPROVED FOR IMPLEMENTATION BY HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI Name	Guidelines on AIS maintenance practices
2	LMI No	LMI/ OGN/OPS/ELEC/042, REV:1
3	Reference Document	COS-OGN/ OPS/ELEC/042, REV:1
4	Superseded Document	COS-OGN/ OPS/ELEC/042, REV:0
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12	Approved by	Sameer Sharma HOP (Simhadri)

TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

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1. INTRODUCTION

Switchyard (Air Insulated/Gas Insulated) in the power plants are one of the important parts working as the interconnection between the Generating Station and the Grid. The generated power from the generating station is transmitted to the grid through the switchyard. Switchyard can be considered as the backbone of the power plant & is inevitable. Air Insulated switchyard is the most common type of switchyard used in power plants, where air acts as a medium to insulate different components of the switchyard from each other.

Over the years, from time-to-time CC-OS has issued several guidelines for safe and reliable operation and maintenance of Switchyard & equipment. The guidelines have been issued in the form of Operation Memorandums, IOMs and technical compliance documents like OIN, OGN, OD etc. For quick and easy reference of the guidelines by station maintenance personnel, it has decided to integrate the recommendations in a single document in the form of this OGN. GIS stations may adopt the maintenance practices as per newly released TC document Ref: COS- OGN/OPS/ELEC/041, Rev1, Dec2024.

2A. SCOPE

This LMI is applicable to 400KV AIS switchyard of NTPC Simhadri and all equipment of EHV System.

2B. REFERENCE DOCUMENTS

COS-ISO-00-OGN-OPSELEC-042 Rev. No: 1 April ,2025

2C. SUSPENDED DOCUMENTS

LMI/ OGN/OPS/ELEC/042, REV:0 Jan 2023

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3.0 RECORDS OF REVISION

Clause	Old practice	New Practice	Remarks
4		Sr no 3 practices updated with responsibility.	Addition
		Sr no 7 practices updated	Revised
		Sr no 14 practices updated	Revised
		Sr no 18 (ii) practices updated	Revised
		Sr no 21 practices updated	Revised
		Sr no 42 practices updated	Revised
		Sr no 44 practices updated	Revised
		Sr no 45 practices updated	Revised
		Sr no 46 practices updated	Revised
		Sr no 49 practices updated	Revised
		Added Safe maint practices sr no 51 & 58	Addition
5.1	Yearly schedule (5.1.1 to 5.1.8)	Yearly/OH schedule	Revised
	3 year schedule for 5.1.3 (05)	Yearly/OH schedule	Revised
	monthly schedule for 5.1.5 (01,02)	Daily schedule	Revised
	3-year schedule for 5.1.8 (05)	Yearly/OH schedule	Revised
		Purity & SO2 checks added in 5.1.3	Addition
5.2	Monthly schedule for 5.2 (02)	Daily schedule	Revised
	OH schedule for 5.2	Yearly/OH schedule	Revised
		DGA freq for AREVA CT added at sr no 14.	Addition
	Monthly schedule for 5.2 (18)	Daily schedule	Revised
5.3	Monthly schedule for 5.3 (01)	Daily schedule	Revised
	OH schedule for 5.3	Yearly/OH schedule	Revised
5.4	OH schedule for 5.4	Yearly/OH schedule	Revised
5.5	OH schedule for 5.5	Yearly/OH schedule	Revised
	SOS schedule for 5.5 (05)	Yearly/OH schedule	Revised
5.6	Quarterly schedule for 5.6 (01)	Yearly	Revised
	OH schedule for 5.6	Yearly/OH schedule	Revised
5.7	OH schedule for 5.6	Yearly schedule	Revised
5.8	OH schedule for 5.6	Yearly schedule	Revised
5.10.		5.10 (02) updated	Revised

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	OH schedule for 5.10.4	Yearly/OH schedule	Revised
5.11	OH schedule for 5.11.1	Yearly/OH schedule	Revised
		Sr no 2 practices added in 5.11.1	Addition
	OH schedule for 5.11.2	Yearly schedule	Revised
	OH schedule for 5.11.3	Yearly/OH schedule	Revised
		Sr no 27 practices added in 5.11.3	Addition
	OH schedule for 5.11.4 to 5.11.12	Yearly/OH schedule	Revised
6		Insulator cleaning Practices & guidelines added in 6.1.1	Revised
		Practice updated in 6.1.2	Revised
7		Breaker SF6 Dew point parameter updated in chart 7.1.2 (f)	Addition
		Practices updated in 7.1.6 (d)	Revised
		Practices updated in 7.4.1 (c)	Revised
9		Table 1 Sr no 1 (C) updated for 765KV	Addition
		Table 1 Sr no 1 (D &E) updated for new OEM	Revised
		Table 1 Sr no 2 (E) limit revised	Revised
		Table 1 Sr no 4 (A) practice updated	Revised
		Table 1 Sr no 5 (A) Vendor added	Addition
		Table 2 updated for SF6 gas parameters	Revised
11		Practices added in 11.2 (XI)	Addition
12		Practices updated in Sr no 2, 3 & 6.	Revised
16		Material Updated in i	Revised
		Material Updated in xviii	Revised
17		B: yearly maintenance activity sr no (e) updated in 17.2	Revised
		3 yearly maintenance record checklists deleted in 17.2	deleted
		Monthly maintenance record checklist deleted in 17.3	deleted
		Monthly maintenance record checklist deleted in 17.4 (B)	deleted
		Limit revised in 17.6 (A)	Revised
		Practices added in 17.8 (A)	Addition
		Checklist for Hotline Washing of Insulator added as 17.14	Addition

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4.0 SAFE MAINTENANCE PRACTICES IN AIS:

Air Insulated switchyard is the most common type of switchyard used in power plants, where air acts as a medium to insulate different components of the switchyard from each other. Since the entire switchyard is widespread with potential electrical hazards, even the slightest of negligence can prove damaging. Regular training and communication are crucial to ensure that the safety norms are followed without any slip-ups. While most of the safety practices arise out of common sense and practical learning, some potential problem areas may skip attention. Hence the practices that can minimize the high-risk safety/security threats commonly encountered in switchyard are mentioned here, which are to be complied by each NTPC stations.

1. The single Line diagram of the switchyard mentioning the detail nomenclature of the bay equipment must be displayed at the entrance of the control room and must be available in the switchyard control room for reference.
2. Identification of the Bay No., Bay Equipment and Marshalling Kiosk (Isolator/Earth Switch/Breaker/CT/CVT etc) including name of the circuit is to be done with proper labelling (Black letter in yellow background) mentioning the equipment nomenclature prefixed by the bay no.
3. Bay wise Isolation and Normalization procedure for all voltage levels of the station is to be prepared and converted in the form of LMI. It is to be ensured that the document is available in the Switchyard Control Room. Records to be maintained by Operation deptt.
4. Earthing should be provided through the system Earth Switches; their interlocks should not be bypassed. In case of additional Earthing required, it should be mentioned in the Permit and in Isolation & Normalization check list. Grid code shall be taken to ensure earth switch close and open at remote substations.
5. Phase Identification (R/Y/B) of the all the bay equipment /Bus post insulators is to be done either by paint/stickers (Red for R phase, Yellow for Y phase & Blue for B phase).
6. Painting of the outdoor equipment including Marshalling Boxes (Isolator/Earth Switches)/Junction Boxes/Breaker MB of switchyard is to be done once in every 06 years during overhauling of the respective Bay (Paint shade for various equipment/MB should be as per the standards adopted in NTPC).
7. All Earth Switch MBs in switchyard may be painted with green colour for easy identification to avoid inadvertent operation of Earth switches. Bus earth switches to be painted with RED colour.
8. Outdoor marshalling boxes:
 - I. Cleaning and blowing of the outdoor Marshalling boxes & terminal tightness checking during respective equipment/Bay OH.

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- II. Control & Power fuses of the Marshalling boxes is to be checked for healthiness.
- III. The AC and DC supply of the Earth switches should always be kept in OFF condition (if required, AC & DC fuses are to be removed at the Local Marshalling Boxes).
- IV. Wherever, there is not auto opening of the Isolator, AC & DC supply to the Isolator should be kept OFF like it is kept in case of Earth switches.
- V. The operation of Isolator is to be done from Remote. In case of exigencies, the operation of Isolator can be done from Local after taking suitable precaution. After closing of the isolator, it is to be ensured that the isolators are properly aligned & after opening of the Isolator, it is to be ensured that, physically, the isolator is open.
- VI. Normally, the marshalling Kiosks (Isolator/Earth Switch/Breaker) are to be kept under lock and key. At certain station, the locking arrangement is made in such a way that, there is common key for the marshalling kiosks (Earth switch/Isolators/Breaker) of a particular bay and with the key to a particular bay, marshalling box of the other bay cannot be opened. This practice will prevent inadvertent opening of any other kiosk, which is not under PTW/not required to be opened. There are vendors who can supply multiple locks with the same key (Godrej is one of them).
- VII. It is to be ensured that, the marshalling Kiosks (Isolator/Earth Switch/Breaker), CT & PT Junction boxes etc. inside switchyard are properly closed and sealed. If required, the door gaskets are to be replaced. Additional gland holes in the Marshalling box, if any is to be closed to avoid entry of moisture and foreign object inside the MB to cause a mal- operation/unwarranted outage.
- VIII. Space heater, wherever provided should be checked for healthiness (cut in & cut out through Thermostat) and kept in service. This shall avoid condensation of moisture inside the marshalling boxes to cause a malfunction/unwarranted outage. Care should be taken such that, the cables inside the marshalling boxes/Junction boxes does not come in contact with the space/panel heaters (which may result in heating and failure of the cable/core).
9. No work should be taken up without proper Permit to Work and without supervision.
10. Senior Authorized persons should only issue Permit to Work inside switchyard.
11. Only qualified and experienced personnel should be given authorization to work in the switchyard. A list of competent persons to work in switchyard should be approved by Head of stations/Head of O&M. The list of competent persons should be available (displayed) in the switchyard Control room.
12. Maintenance work in switchyard is to be avoided during night hours. In case of urgent requirement, work can be executed after taking permission from the head of O&M.
13. Normally, the switchyard area is cordoned off by provision of metallic/non-metallic

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fences along the switchyard. It should be ensured that there is no opening anywhere to avoid trespassing, deliberate scaling, and entry of stray animals inside the switchyard premises. The entry gates provided for entry into the switchyard is to be kept under lock and key. The key is to be kept at the switchyard control room.

14. One of the primary objectives in grounding of metal parts at an outdoor substation/Switchyard is to establish an equipotential grid to which all conductive parts are connected. Earthing of the Fences around the switchyard is important to avoid dangerous touch potentials and fatal electric shocks. Normally the earthing of the switchyard fences is done at multiple location. Hence it is to be ensured the earthing of fences are healthy (Earth resistance may be measured at random basis once in 01 year). The earthing strips of the Fence may be painted with green to avoid deterioration due to corrosion.
15. In case of execution of any work (either construction or de-weeding or any other work) inside a charged switchyard, the following things are to be complied:
 - i. proper PTW is to be taken, and the work is to be done under the supervision of the respective maintenance group executing the work.
 - ii. A register is to be maintained at the switchyard control room mentioning the name & details of the persons working including their signature at the time of entry into and exit from switchyard. Special larger size ID card may be introduced which should be hung by all individuals including our own employees entering switchyard. The same may be issued by Switchyard control room on daily basis and returned daily and the records may be maintained in a register.
 - iii. The permit/LWC required for any such activity is to be taken on daily basis. PTW may be suspended / reissued / Transferred strictly. The option of "Suspend / Reissue / Transfer" has been activated recently in SAP in all NTPC Stations/projects.
 - iv. The key for access to the switchyard should be given to the person holding the PTW/LWC.
 - v. The PTW holder shall cancel the PTW/LWC and return the key to the switchyard control room personnel. The switchyard control room engineer should ensure that the access gate to the switchyard is closed.
 - vi. Job safety analysis is to be carried out for each of the Jobs planned to be executed. Pep- talk should be given to the workers about the risks and hazards and the guidelines including use of proper PPE (a standard risk/hazards including guidelines for working in switchyard is to be prepared) and undertaking from the workers may be taken in this regard.
 - vii. Construction materials handling training to be imparted by the competent person of EMD.
 - viii. The construction materials storage area must be clearly demarcated by EMD and the same to be ensured by the EIC of the construction contractor.
16. The switchyard is not intended to support visitor traffic and hence to reduce the possibility of any untoward incidents/accidents, the visitors (including the

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employees, who are not authorized to work in switchyard) are to be prohibited for entering the switchyard. In unavoidable circumstances, visitors accompanied by authorized personnel can be allowed access but with necessary precautions and PPE.

17. It is important to monitor and guide vehicle movements inside the switchyard to ensure that there are sufficient clearances between objects and the overhead lines and elevated equipment within the switchyard. Vehicle drivers must proactively ensure safe navigation as well. Similarly, the operations of the mobile hoisting equipment, Cranes, Hydra, Concrete mixers, Filter Machines, & Jetting Machines etc. also need to be supervised and controlled by the experienced personnel.
18. There should be a pool of identified and competent drivers (having experience of working in the switchyard), who shall be allowed to work inside the switchyard. The competency test of such drivers to be conducted in presence of head, EMD for acceptance. The tests may be micro control capability like close braking capability, capability to move in small distance like a foot, ability to stop lifting, raising of booms etc. at the desired locations, close drive along the road and in sharp turning (this may be carried out by keeping bricks on the road etc.)
 - i. Before start of work on any day, healthiness vehicles of all types being deployed (Like Hydra, crane, Mixers, Filter machines, Tractors etc.) may be ensured.
 - ii. Alcohol test of Drivers of heavy vehicles (Hydra, Crane, JCB etc) may be carried out before allowing entry into switchyard.
 - iii. The speed limit of any vehicle inside switchyard should not be more than 10 KMPH.
 - iv. The fitness certificate of the vehicle is to be ensured.
 - v. The driver of the vehicle should have authorization to drive the type of vehicle.
 - vi. Pep-talk should be given to the driver about the risks and hazards and the guidelines (a standard risk/hazards including guidelines for vehicle movement inside switchyard is to be prepared) and undertaking from the driver is to be taken.
19. All equipment (Filter machine/Hydra/Crane/Mixers etc.) used for any work inside the switchyard should be properly earthed (double) to the nearest earth point.
20. Storage of equipment and other materials may create an unwarranted risk. It not only increases footfall in the area as materials get shifted in and out of the facility but also lures thieves on the lookout for expensive electrical components/ equipment. Not everyone entering the substation in such cases are fully aware of the dangers they are exposed to and accidents that they can accidentally trigger. Hence the equipment storage in switchyard is prohibited to prevent undue accidents. However, if storage of spares inside the switchyard is inevitable, then a suitable location away from the active parts of the switchyard shall be identified for storage of such spares with clear cut demarcation and labelling.

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21. No loose materials (GI sheet, metallic wires or any other materials, which may fly due to high- speed wind) should be left in the switchyard. Bird nest should be removed from all gantry structures & equipment to avoid any fault due to fall of metallic wires on live conductors.
22. The switchyard area should be properly illuminated to maintain an average lux of 30 to 50 and for the control room should be average 300 Lux.
23. Cleaning of Cable trenches inside the switchyard is to be ensured. It is also to be ensured that there is no water logging in the cable trenches. If required, periodic dewatering of the cable trenches is to be done.
24. The storm water drains of the Switchyard should be clean. Before the onset of the monsoon the cleaning of storm water drains is to be done to avoid water logging.
25. It is to be ensured that, the cable trench and the drains are provided with the PF slippers and the healthiness of the PF slippers are to be ensured.
26. The entry point of control/CT secondary cables from switchyard to the Control Room should be kept sealed so that animals/reptiles etc. may not enter inside the control room through the cable trenches.
27. Cleanliness of the Relay/Control panel and cable trenches inside the switchyard control room is to be ensured
28. Emergency DC light in switchyard control room is to be ensured (if not envisaged, then provision of DC light is to be commissioned at the earliest). Periodic checking (On monthly basis) of the DC emergency light is to be carried out as it is done for the emergency light checking of Unit control Room/Central Control Room.
29. It is to be ensured that, the communication system (hotline communication system with respective RLDC) is available in the switchyard Control Room/Shift Charge Engineers Room.
30. Provision of either DG set/supply from emergency MCC of the unit for charging the switchyard ACDB/MCC in case of total black out situation of the station (This will help for starting of the compressors required for pneumatic operated breaker as well as supply to the battery chargers of switchyard).
31. Fire Fighting system:
 - i. Ensure availability of Fire detection & protection system in the switchyard control room.
 - ii. Fire protection system for outdoor switchyard is not envisaged. However, fire hydrant points in adequate nos. are generally provided. Availability of these fire hydrant points are to be ensured.
 - iii. Ensure 100 % availability of Sprinkler system (normally provided for Transformers & reactors) including routine checking of sprinkler system. Actual spray should be done when the corresponding equipment is under shut down with proper isolation.

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- iv. Proper access road for movement of fire tenders in case of emergency.
- v. Mock practice for movement of fire tenders inside switchyard is to be done on quarterly basis.
- 32. It is to be ensured that there is no water leakage inside the switchyard.
- 33. The oil soak pit of Transformers and Reactors inside the switchyard should be clean. Dewatering pump may be provided at the oil soak pit for dewatering.
- 34. The gravels in the switchyard should be uniformly spread.
- 35. It is highly recommended not to carry/use an umbrella inside a switchyard because of the following reason In switchyard of 220 kV Class and above, because of the induction effect the hair on the hands will rise automatically, which is known as the goose bump effect. During rainy season, the effect is much more due to moisture present in the air because of rain. the induction may cause a direct effect on the tip of the umbrella due to the sharp metallic point, which is highly sensitive or conductive in nature. Due to induction, static charge may accumulate at the tip of the umbrella and may cause severe shock to the person holding it
- 36. Display Board is to be placed with Do's and Don'ts at the entry of the switchyard.
- 37. Vegetation management in switchyard is necessary to reduce the risk and to ensure a safe and reliable operation of the switchyard equipment and system. Unmanaged vegetation may also provide a habitat for unwanted animals that can cause further damage to the switchyard equipment and the safety of working personnel. Birds and snakes regularly use tall grasses and thick shrubs as places to hide or build nests. Hence following are to be complied with respect to vegetation management of the switchyard.
 - i. The inside of switchyard should be free of any weeds or shrubs and the yard should ideally be covered with gravel which conceals and protects a grid of wires that functions as the grounding for high voltage equipment. Should there be vegetation in the gravel it could interfere with the ground grid, seriously compromising the safety functions of the grid and posing an electrical hazard to working personnel. It is to be ensured that there is no tree inside the active portion of the switchyard.
 - ii. Regular deweeding of switchyard is to be carried out for safety of the operating personnel.
 - iii. The area outside of the Switchyard should also be kept clear of all vegetation. Nearby trees can act as a fuel source for fire should an electrical spark or fire occur. Overhanging branches or large old trees can fall on to equipment and damage the site. It's advisable to remove/Trim these trees before they become too big and pose a serious risk to the switchyard as well to the working personnel.
- 38. It is to be ensured that there is no accumulation of dry grass/shrubs/herbs (which is a potential threat for fire hazards) inside and around the switchyard including the cable trenches.

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39. Daily walk down checklist of the switchyard is to be done for monitoring the Breaker Air pressure/ Breaker SF6 gas pressure/CT oil level/PT oil level/Breaker Hydraulic pressure/ Leakage current of LA counter/Isolator/Relay & control panel/Kiosk/Bushing oil level of Transformers and Reactors etc located inside the switchyard. Depending upon the severity of the abnormalities, appropriate action is to be taken. Daily monitoring checklist must be followed as mentioned in this OGN.
40. Fortnightly measurement of secondary Voltage of all the CVTs in the switchyard. This shall help in early detection of any deterioration in the respective CVT and timely action for replacement can be taken to avoid failure.
41. Monthly Thermal scanning of the complete switchyard (Clamps/Connector/Joints/Equipment) is to be carried out for any abnormalities. The data must be maintained in system. Depending upon the severity of the hot spot temperature, appropriate action is to be taken (+20°C above ambient is to be monitored frequently & + 50°C above ambient is to be attended at the earliest).
42. DCRM (Dynamic Contact Resistance Measurement) is to be done during every OH. Overhauling of EHV Breakers are to be planned as per OEM Recommendations/10years/Deviation in the DCRM results, whichever is earlier. Servicing of the mechanical cubicle of the Breaker (with spring operated mechanism) is to be done once in every 06 years. Replacement may be planned after 25 years of service or in case of obsolesce (as certified by OEM).
43. Switchyard bay equipment & relay testing to be carried out as per procedures & guidelines mentioned in this OGN. Stations are instructed to adhere the testing & checklist schedule as per given guidelines.
44. Earth resistance measurement to be checked on annual basis at each bay equipment including GT, ICTs & reactors earth pit. Record to be maintained & if required necessary action to be taken. It is advised to carry out earthing assessment of complete electrical system once through third party Vendor & corrective action must be taken.
45. PID (Punctured Insulator Detection) test on String and Suspension Insulators inside switchyard and other areas are to be carried out once in every 10 years. Rectification of the defective ones is to be planned accordingly after taking suitable shut down. However it is advised to use Long rod type polymer Insulator instead of porcelain type for highly polluted /coastal region. Also Hot line washing to be done at least twice in a year or as per condition for cleaning of Swyd equipment insulators (except LA & Bushings) by certified Vendor. SOP to be adhered in-line with Practices & Checklist mentioned in clause 6.1.1 & 17.4. Station SOP to be shared with COS & Site chemistry prior to work.
46. Integrity of the Switchyard structure including their civil foundation (Gantry and Lighting sum lightning mast) is to be checked and if required necessary action is to be taken for ensuring the healthiness. Switchyard GI structures to be monitored in regular interval & healthiness to be ensured. If rusting observed on structures & hardware, GI/zinc oxide & zinc phosphate paints or other as recommended by

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vendor may be applied.

47. SOP (Standard Operating Procedure) is to be prepared for taking Transfer Breaker into service from Main Breaker & for taking Main Breaker into service from Transfer Breaker (applicable for switchyard having Transfer Bus coupler arrangement). It is also instructed to prepare SOPs for bay Isolation/Normalization for every bay of switchyard. SOPs have been provided here as per NTPC general switchyard scheme.
48. Switchyard Operations under Emergency Conditions & Switchyard Bus & Line charging instructions mentioned in this OGN are to be followed by the stations.
49. It is to be ensured that the Switchyard equipment (CT & CVTs), which have service life more than 20 years & LAs aged by more than 15 Years in service are replaced as per COS guideline Ref: OS/GM/ELEC/702, dated: 18.07.2017. Instrument Transformers (CT/CVT) & LAs should have Polymer based Insulators.
50. Other Guidelines/Recommendations related to switchyard issued by COS-Electrical time to time in form of OGN/OIN/IOM/KT Meet/HOD MEET are to be complied for safety and reliability of the system.
51. It is to be ensured that no fire water/ash pipelines should be available near to the Switchyard EHV conductors. If provided, suitable canopy to be installed to prevent water hitting on live conductors or lines may be re-routed.
52. To minimize mechanical stress & subsequent hot spots at GT LA dropper conductor connections it is recommended to install additional loop conductor with PG clamp/ C-Wedge connectors.
53. To secure the Switchyard SCADA network, it is instructed to maintain the SCADA system with latest OS (operating system) along with Anti-virus & firewall. SCADA system should be Cyber secure. ABT network should be separated from the SCADA network.
54. Current transformers (CT) should be kept on vertical position at store.
55. Power supply sources of Station RTU (Remote terminal Unit) for data transmission to RLDC should be from DCDB.
56. Wherever MOU has been made with other transmission utilities (PGCIL, SEBs etc) for O&M of bays, Transformer/Reactors & transmission lines; charges should be levied as per the guidelines of CEA.
57. Electrical installation inspection should be carried out by CEA person on 5 years frequency as per the guidelines of CEA.
58. Calibration of testing kits to be done, calibration schedule should be 3 years.

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5.0 WITCHYARD EQUIPMENT MAINTENANCE SCHEDULE:

5.1 CIRCUIT BREAKERS

- SF6 Circuit Breakers

01.	Checking tightness of foundation bolts	Yearly/OH
02.	Cleaning of breaker interrupter, support insulators, PIRs and grading capacitors	Yearly/OH
03.	Checking of oil leaks from grading capacitors	Monthly
04.	SF6 gas leakage test with leak detector	Yearly//OH /SOS
05.	Dew point, purity & SO2 measurement of SF6 gas	Yearly/OH

- Pneumatic operating mechanism with unit compressor

01	Checking of air compressor for oil level, oil quality, air filter, V-belt tension	Quarterly
02	Maintenance of air dryer, if provided	Half Yearly
03	Functional checking of auto-starting of air compressors and dryers	Yearly/OH
04	Checking of air pressure drop during duty cycle operation	Yearly/OH
05	Overhauling of compressors	SOS/OEM guideline

- Air Blast Circuit Breakers with central compressed air system

01.	Checking of tightness of foundation bolts	Yearly/OH
02.	Cleaning of breaker interrupter, support insulators, PIRs and grading capacitors	Yearly/OH
03.	Checking of oil leak from grading capacitors	Monthly
04.	Air (pressure) leakage check	Yearly//OH /SOS
05.	Checking of air pressure drop during duty cycle operation	Yearly/OH

- Central compressed air system

01.	Checking of air compressor for oil level, oil quality, air filter, V-belt tension	Quarterly
02.	Maintenance of air dryers	Half yearly
03.	Functional checking of auto starting of air compressors and dryers	Half Yearly
04.	Dew point measurement of operating air at the outlet of air dryer	Yearly/OH
05.	Overhauling of compressors	SOS/OEM guideline

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- Control cabinet

01.	Checking of tightness of all cable terminations in MB	Yearly/OH
02.	Checking of door sealing gaskets and replacement, if necessary	Yearly/OH
03.	Repainting of metallic surfaces	SOS
04.	Checking of space heater (before monsoon)	Yearly/OH
05.	Checking of pole discrepancy relay	Yearly/OH
06.	Checking of pressure settings	Yearly/OH
07.	Checking of healthiness of operation counter	Yearly/OH
08.	Functional checks, duty cycle operation including rapid re-closing (O-0.3s-CO)	Yearly/OH
09.	Checking of all operation lock-outs including SF ₆ density monitor	Yearly/OH
10.	Checking of all interlocks	Yearly/OH
11.	Checking of remote operation, protection trip and alarms	Yearly/OH

- Measurement/ testing

01.	Checking of close/trip coil currents and alarm circuit healthiness	Yearly/OH
02.	Capacitance and tan measurement of grading capacitors	Yearly/OH
03.	CB operating timings (Main, PIR.)	Yearly/OH
04.	Static contact resistance measurement	Yearly/OH
05.	Dynamic contact resistance (DCRM), contact travel, contact speed, contact wipe, arcing contact length	Yearly/OH

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5.2 CURRENT TRANSFORMERS (CT)

01.	Checking of bellow expansion (level high/low/normal) and freeness of bellow expansion	Yearly//OH
02.	Visual inspection of CT for oil leakage/seepage from: (i) Dead tank gasket joint (ii) Drain tap (iii) Primary conductor sealing gasket Gauge glass (iv) Metal bellows Schroeder valve Insulator cemented joint (v) Secondary terminal board/marshalling box	Daily
03.	Thermo scanning of CT body and primary connections	Monthly
04.	Checking of primary connection strips, if provided externally	Yearly/OH
05.	Measurement of Tan and capacitance (C1)	Yearly/OH
06.	Tan delta tip up test	Yearly/OH
07.	Checking the tightness of PF terminal earthing connection. Check CT marshalling box sealing and gasket effectiveness.	Yearly/OH
08.	IR measurement (DAR)	Yearly/OH
09.	N ₂ pressure checking (if applicable)	Once in 02 Years
10.	Checking of primary connection strips, if provided internally	SOS
11.	Measurement of CT secondary resistance	Yearly/OH
12.	Magnetization characteristics	Yearly/OH
13.	CT ratio test	Yearly/OH
14.	DGA and testing of other parameters of oil (Limits as specified in IEC60599)	Within warranty period and thereafter 10Y or SOS. On 5 yr freq for AREVA dead tank CTs
15.	Checking of burden on the secondary winding	SOS
16.	On line acoustic PD measurement	SOS
17.	RVM test	SOS

Marshalling box and terminal box

01.	Checking of oil leakage from terminal box	Daily
02.	Checking of healthiness of gaskets	Yearly/OH
03.	Checking of space heater and illumination	Yearly/OH
04.	Checking the tightness of all connections	Yearly/OH
05.	Cleaning of marshalling box and terminal box	Yearly/OH

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5.3 POTENTIAL TRANSFORMERS/CAPACITANCE VOLTAGE TRANSFORMER/COUPLING CAPACITORS

01.	Checking of oil leaks	Daily
02.	Measurement of voltage at Control room panel	Fort Nightly
03.	Visual checking of earthing HF point (in case it is not being used for PLCC)	Yearly/OH
04.	Checking for any breakage or cracks in cementing joint	Yearly/OH
05.	Cleaning of CVT capacitor stacks and tightness of terminal connections	Yearly/OH
06.	Thermal scanning of Capacitor stacks & EMVT tanks	Monthly
07.	Capacitance and Tan delta measurement	Yearly/OH
08.	Testing of EMU tank oil for BDV (if oil found discolored)	SOS
09.	Checking for rust and painting	SOS

- CVT/PT Marshalling box

01.	Checking of space heater and illumination	Yearly/OH
02.	Checking and tightness of all connections including earth connections	Yearly/OH
03.	Cleaning of marshalling box and junction box	Yearly/OH
04.	Checking of healthiness of gaskets	Yearly/OH

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5.4 ISOLATORS AND EARTH SWITCHES

- Main contacts

01.	Cleaning and lubrication of main contacts, pins and bearings	Yearly/OH
02.	Checking of tightness of bolts, nuts and pins, etc.	Yearly/OH
03.	Cleaning of support insulators and checking of insulator cracks, if any	Yearly/OH
04.	Checking of interlocks	Yearly/OH
05.	Checking of earth connection of structure	Yearly/OH
06.	Operation check of isolators	Yearly/OH
07.	Checking of alignment	Yearly/OH
08.	Main contact resistance measurement	Yearly/OH

- Operating mechanism

01.	Checking and lubrication of linkages including transmission gears	Yearly/OH
02.	Checking and tightening of stopper bolts	Yearly/OH
03.	Cleaning of auxiliary switch contacts and greasing with silicon grease	Yearly/OH
04.	Lubrication of operating mechanism hinges, lock joints on levers, Bearings	Yearly/OH
05.	Checking of all mounting bolts for tightness	Yearly/OH
06.	Checking of healthiness of door gaskets	Yearly/OH
07.	Checking of earth connection of MOM box	Yearly/OH
08.	Checking of tightness of electrical connections	Yearly/OH
09.	Checking of space heaters and illumination	Yearly/OH

- Earth Switc

01.	Checking and alignment of earthing blades	Yearly/OH
02.	Cleaning of contacts and lubrication	Yearly/OH
03.	Operation of earthing switch	Yearly/OH
04.	Checking of aluminum / copper flexible connectors	Yearly/OH
05.	Checking of earth connections of earth switch, structure and MOM box	Yearly/OH
06.	Checking of tightness of bolts, nuts and pins etc. and lubrication of pins and bearings	Yearly/OH
07.	Contact resistance measurement	Yearly/OH

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- Marshalling box/ operating mechanism

01.	Visual check of auxiliary contacts, cleaning and lubricating with silicon grease	Yearly/OH
02.	Cleaning and terminal tightness in MOM box	Yearly/OH
03.	Checking of space heaters and illumination	Yearly/OH
04.	Checking of healthiness of gaskets	Yearly/OH
05.	Checking and tightening of stopper bolts	Yearly/OH
06.	Checking of all mounting bolts for tightness	Yearly/OH
07.	Lubrication of operating mechanism	Yearly/OH

5.5 SURGE ARRESTER/LIGHTNING ARRESTER

01.	Checking of leakage current (Third harmonic resistive current)	Monthly
02.	Testing of counters and leakage current meters	Yearly/OH
03.	Cleaning of insulator	Yearly/OH
04.	Checking of earth connections between surge arrester, surge monitor and earth	Yearly/OH
05.	Measurement of capacitance and Tan delta and IR of each stack	Yearly/OH

5.6 BUS-BAR, JUMPERS, CONNECTORS, CLAMPS, SWITCHYARD ILLUMINATION, ETC.

01.	Measurement of station earth resistance	Yearly
02.	Cleaning of insulators	Yearly/OH
03.	Checking of insulators for cracks	Yearly/OH
04.	Thermal scanning of all conductor joints, terminal Connectors/clamps	Monthly
05.	Healthiness monitoring of Conductors, Jumpers & Earth wires through Binocular	Monthly
06.	Checking of earthing connection of all structures	Yearly/OH
07.	Attending the hot spots	SOS
08.	De-weeding of switchyard	SOS
09.	Repainting, rust removal of all structures, equipment, etc.	SOS
10.	Checking of switchyard lighting	SOS
11.	Checking of DC emergency lighting	Monthly

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5.7 WAVE TRAPS

01.	Tightness and cleanliness	Yearly
02.	General inspection / cleaning of tuning unit	Yearly
03.	Thermal Scanning of joints	Monthly
04.	Cleaning of Bus post insulators (if provided)	Yearly
05.	Repair of bird guard	SOS

5.8 PLCC SYSTEM

01.	Checking of Return Loss	Yearly
02.	Power supply measurements	Yearly
03.	Transmitter checks	Yearly
04.	Receiver checks	Yearly
05.	Checks for Alarms	Yearly
06.	Reflex Test	Yearly
07.	LMU composite/ Return loss	Yearly
08.	PLCC protocol checking with remote end substation. Simulation of contacts for carrier & DT transmit & receive.	Yearly

5.9 TELEPHONE EXCHANGE

01.	Maint. of EPAX as per recommendations of manufacturers	SOS
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5.10 FIRE PROTECTION SYSTEM

- Fire alarm system

01.	Sequence test for annunciation in control room panel	Monthly
02.	Smoke test (Control room & Kiosk)	Monthly
03.	Cleaning	Monthly
04.	Battery electrolyte level checking	Monthly

- Hydrant system

01.	Checking of pressure of the hydrant system at the remotest end, auto-starting of pumps, diesel engine, etc.	Yearly
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- Deluge system

01.	Operation of deluge system, checks outlet pressure, check alarm,	Half yearly
	check starting of diesel/electrical pump.	

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- Electrical panels

01.	Cleaning and Tightening of terminals	Yearly/OH
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- General

01.	Greasing of all valves	Half Yearly
02.	Painting of pipes, air lines, marshaling box	SOS

- Fire extinguishers

01.	Re-filling of fire extinguishers	SOS
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5.11 ELECTRICAL PROTECTIONS MAINTENANCE SCHEDULE:

- Control and relay panel

Sl. No.	Activities	Frequency
1	Testing of DR/EL/SER with time synchronization unit	Monthly
2	Healthiness monitoring of Relays/DR/EL/SER/GPS	Daily
3	Calibration of Commercial Energy meters	Every Five Years/ As per CERC guidelines
4	Cleaning with air blower and vacuum cleaner	Yearly/OH
5	Tightening of cable/ wire connections	Yearly/OH
6	Checking of panel earthing	Yearly/OH
7	Checking of voltage (in service) for relays	Yearly/OH
8	Checking of DC logic circuits for trip and annunciations including timers by simulation	Yearly/OH
9	Calibration of panel meters (Indicating/recording instruments along with the transducers), if any	Once in Every Four years

- Distance protection

Sl. No.	Activities	Frequency
1	Reach check for all 4 zones*	Yearly
2	Time measurement	Yearly
3	Power swing blocking check	Yearly
4	Switch on the fault (SOTF) check	Yearly
5	Level detectors of pps.	Yearly
6	Fuse failure check	Yearly
7	Polarization check	Yearly
8	Negative phase sequence (NPS) detector check	Yearly
9	VT fuse failure check	Yearly

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* Includes Z1, Z2, Z3 and Z4(reverse)

Note

- The above schedule for Distance relay is generic in nature and the manufacturer's maintenance instruction to be referred for any particular make of relay testing.
- As there are a number of different unit protections, the manufacturer's maintenance instruction to be referred for maintenance
- Whenever relays are tested on-line, proper isolation of the relay under test to be ensured.
- The other protection to be healthy and in service before taking any protection for on-line testing.

• Common tests for protections

01.	Trip contacts check	Yearly/OH
02.	Annunciation check	Yearly/OH
03.	Check for carrier send & receive with remote end	Yearly/OH
04.	Check for Direct trip send & receive with remote end	Yearly/OH
05.	Over voltage relays	Yearly/OH
06.	Local breaker back-up check	Yearly/OH
07.	STUB protection check	Yearly/OH
08.	Fault locator initiation check	Yearly/OH
09.	DR. EL initiation check	Yearly/OH
10.	Auto reclose check	Yearly/OH
11.	DC logic	Yearly/OH
12.	Reactor back up impedance	Yearly/OH
13.	Auxiliary relays healthiness	Yearly/OH
14.	Auxiliary relays (Buchholz, PRD, etc)	Yearly/OH
15.	Reactor differential protection	Yearly/OH
16.	REF protection	Yearly/OH
17.	DC logic	Yearly/OH
18.	Over fluxing relay	Yearly/OH
19.	Over load	Yearly/OH
20.	Directional over current	Yearly/OH
21.	LBB	Yearly/OH
22.	Auxiliary relays (Buchholz, PRV, etc.)	Yearly/OH
23.	Fuse failure check	Yearly/OH
24.	Transformer differential protection	Yearly/OH
25.	Restricted earth fault	Yearly/OH
26.	DC system earth fault relay	Yearly/OH

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27.	Bus bar & feeder trip relay functional checking	Yearly/OH
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- Bus bar protection

01.	Primary injection test *	SOS
02.	Protection stability and sensitivity checks *	SOS
03.	Relay and DC logic check	Yearly/OH
04.	Main Central Unit (MCU) relay testing for all Bus	Once in 5 years

(*->To be done whenever the protection AC circuits are disturbed like addition of new feeder)

- Differential relays

01.	Trip contacts check	Yearly/OH
02.	Pick up current at the fixed / selected setting	Yearly/OH
03.	Operation of high set element / instantaneous unit at the fixed / selected setting	Yearly/OH
04.	Operation of the relay at the selected restraint bias setting.	Yearly/OH
05.	Checking of 2nd & 5 th harmonic current restraint feature	Yearly/OH
06.	Operation of alarm and trip contacts.	Yearly/OH

- Under voltage relay

01.	Starting and pick up of the relay as per plug setting	Yearly/OH
02.	Relay operating time as per relay characteristic	Yearly/OH
03.	Operation of alarm and trip contacts	Yearly/OH
04.	Verification of input voltage on relay terminals	Yearly/OH

- Over voltage relay

01.	Starting and pick up of the relay as per plug setting	Yearly/OH
02.	Relay operating time as per relay characteristic	Yearly/OH
03.	Operation of high set element / instantaneous unit at voltage setting, if applicable	Yearly/OH
04.	Operation of alarm and trip contacts	Yearly/OH
05.	Verification of input voltage on relay terminals	Yearly/OH

- Neutral displacement relay

01.	Starting and pick up of the relay as per plug setting	Yearly/OH
02.	Relay operating time as per relay characteristic	Yearly/OH
03.	Operation of alarm and trip contacts	Yearly/OH

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04.	Verification of continuity of input circuit (for RVT / NCT secondary circuit in case of capacitor banks, under shutdown).	Yearly/OH
05.	Verification of open delta voltage input by by-passing PT secondary supply one phase at a time (in case of 3 nos. single phase PT"s).	Yearly/OH

- Over current and earth fault relay

01.	Starting and pick up of the relay as per plug setting.	Yearly/OH
02.	Time of operation as per relay characteristic	Yearly/OH
03.	Operation of high set element / instantaneous unit at current setting, if applicable.	Yearly/OH
04.	Operation of alarm and trip contacts	Yearly/OH
05.	Verification of input currents	Yearly/OH
06.	Verification of directional feature, if applicable.	Yearly/OH

- Over flux relay

01.	Operating of over flux alarm as per relay setting by varying the voltage and frequency one at a time	Yearly/OH
02.	Operating of over flux trip features as applicable for the following: - o IDMT characteristic - o Instantaneous element Fixed time setting	Yearly/OH
03.	Operation of alarm and trip contacts	Yearly/OH
04.	Verification of input voltage on relay terminals	Yearly/OH

- Local breaker back up protection, restricted earth fault (REF) and other instantaneous current operated relays

01.	Starting and pick up of the relay as per plug setting.	Yearly/OH
02.	Pick up value of the relay at the selected setting	Yearly/OH
03.	Operating time of the relay	Yearly/OH
04.	Operation of alarm and trip contacts	Yearly/OH
05.	Verification of input currents	Yearly/OH

- Fuse failure relays

01.	Remove main fuse of each phase voltage input to the distance protection scheme one by one in the relay panel Checking that the "VT Fuse Fail Alarm" is received. Checking that the distance protection does not operate	Yearly/OH
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6.0 SWITCHYARD EQUIPMENT MAINTENANCE PROCEDURES

6.1 GENERAL INSTRUCTIONS FOR MAINTENANCE OF SWITCH YARD EQUIPMENT

6.1.1 Insulator Cleaning

The insulators of the Circuit breaker / CT / CVT / Isolator/ BPI shall be cleaned from salt and dirt/dust deposition together with the cleaning of the other insulators in the switchyard.

1. For installation with higher atmospheric/saline pollution, cleaning frequency may be increased and it may be suitably protected against pollution. For such installations (e.g. Vallur, Simhadri, Dadri, Jhajjar etc), Hot line washing to be done at least twice in a year or as per condition (Preferably on pre-monsoon and post-winter). For severely polluted stations, increase the frequency to include washing before winter as well or as per site condition (except LA & Transformer Bushings) through certified Vendor. SOP to be shared with COS & site chemistry prior to work. Work to be carried out as per checklist mentioned in clause

Purpose: Pollution level, ash and dust deposition in switchyard area may lead to equipment failure. Before start of foggy season hot line washing is to be carried out to avoid insulation failure due to pollution etc.

Scope under live condition:

1. Cleaning of all insulators in EHV Switchyard and Transformer Yard
2. Bus Post Insulators
3. HCB Isolator insulators
4. CT, CVT
5. Circuit Breaker insulators
6. Switchyard Gantry string and suspension insulators.

Exclusion under live condition:

1. Lightning Arrestors
2. Power transformer bushings
3. EHV Dead tank Current Transformers (as per station feedback)
4. Any leaking Equipment insulator are not to be cleaned under live line conditions.

Operating principle, Safe practices & Precautions:

- Whenever a jet of water at very high pressure is released in air, the water particles split due to air resistance.
- At a sufficient distance the jet moves in the form of small particles, separated by air, and the water path is not continuous. Also due to triplex piston (3 Pistons) pump there will be breaking of water flow.
- When the water has sufficient electrical resistivity, there is negligible leakage current encountered when the nozzle is held at a predetermined distance from the live point.
- The high-pressure jet is produced by means of a pump. The demineralized (DM) water with conductivity of less than 5 micromhos per cm is used for washing purpose.
- Nozzle is held at a sufficient predetermined distance minimum 10 feet for 110

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- KV minimum 12 feet for 220 KV and min. 15 Ft for 400 KV from the live point, so that negligible leakage current is encountered.
- For safety, conductivity of water is continuously monitored by conductivity meter & leakage current is continuously monitored by leakage detector.
 - Working agency should have experience of Hotline washing job & certified through government body.
 - Availability of hotline high pressure washing pump kit in good working conditions.
 - There should not be rain/drizzling and wind speed should not be more than 35km/hrs.
 - Check that a valid work permit is issued by client to carry out hotline washing.
 - Electric supply source shall be adequate rating as per pump rating and from ELCB having 30 mA rating.
 - Ensure adequately equipped First aid box should be kept ready throughout the work period.
 - Ensure that all staff wear the safety PPEs. It shall be in good condition.
 - Remove wristwatch, ornaments, Mobile phone, and other things from pockets and deposit them with the Site Supervisor / Site Engineer.
 - Before starting the work check the wind direction.
 - Earth wire should be minimum 6mm copper, multistrand wire.
 - Conductivity meter and leakage detector healthiness should be ensured throughout the time of usage.
 - Check for water leakages at joints of hose. Rusty water may enter in system
 - Check that door of any panel, Junction box etc. should not be open
 - Ensure that no equipment bushing is broken/ cracks etc.
 - Ensure that no wire hanging on clamp, bushing etc.
 - Ensure that no Oil leakage in any equipment (CT/CVT).
 - Wash the insulator petticoats thoroughly by moving the jet from bottom to top & then again slowly from top to bottom. After washing the 1/4th length of bottom insulator length from all sides, jet to be applied for next 1/4th length of insulator. Subsequently for 3/4th & full length. From Top to Bottom, washing should be done in similar fashion.
 - Check water level in tank for sufficient water.
 - No water leakage from joints of hose pipe.
 - Delivery pressure should be between 800 psi to 1200 psi
 - Continuous monitoring to be ensured by NTPC personnel during Hot line washing works.
 - During offline washing, low pressure spray to be applied on insulators particularly on LAs & Transformer Bushings.
 - Apply chemical coatings (Silicone/Petrolatum, etc.) on transformer bushings, LAs, and other porcelain insulators.

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6.1.2 Rust Protection

Some parts of the operating mechanism are made of steel and are surface treated against rust. In spite of the good rust protection, minor corrosion will occur after some years, especially when the breaker / isolator is standing in strong corrosive surroundings. The rust stains shall be sand papered away and new rust protection shall be painted or sprayed by GI /Zinc oxide & Zinc phosphate paints. Rusted hardware should be replaced with new spare.

6.1.3 Lubrication

For lubrication, the lubricants recommended by manufacturers shall primarily be used. The bearings of the breaker and operating mechanism of isolator and breaker are to be lubricated with grease and they normally do not need lubrication before the major overhauls. Parts in mechanism such as arms, links and link gears are lubricated with grease

G. These bearings shall be regularly lubricated with a few drops of oil B. The teeth in the gear shall be lubricated with grease G. Dryness of driving mechanism may lead to mal operation and failure.

6.1.4 Treatment of contact surfaces

The contacts of breaker / isolator / ground switch shall be treated according to the following directions:

- Silvered contact surfaces: Silvered contact surfaces shall be cleaned, if necessary, with a soft cloth and solvent (trichloroethane). Steel brushing or grinding is not allowed.
- Copper surface: Copper surfaces shall be clean and oxide free. If necessary, they shall be cleaned with cloth and solvent (Trichloroethane) or steel brushing - After steel brushing, the surface shall always be cleaned from loose particles and dust.
- Aluminum surfaces: Aluminum contact surfaces shall be cleaned with steel brush or emery cloth. The surface is very thoroughly cleaned from particles and dust with a dry cloth. After this, a thin layer of Vaseline is applied. This shall be done within 5 minutes after the cleaning. The joint shall be assembled within 15 minutes.

6.1.5 Moving Contact Surfaces

- i. Silvered: Cleaned if necessary, with soft cloth and solvent (trichloro ethane). No steel brushing.
- ii. Non-silver coated: Cleaned as silvered surfaces, can be steel brushed.
- iii. After steel brushing, they shall be thoroughly cleaned from loose particles and dust.
- iv. Lubrication: Lubricant - Grease K is applied in a very thin layer on the surfaces of the male contact and the puffer cylinder. The superfluous grease is carefully removed.

6.1.6 Safety

Every maintenance/ testing work needs to be carried out with proper clearance in

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form of Permit to Work (PTW). The person who is authorized to issue such clearance and the person who is authorized to receive such clearance independently ensure that proper isolation has been carried out and earthing provided at appropriate locations. Special Sanction for Test (SFT) is required when it is required to remove earth connection for performing tests.

6.1.7 Precautions

Before performing a test, it is very much required for the maintenance engineer to have knowledge of the equipment and the testing instruments used and to apply all the requirements to safeguard the equipment, testing instrument and more important human life. The boundary of work area is to be identified and caution is to be exercised not to inadvertently contact the live neighboring components. Proper isolation of AC and DC power supply is to be ensured to avoid electric shock. Always ensure proper ladder and other safety equipment while climbing to higher places. Make shift arrangements may lead to accident.

6.1.8 Analysis of Test Results

All test results to be carefully analyzed preferably compared with earlier test result and pre-commissioning test results to arrive at conclusion. A trend analysis is more important to identify the evolving defects and to take corrective action.

6.2 CIRCUIT BREAKERS

Circuit breakers basically consist of two main parts, the interrupting chambers and the operating mechanism. The interrupting chambers normally do not require routine preventive maintenance other than cleaning but operating mechanism do require proper upkeep.

a) Interrupting Chamber

Circuit breaker interrupting chamber is an enclosed unit mostly filled with oil or SF₆ gas. Lower voltage Circuit breakers have vacuum interrupting chambers also. There is stress on the contacts during fault current interruption and damages may happen in arcing contacts or main contacts. The breaker interrupting chamber is recommended to be opened only based on condition monitoring tests or as per advice of the manufacturers.

In SF₆ circuit breaker, whenever there is fault interruption, there is formation of metallic fluoride inside the arcing chamber. This fine white/ gray powder has high dielectric strength when it is dry. Moisture absorbent material is provided in circuit breaker to keep the gas dry. The absorbent material is to be changed as per advice of manufacturers. The absorbents are normally packed in air tight sealed container and it should not be exposed to atmosphere for long time (not more than 15 m) as it may absorb moisture from air. SF₆ pressure and dew point also to be monitored since these parameters are vital for successful fault clearance by the breaker. If there is loss of SF₆ gas pressure, then leakage test with leak detector to be carried out to identify the leak area for rectification.

In minimum oil circuit breakers, breather is provided to keep the oil free from

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moisture. Breather silica gel need to be monitored and changed when ever required. The level and quality of oil are also to be monitored.

b) Operating Mechanisms

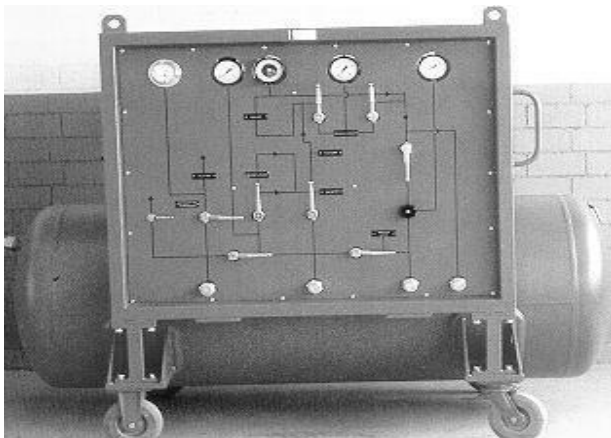
Normally circuit breakers have pneumatic, hydraulic and spring operating mechanisms. As operating force is required for closing and tripping of circuit breakers, there can be combination of these mechanisms in one circuit breaker. Since operating mechanisms have a number of moving parts, they need more maintenance such as greasing, lubrication, cleaning, setting of limit switches, etc. Compressors/ oil pumps/ spring charging motors also require maintenance. Other maintenance on particular operating mechanism such as air compressor maintenance, nitrogen priming pressure checking in hydraulic mechanism, checking of over travel, checking of gaps in operating plunger of close/ trip coils etc. are to be carried out as the case may be and as specified by the manufacturers.

c) SF6 Gas

Most of the higher voltage circuit breakers adopt SF6 in interrupting chamber. The density of SF6 gas is about five times that of air and heat dissipation is also much more than air. At atmospheric pressure, dielectric strength of SF6 gas is about 2.4 times that of air and at about 3 kg/cm², it is same as that of oil. As SF6 is Green House gas, it needs to be handled carefully and should not be let to the atmosphere.

d) Emptying and re-filling of gas

The breaker is evacuated by means of the gas treatment equipment that purifies and also compresses the gas for storage, so that it can be reapplied. For economic and ecological reasons, SF6 contained in electrical equipment, should not be vented into atmosphere. Prior to the gas removal, the quality of the SF6 gas should be tested.



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SF₆ gas handling plant

Operational contamination should be absorbed with suitable filter unit provided in the gas handling plant. Such filters/ sieves should already be installed into the SF₆ gas maintenance/ handling unit. When SF₆ is suctioned from a gas compartment, the gas is passed automatically through filters, which dry and purify the gas.

e) Evacuation of SF₆ gas Circuit Breakers

After maintenance/ overhaul of the circuit breaker, it should be evacuated by vacuum pump before filling in the SF₆ gas so that SF₆ gas does not mix with ambient air and also humidity and dust particles are removed from the Breaker. With vacuum pump, a final vacuum must be reached less than 5 mbar.

f) Grading Capacitors

Grading capacitors are provided in multiple break Circuit Breakers to equalize potential difference across the breaks. If grading capacitors are provided, it is to be ensured that they are properly connected and there is no leakage of oil from the grading capacitors. The insulator porcelain to be kept clean and condition monitoring tests are to be carried out to identify the healthiness.

g) Pre-insertion Resistors

Pre-insertion Resistors (PIR) are provided in circuit breakers to limit surges during switching. Unless any abnormality is noticed during condition monitoring tests, no internal inspection of the PIR contacts of resistors is required. The insulators are to be kept clean and ensuring proper tight connection with the circuit breaker.

6.3 CURRENT TRANSFORMERS

6.3.1 Visual Inspection

Current transformers are normally filled with oil and have oil impregnated paper insulation for both primary and secondary winding. Careful inspection is to be made for any trace of oil leakages. Oil leakages are more prone through cemented joints or secondary terminal box due to improper sealing of terminal studs. As CTs have less oil quantity small leakage may lead to exposure of paper insulation and subsequent moisture absorption.

If bellows are provided in CTs, the position of bellow lower or higher than normal range indicates either leakage of oil or expansion due to internal gas generation. Both the conditions are serious for the life of the CTs and immediate action to be initiated for rectification.

Visual inspection is also to be carried out on the healthiness of terminal connections, condition of porcelain, development of cracks, chippings, cleanliness of insulator surface etc.

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6.3.2 Maintenance of Gaskets

Marshalling boxes, CT terminal boxes are to be properly sealed to prevent any dust, rain water and insects. Door gaskets are to be changed periodically to give proper sealing. All door bolts/ latches are to be properly tightened and never left loose.

6.3.3 Secondary Terminals Connection

Stud type terminals with ring type thimbles are to be used in Marshalling box cable terminals. This gives better grip even if more than one wire is connected to one terminal. The tightness of wires may get loosened due to vibration, climatic condition, it is required to check tightness of terminals periodically to avoid mal operation/ non-operation due to improper contacts. All terminals of unused CT secondary terminals are to be properly shorted to avoid development of abnormal voltage and subsequent failure of CTs. The tan δ test tap is to be properly earthed to avoid damage to insulation. It is to be ensured that all extra holes at Marshalling boxes/ terminal boxes are properly plugged and kept vermin proof.

6.3.4 Primary Terminals

Proper connection of primary terminal can be ascertained by thermal scanning. If Thermal scanning is not carried out, physical checking of terminal connection is to be checked with proper torque. All corona shields to be provided and any damage to corona shield to be replaced with new one. As CT primary carried heavy current, any loose joint may lead to arcing and welding of terminal connectors.

6.4 CAPACITANCE VOLTAGE TRANSFORMERS/ POTENTIAL TRANSFORMERS/ CAPACITOR COUPLING

6.4.1 Visual Inspection

The bellows provided in most of the CVTs are not visible from outside. PTs/ CVTs and CC are also oil filled equipment and oil leak is to be observed. If oil leak is observed in any one stack, the entire CVT is to be replaced. CVTs are tuned units and replacement of any one stack is not recommended to avoid phase angle errors.

6.4.2 Electro-Magnetic Unit

Electro-Magnetic Unit (EMU) of CVT houses the secondary transformer, Compensating reactor and Ferro resonance suppression circuit. The color of oil indicated through the gauge glass gives some indication of the healthiness of the internal components. Any abnormal heating may also be observed through Thermal scanning.

6.4.3 Secondary Voltage

Deviation in secondary Voltage of CVT is clear indication of failure of capacitor elements. Necessary action should be taken to replace CVT if secondary voltage in any one CVT is abnormal (+2V and -4V). Continuing the equipment in service beyond this stage may lead to failure/ bursting of CVTs.

6.4.4 Other Maintenance

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Maintenance of Marshalling box gaskets, tightening of secondary terminal connections and tightening of primary terminal connections, etc., are also to be ensured for healthy operation. It is to be ensured that all extra holes at Marshalling boxes/ terminal boxes are properly plugged and kept vermin proof.

The anti-condensation heater and the thermostat are to be kept in working condition to keep inside of the panel dry.

6.5 ISOLATORS / DISCONNECTORS

Isolators have main current carrying arms and operating mechanism for connection and disconnection. As isolators are off-line devices, they are normally air break devices. Normally horizontal double break, Horizontal center break, Pantograph, Vertical break Isolators are in use for EHV isolations.

The alignment of isolator is very important for smooth operation. The limit switches, the healthiness of auxiliary contacts needs to be checked periodically. The main contacts are to be inspected and to be made smooth if any pitting marks seen. The corona shields to be kept smooth and shining and checked for tightness of fitting. Damaged corona rings to be replaced. All moving parts to be lubricated for smooth operation. The gear mechanism and motor normally need not require any maintenance and manufacturer's recommendation to be referred for maintenance of gears.

6.6 EARTH SWITCHES

The earth switch is a safety device and smooth operation of the earth switch is to be ensured by proper alignment. The earth blade contact point to be cleaned properly for proper contact and contact resistance to be measured to ensure healthiness. The earth connection from blade to earth is to be carefully checked. All the joints to be tightened and flexible copper braid connections are provided the healthiness is to be ensured. All moving parts to be lubricated for smooth operation.

6.7 LIGHTING ARRESTER/SURGE ARRESTERS

Surge arresters are to be maintained to give protection to other connected switchyard equipment. Cleaning of porcelain insulators is very much required for uniform voltage distribution. Voltage grading rings are to be properly positioned and checked for tightness and any damaged rings to be replaced. Healthiness of surge monitors are to be checked and if found defective to be replaced with healthy monitor. Healthiness of earth connections to be checked as it plays a vital role on the operation of the surge arrester. Normally it is not recommended that if one stack fails it is replaced with healthy stack. It is always a good practice to change the entire arrester as the stressed stacks will start failing along with the new stack.

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7.0 SWITCHYARD EQUIPMENT TESTING PROCEDURES

7.1 CIRCUIT BREAKERS

7.1.1 CAPACITANCE AND TAN δ OF CB VOLTAGE GRADING CAPACITOR

Circuit Breaker grading capacitors have oil filled paper capacitor elements. The healthiness of the capacitor elements could be checked by measurement of Capacitance and Tan δ .

a) Test Procedure

Connect LV cable to the middle of the double interrupter. Connect HV cable to the other end of the Grading capacitor to be tested. The opposite end of the other grading capacitor must be grounded using earth switch. Before applying HV, interference is to be nullified using Interference suppression unit (ISU).

Measurements must be taken in UST Mode only. Disconnect the HV cable and connect the same to the other grading capacitor and ground the previous grading capacitor. Now the second grading capacitor is ready for testing. Standard procedure (as specified by kit supplier) for measuring capacitance and tan delta in charged switchyard/induced voltage conditions should be followed. Carry out the measurements in main and reverse mode/polarity and then compute the average value.

b) Evaluation of Test Results

Increase in the value of tan δ will indicate Chemical deterioration due to age and temperature, including certain cases of acute deterioration due to localized overheating, contamination by water, carbon deposits, bad oil, dirt and other chemicals or ionization. The acceptable value is 0.007 but any increase in tan δ and any change in capacitance value is to be analyzed.

7.1.2 DEW POINT MEASUREMENT OF SF6 GAS/ OPERATING AIR FOR CBS

To measure Dew point of SF6 gas and operating air used in circuit breakers.

Dew Point is the temperature at which moisture content in SF6 gas/air starts condensing.

a) Dew Point at Rated Pressure of CB

Dew Point when measured keeping regulating valve in service at the outlet of dew point kit to allow required flow rate of gas/air, is called dew point at rated pressure of CB.

b) Dew Point at Atmospheric Pressure

Dew point when measured by regulating the gas flow at the inlet of dew point kit and keeping outlet regulating valve (if provided) in fully open condition so that flow rate of gas/air is maintained as required, is called dew point at atmospheric pressure.

c) Test Equipment

Dew point measuring kit and associated accessories.

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d) Testing Procedure

All the joints/connectors should be dust and moisture free. If required, same should be cleaned by clean cloth. Dry the joints and pipe by dry air. Make the connections to the kit from CB pole ensuring that regulating valve is fully closed at the time of connections of the Dew point kit. By regulating the flow rate of SF₆ gas (0.2 liter/min to 0.5 liter/min - ref. IEC 480), the value of dew point is observed till it becomes stable. If the regulating valve is provided at outlet of the dew point kit then dew point for rated pressures is to be monitored.

e) Evaluation of Test Results

Measurement of Dew point of SF₆ gas is an adequate parameter for condition monitoring SF₆ gas in a CB. Dew point measurement of SF₆ gas in a CB indicates the change in the Value of dielectric properties of SF₆ gas. The dielectric properties of SF₆ gas do get changed with time due to mixing of impurities like moisture; decomposition products of SF₆ gas.

There are two sources of moisture ingress in SF₆ gas after it is filled in CB and during service period. Exudation of moisture contained during manufacturing from insulation materials used in Circuit Breakers and permeation of moisture through sealed sections, i.e., gaskets, 'O'-Rings, etc.

f) Sulphur Hexafluoride Gas

During Arc interruption in CBs, decomposition of SF₆ gas takes place which in presence of moisture, may result in deterioration of Organic Insulating materials inside interrupting chamber and corrosion of metals due to formation of hydro fluorides. Therefore, to avoid dielectric failure of CBs, monitoring of moisture content in SF₆ gas is very important.

Dew Point of SF₆ gas varies with pressure at which measurement is being carried out. This is since Saturation Vapor Pressure decreases with increase in Pressure of the SF₆ gas. Hence, dew point of SF₆ gas at higher pressure is lower than dew point at atmospheric pressure. Therefore, if measurement is done at a pressure other than the atmospheric pressure, same is to be converted to the atmospheric pressure Dew point as recommended by manufacturers are given in Table below:

Dew point recommended by manufacturers

Sl. No	Make of CB	Dew point at rated pressure	Dew point at atmospheric pressure	Remarks
01.	BHEL	- 15 °C	- 36 °C	At the time of commissioning
		- 7 °C	- 30 °C	During O & M
		- 5 °C	- 27 °C	Critical
02.	CGL	- 15 °C	- 35 °C	At the time of commissioning
		- 10 °C	- 31 °C	During O and M
03.	ABB	- 15 °C	- 35 °C	At the time of commissioning

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			- 5 °C	- 26 °C	During O and M
04.	GE/ASLTO M		- 15 °C	- 36 °C	At the time of commissioning
			- 7 °C	- 30 °C	During O and M
			- 5 °C	- 27 °C	Critical
06	SIEMENS		- 15 °C	- 35 °C	At the time of commissioning
			- 10 °C	- 30 °C	During O and M
			- 5 °C	- 25 °C	Critical

7.1.3 MEASUREMENT OF CIRCUIT BREAKER OPERATING TIMINGS INCLUDING PRE- INSERTION TIMINGS

To measure Operating Timings of Circuit Breakers to evaluate the operational performance.

a) Test Equipment

Circuit Breaker Operational Analyzer and associated accessories.

b) Testing Procedure

Make connections according to standard set up of CB analyzer. It is to be ensured that R, Y, B phase marking cables are connected with the proper place in the CB analyzer and color codes are to be maintained for all the three poles of CB.

Give closing command to closing coil of CB and note down the PIR and main contact closing time. Take the printout from the analyzer. Give tripping command to trip coil-I of CB and note down the main contact tripping time. Give tripping command to trip coil-II of CB and note down the main contact closing time. Note down the timings for 'C-O', 'O-C' and 'O-C- O' by giving respective commands.

c) Evaluation of Test Results

- Closing timings

Closing time of EHV CBs should not exceed the permissible limits and discrepancy should not be there between main contacts and PIR contacts. Discrepancy in operating times of PIR and main contacts should not exceed the permissible limits. In any case, main contacts should not close prior to closing of PIR contacts and PIR contacts should not open prior to closing of main contacts. In case, contact bouncing is observed in operating timings for PIR and main contacts, same is to be rectified.

- Tripping timings

Trip time for EHV CBs should not exceed the permissible limits. Discrepancy in operating timings for Pole to Pole and Break to Break should not be allowed beyond permissible limits which is 3.33 ms and 2.5 ms respectively. In case, operating times of different poles/breaks are not within permissible limits, Dynamic Contact Resistance measurement to be carried out to see the healthiness of main contacts.

- C-O Timings

C-O timings should be within permissible limits as specified by different manufacturers.

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7.1.4 MEASUREMENT OF DYNAMIC CONTACT RESISTANCE OF EHV CIRCUIT BREAKERS

To monitor condition of CB main and arcing contacts without opening the interrupter.

a) Test Equipment

DCRM kit with CB operational analyzer

b) Testing Procedure

CB should be in open position, Isolator of both sides of CB should be in open position and Earth switch of one side of CB should be in open position. Make connections according to standard set up of CB analyzer along with DCRM kit.

The tightness of connections at CB flanges is most important to ensure error free measurement). CB during CO operation generates lot of vibrations and failure of connections during this period can dramatically change the dynamic signature of CB resistance.

Incoming and outgoing flanges of CB to be cleaned with polish paper to remove paint, oxidation, etc., at points where current clamps are mounted. Select this point of connection, as close as possible to the end of porcelain insulator to ensure that minimum resistance is offered by flanges, bolts, terminal connectors, etc.

c) Evaluation of Test Results

Condition of contacts can be monitored by comparing DCRM signatures taken at the time of pre-commissioning and subsequent service. If there are wide variations in resistance value from pre-commissioning value, it indicates erosion of contacts. DCRM is also helpful in monitoring length of arcing contacts by measuring contact travel and dynamic contact resistance. The contact travel from the point of opening of main contacts to opening of arcing contacts is length of arcing contact. During service due to erosion of contacts, length of the same gets shortened to the extent that it may lead to commutation failure, hence, monitoring length of arcing contacts is very important.

7.1.5 MEASUREMENT OF CONTACT RESISTANCE OF CIRCUIT BREAKER

To measure contact resistance of circuit breaker main contacts.

a) Test Equipment

Micro ohm meter of 100A and above with associated accessories.

b) Testing Procedure

CB should be in close position and Isolators on both side of CB should be in open position. The ohm meter operates on the four-wire measurement principle. To measure the contact resistance, connect the respective leads and adjust the variac so that 100 amperes current flow through the CB contacts. Value of contact resistance is directly displayed on the digital LED display screen.

c) Evaluation of Test Results

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Contact resistance of the CB contacts indicates wear out and misalignment of the main contacts. If the value of contact resistance exceeds the permissible limit, i.e.,

150 micro-ohm this could result in overheating of the contacts. Therefore, the problem of high contact resistance should be attended immediately by making proper alignment of contacts or by replacing arcing/main contacts.

7.1.6 CHECKING OF POLE DISCREPANCY RELAY AND TIMER FOR CIRCUIT BREAKERS

a) Definition

Pole discrepancy is defined as the difference in closing and opening timings of different poles of CB.

b) Test Equipment

Standard timer, testing leads, etc.

c) Testing Procedure

Isolators on both sides of CB to be in open position and Earth switch in closed position.

(i) When CB in open position

Closing command is extended to close one pole, say R-Pole, of CB. After closing R-Pole of CB, this Pole should automatically open after set time (as per pole discrepancy timer settings). Repeat the test for remaining two poles of CB.

(ii) When CB in closed position

Tripping command is extended to trip one pole, say R-Pole, of CB. Remaining Y and B- Poles of CB should automatically open after set time. Repeat the same test for remaining two poles of CB.

(iii) Timer checking

Connect standard timer in parallel with CB timer. Extend 220 volt DC to both the timers and note down the timings at which output contacts of timer (which are going to pole discrepancy trip relay) gets closed. If there is a difference between timing readings of two timers, adjust the CB timer's settings accordingly.

d) Evaluation of Test Results

Permissible value of pole discrepancy between two poles of CB is 3.33 msec. However, timer is set to trip the remaining poles taking into consideration the adopted auto reclose scheme. For Unit/ICT/ST breakers timer setting should be 500ms. For Line breakers PD timer setting should be 1.5 sec (Main CB) & 2.5 sec (Tie CB).

7.1.7 OPERATIONAL LOCKOUT CHECKING FOR CIRCUIT BREAKERS

Operational lockouts are provided in circuit breakers to prevent the operation of the breaker during deficiency in the circuit breaker or its operating mechanism. The lockouts provided in CBs is to be tested to check for any drift/ non-function as they are safety devices to prevent improper operation leading to failure.

a) Low Pressure Alarm

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Close isolation valve between CB Pole and density monitor. Start releasing SF₆ gas from density monitor till the low-pressure gas alarm contacts are actuated which is detected by Multimeter. Note down the pressure and temperature at which the contacts get actuated.

b) Operational Lockout Alarm

Continue releasing SF₆ gas from isolated zone till the operational lockout alarm contacts are actuated which are detected by Multimeter. Note down the pressure and temperature at which the contacts get actuated. Gas is to be taken in to gas evacuating unit and after checking gas to be filled to the rated pressure in the CB.

c) Pneumatic Operating System Lockout

(i) Compressor START/STOP Switch

Close the isolating valve of CB. Release air into atmosphere from the compressor. Note down the value of pressure at which compressor starts building up air pressure and pressure at which compressor stops.

(ii) CB auto reclose lockout

Close Isolation valve between pneumatic system and pressure switches. Release air from the isolated zone to atmosphere. Note down pressure at which A/R L/O contacts of pressure switch get actuated which are detected by multi meter. The leads of the multi meter should be connected to the contactor where the AR L/O of CB is made.

(iii) CB closing lockout

Release air from the isolated zone to atmosphere. Note down pressure at which CB Closing L/O contacts of pressure switch get actuated which are detected by multi meter.

(iv) CB operational lockout

Release air from the isolated zone to atmosphere. Note down pressure at which CB Operational L/O contacts of pressure switch get actuated which are detected by Multimeter.

(v) Mechanical Closing Interlock (MCI) (FOR ABB CBs ONLY)

CB should be in closed position. Release air from pneumatic system of CB to atmosphere and observe whether CB poles start opening, if so, note down the pressure at which tie rod starts coming down, the movement to stop after a few milli meter. The MCI is provided to prevent opening of the CB. Non operation of MCI is to be investigated.

(vi) Hydraulic operating system lockout

By opening pressure release valve, note down the pressure at which pump starts building up oil pressure and pressure at which pump stops.

(vii) CB auto reclose lockout

Close Isolation valve between hydraulic system and pressure switches. Release oil from the isolated zone to oil tank. Note down pressure at which A/R L/O contacts of pressure switch get actuated which are detected by multi meter.

(viii) CB closing lockout

Release oil from the isolated zone to oil tank. Note down pressure at which CB Closing L/O contacts of pressure switch get actuated, checked by

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Multimeter.

d) Evaluation of Test Results

(i) SF₆ gas pressure lockout

All the SF₆ gas pressure switches settings should be checked and corrected with ambient temperature. Settings of SF₆ gas pressure switches should be within ± 0.1 bar/ kg/cm² of the set value (after considering the temp. correction factor).

(ii) Air pressure lockout

All the air pressure switches settings should be checked and corrected and should be within ± 0.1 bar/ kg/cm² of the set value.

(iii) Oil pressure lockout

All the oil pressure switches settings should be checked and corrected and should be within ± 0.1 bar/ kg/cm² of the set value.

7.2 CURRENT TRANSFORMERS

7.2.1 CAPACITANCE AND TAN DELTA MEASUREMENT

Capacitance and tan δ measurement on CTs to evaluate the healthiness of solid insulation. As most of the CTs have oil impregnated paper as solid insulation, capacitance and tan δ measurement indicates the amount of deterioration. The testing is carried out similar to the measurement carried out for transformer bushings.

a) Precautions

CT is to be isolated from supply and proper isolation to be done before carrying out this test. CT terminal Jumper is to be disconnected from HV terminal of CT if test tap is not provided. If test tap is provided, test tap should be disconnected from ground before measurement and to be grounded after the measurement is completed. Measurement is to be carried out at 10 kV or 12 kV level with highest sensitivity level. As CTs are in high induction area, testing instrument should be able to suppress the interference.

b) Test Procedure

(i) CTs with test taps

Tan delta tap should be disconnected from ground. High voltage from tan delta kit should be applied to primary (HV) Terminal and LV should be connected to the Tan delta test tap. Before applying HV, interference is to be nullified using Interference suppression unit (ISU). Measurements must be taken in UST – Mode. Standard procedure (as specified by kit supplier) for measuring capacitance and tan delta in charged switchyard/induced voltage conditions should be followed for measurement of capacitance and tan delta values. Measurements are to be carried out in main and reverse mode and then the average value to be computed.

(ii) CTs without test tap

Jumpers are to be opened before carrying out measurements. High voltage

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from tan delta kit should be applied to primary (HV) Terminal and LV should be connected to the ground/earth. Before applying HV, interference is to be nullified using Interference suppression unit (ISU). Measurements must be taken in GST – Mode. Standard procedure (as specified by kit supplier) for measuring capacitance and tan delta in charged switchyard/ induced voltage conditions should be followed.

Measurements are to be carried out in main and reverse mode and then the average value to be computed.

c) Evaluation of Test Results

The permissible values of capacitance and dissipation factor of class A insulation, e.g., Paper insulation impregnated in oil is 0.007 at 20°C. Tan delta value in no case should exceed the permissible value. It is still more important to monitor the rate of change of Tan delta and capacitance. Tan delta value should not increase by more than 0.1% per year and capacitance value by 1% per year. So before taking any decision about condition of CVT above facts need to be considered. Change in capacitance values can be within +10%, -5% of the original value.

An increase in only tan δ value (not appreciable change in capacitance value) indicates deterioration of cellulose insulation whereas increase in both tan delta and capacitance values indicates entry of moisture in the insulation.

Main reason for increase of tan δ value is because of presence of inherent air voids which are created during manufacturing process. In course of application of high voltage, these voids are ionized which result in deterioration of insulating properties of the insulation.

7.2.2 INSULATION RESISTANCE MEASUREMENT FOR CURRENT TRANSFORMERS

IR measurement of all EHV Current Transformer between HV (Primary) Winding and Test tap (for CTs having test taps) and between HV and Earth (for CTs not having test taps).

a) Test Equipment

5 kV Insulation tester and associated accessories like test leads etc. Motorized insulation tester is preferred for better results.

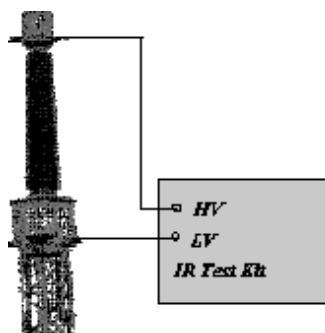
b) Isolations Required

- Associated Circuit Breaker to be in open position.
- Isolators from both sides of CT should be in open position.
- Earth switch should be open at the time of IR measurement.

c) Testing Procedure

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Connections are to be given as shown in the figure below. HV terminal of the instrument is to be connected to the Primary terminal of CT by using crocodile clip for firm grip and



LV terminal to be connected to ground. A test voltage as specified is applied and successive readings are taken. Values of IR are to be recorded after 15 seconds, 60 seconds and 600 seconds. Ambient temperature and weather conditions are to be recorded.

d) Evaluation of test Results

Changes in the normal IR value of CT indicate abnormal conditions such as presence of moisture, dirt, dust, crack in insulator of CT and degradation of insulation. Changes in the IR value of CT are also based on the weather conditions. It is advised to carry out IR measurement during sunny and dry weather preferably. Insulation Resistance changes with deterioration in insulating properties. Absolute value of IR is important to monitor but the rate of change is equally important. The value of IR should not be less than 1000 MΩ. If readings of IR increase with time, the insulation is good. However, if readings remain same over the time span, insulation is contaminated. This is because charging current and absorption currents subside with time and only conduction current remains. This indicates that total current taken by insulation changes with time. However, if there is no appreciable change in the total current drawn by the insulation, it is an indication of domination of conduction current over charging and absorption currents. Factors to be monitored for these abnormalities are briefly given below:

Dielectric absorption ratio = (IR value after 60 seconds) / (IR value after 15 seconds)
Polarization Index = (IR value after 600 seconds) / (IR value after 60 seconds)

If Dielectric Absorption Ratio is above 1.5 then insulation quality is assumed to be good. If Polarization Index is more than 1.3 then also insulation quality is assumed to be good.

7.2.3 . MEASUREMENT OF SECONDARY WINDING RESISTANCE FOR CURRENT TRANSFORMERS

To measure secondary winding resistance for Current Transformers to check the healthiness of secondary winding

a) Test Equipment

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Digital or Analog ohm meter with leads

b) Isolations Required

CB to be in open position and Isolators from both sides of CT should be in open position. Test links to be opened in the CT MB prior to measurement of secondary resistance. It is to be ensured that associated CTs are not in charged condition. For example, Main and Tie CTs for differential relays should not be in charged condition. If any earth is provided in the secondary circuit of CT, same is to be removed prior to measurement.

c) Testing Procedure

Connect leads of ohm meter between different terminals of CT secondary cores. Select the range of ohm meter as per pre-commissioning/factory test results. Record the winding resistance values.

d) Evaluation of Test Results

Value of secondary winding resistance should be within acceptable limits. Extreme low value of resistance indicates turn to turn shorting whereas if the value is high, it indicates some loose connection which is to be identified and tightened before repeating the measurement.

7.2.4 MAGNETIZATION CHARACTERISTICS OF CORES FOR CURRENT TRANSFORMERS

To carry out magnetization characteristics of CT cores for current transformers.

a) Knee Point Voltage

Knee Point Voltage is defined as the voltage at which a 10 % increase in flux density (Voltage) would cause 30% to 50% increase in exciting ampere-turns (current).

b) Test Equipment

Voltage source of 5 kV, Voltmeter of range 0 to 5 kV, Ammeter of range 0 to 100 Amps, testing leads/cables, etc.

CB and Isolators from both sides of CT should be in open position. Test links should be opened in the CT MB prior to measurement of secondary resistance. It should be ensured that associated CTs are not in charged condition. For example, main and tie CTs for differential relays should not be in charged condition. If any earth is provided in the secondary circuit of CT, same is to be removed prior to measurement. Applied voltage to the CT core should not exceed the rated Knee Point Voltage of the CT.

c) Test Procedure

Voltage is injected in the secondary after making proper connections. Applied voltage is increased from zero to rated Knee Point Voltage in steps of 25%, 50%, 75% and 100%. Measure the current drawn by the CT secondary core at respective applied voltages and record the test results.

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d) Evaluation of Test Results

The magnetization test is conducted to see the condition of turns of CT secondary. This test shall give indications regarding shorting of turns CT secondary winding. Magnetization characteristic also indicates the suitability of CT for keeping it in service or not.

Knee Point Voltage is normally defined as the voltage at which 10% increase in the applied voltage causes 30% to 50% increase in secondary current. The magnetization current at rated Knee Point Voltage should not be more than the specified/designed value. A curve can be drawn between applied voltage and magnetizing current.

From the curve it can be implied that up to rated KPV (Knee Point Voltage), the VI curve is almost a straight line. However, if this line is not linear, this indicates that the magnetizing characteristics are not desirable. If the slope of the curve starts increasing, it indicates that magnetizing induction becomes low and total primary current is utilized in exciting the core alone. Consequently, output of CT secondary disappears.

7.3 CAPACITIVE VOLTAGE TRANSFORMERS (CVTs)

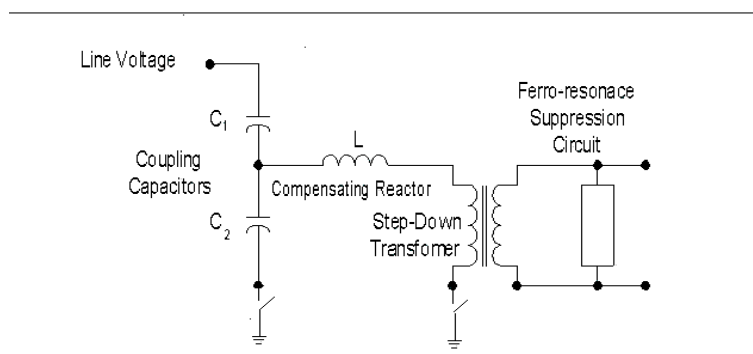
7.3.1 MEASUREMENT OF SECONDARY VOLTAGE

To evaluate the healthiness of capacitive voltage transformer by measurement of secondary Voltage with respect to other phase CVTs and also CVT of same phase installed in another feeder.

As CVTs are made up of capacitor elements, any puncturing of element will alter the secondary voltage of CVT. Capacitor elements in CVTs are designated as C1 and C2 as shown in the diagram below. In 400 kV CVTs, there are about 280 - 300 elements in C1 and C2 out of which C1 will have about 260 to 280 elements and C2 will have about 15 to 20 elements leaving ratio of C1/ C2 to about 20 as tap voltage. It is estimated that puncturing of one capacitor element in C1 side is likely to increase secondary voltage by about 0.35 – 0.45% (0.22 – 0.28 V) and failure of one capacitor element in C2 side is likely to decrease secondary voltage by 5 – 6% (3.2 - 3.8 V)

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CVT Basic Circuit



CVT basic circuit

a) Testing Procedure

Periodic measurement of secondary voltage is to be carried out to find healthiness of the capacitor elements. The measured voltage is to be compared with other phases of the same CVT and in case of doubt, simultaneous measurement to be carried out with another feeder/ Bus CVT. A precision multi meter of not less than accuracy of 0.5 class for AC voltage is to be used for this purpose.

b) Evaluation of Test Result

Any drift in secondary voltage of CVT must be analyzed and action taken accordingly. There may be difference in secondary voltage due to unbalance in primary voltage also. This will be marginal and simultaneous measurement with another healthy CVT could help identify the problem. The following table gives a broad guideline for action to be taken to identify the defective CVTs and to eliminate the defect from service.

Sl. No.	Drift in secondary voltages	Condition of CVT	Measurement frequency
01.	Up to 0.5 volts	Healthy	Six Monthly
02.	0.5 to 0.8 volts	Needs monitoring	Three Monthly
03.	+0.8 to +/-1.2 volts	Needs close monitoring	Monthly
04.	+1.2 to +/-2.0 volts	Needs close monitoring	15 days
05.	Above +/-2.0 volts	Alarming/ Critical	Needs replacement*.
06.	-0.8 to – 4.0 volts	Needs close monitoring	15days
07.	Less than - 4.0	Alarming	Needs replacement*

* Before replacement of CVT, voltage drift is to be reconfirmed by simultaneous measurement of voltage with respect to other feeder CVT of same phase.

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7.3.2 CAPACITANCE AND TAN DELTA MEASUREMENT

CVT condition monitoring based on capacitance and $\tan \delta$ measurement. Many equipment insulation failures can be identified in advance by carrying out $\tan \delta$ and capacitance measurement. Changes in the value of capacitance indicate abnormal conditions such as presence of moisture, layer short circuits or open circuits in the capacitor elements of CVT stacks

a) Testing Instruments

Measurement is to be carried out by 10 kV or 12 kV test kit. As there will be induction at switchyard, kits with induction suppression facility to be used.

b) Test Procedure

Jumper is to be disconnected from HV terminal. PLCC connection/ earth connection is to be removed from HF point/ bushing and earth connection to be removed from neutral point of EMU tank.

Before carrying out the measurement, the insulator petticoats of CVTs should be cleaned from moisture, sand, dust particles or salt deposition etc. otherwise measured values shall be affected greatly by these factors. All safety precautions pertaining to the high voltage kit used to be observed.

HF point should be disconnected from ground/ PLCC. Neutral/ LV of Intermediate voltage transformer should be disconnected from ground. High voltage from $\tan \delta$ kit should be applied to primary (HV) Terminal and LV should be connected to the HF point of the CVT. Before applying HV, interference is to be nullified using interference suppression unit (ISU). Measurements must be taken in UST– Mode. Standard procedure (as specified by kit supplier) for measuring capacitance and $\tan \delta$ in charged switchyard/induced voltage conditions should be followed. Measurements are to be carried out in main and reverse mode/polarity and then the average value to be computed.

c) Evaluation of Test Results

Following norms are proposed for evaluation of the test results: Change in capacitance value above 6%, CVT need to be replaced.

$\tan \delta$ values more than 0.003 from pre-commissioning value needs replacement.

Sl. No.	Parameter	Change of the measurement parameter from the pre-commissioning value	Remarks
01.	Capacitance	Up to 2%	Normal
		+/- 2% to +/-3%	Yearly measurement
		Above 6%	Needs replacement
02.	Tan Delta	Up to +/- 0.002	Normal
		+0.002 to +0.003	Yearly measurement
		Above +0.003	Needs replacement

Note: If only $\tan \delta$ values exceed +0.003 (increase from the pre-commissioning values) and other parameters like drift in secondary voltages and

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capacitance values are within acceptable norms, the Test results are to be reviewed for further action by experts.

7.4 ISOLATORS

7.4.1 MEASUREMENT OF CONTACT RESISTANCE OF ISOLATOR MAIN CONTACTS

To measure contact resistance of isolator main contacts to ensure the healthiness of the current carrying parts are healthy.

- Test Equipment

Micro ohm meter of 100 A or more current injection kit with associated accessories.

- Test procedure

CB is to be in open position and other side isolator of CB also to be in open position. High voltage plugs should be free from moisture during installation and operation. At the time of connections, both sides of isolator should be earthed by closing earth switches or by temporary earths. After making the connections, earthing is to be removed.

The ohm meter operates on the four-wire measurement principle. To measure the contact resistance, connect the respective leads and adjust the variac so that app. 100 amps current flow through the contacts. Value of contact resistance is directly displayed on the digital LED display screen.

By using four terminal method, we can nullify the resistance of test leads if input impedance of measuring device (IC) is very high.

- Evaluation of Test Results

Contact resistance of the main contacts indicates wear out and misalignment of the main contacts. If the value of contact resistance (between both arm terminal assemblies) exceeds the permissible limit, i.e., 300 micro ohm this could result in overheating of the contacts. Therefore, the problem of high contact resistance should be attended immediately by making proper alignment of contacts or by replacing finger contacts.

7.5 SURGE ARRESTERS

Surge arrester plays a very vital role in improving power system reliability by diverting surges to earth and thus protecting costly switchyard equipment. Proper insulation co- ordination is necessary for enhancing life span of costly Main equipment. Since surge arresters protect other equipment therefore monitoring the health of surge arrester is essential.

Surge arresters are monitored basically using three tests to evaluate its condition. Leakage current monitoring, insulation resistance measurement and thermal scanning could indicate any deterioration in surge arresters. Out of these, leakage current measurement and thermal scanning are on-line measurement and shut down is required for IR measurement. As system

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availability is crucial for utilities, more emphasis is to be given for on-line measurements.

7.5.1 LEAKAGE CURRENT MEASUREMENT

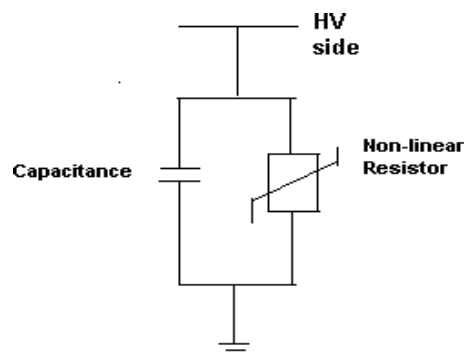
As per IEC-60099(5), techniques for the health monitoring of surge arresters in service the following condition monitoring methods are suggested:

- a) Total leakage current measurement
- b) Watt loss measurement
- c) Insulation resistance measurement
- d) Third harmonic resistive current monitoring.
- e) Third harmonic resistive current (THRC) monitoring with compensation for third harmonics in system voltage.

Total leakage current measurement is not sensitive to the deterioration of the arrester discs as total leakage current measurement do not indicate severity of ZnO disc degradation as resistive current is only 20-30% of the total leakage current and a sharp increase in resistive current due to degradation of arrester's discs do not affect the total leakage current considerably. Watt loss measurement is a useful method, but it involves a separate voltage source and test results are affected by phase shifts introduced by CVT/PT. The harmonics present in the system voltage may create capacitive harmonic currents, which may be comparable with the harmonic currents generated by the non-linear resistance of the surge arrester. As a result, the error in the measured harmonic current may be considerable. The error in the measured third harmonic current may be up to 100% with 1 % third harmonic present in the system voltage. Hence, reliable monitoring of surge arresters is possible through such test kits which provide for compensation for harmonics present in the system voltage.

a) Testing Principle

The zinc oxide elements discs are made by mixing zinc oxide with small number of additive materials such as Bi_2O_3 , CoO , MnCO_3 and granule mixture, then forming and firing. The equivalent circuit of surge arrester is parallel combination of capacitance and variable resistance (as shown in picture). The current flowing through the zinc oxide discs is the total leakage current (I_T) is having capacitive leakage current (I_C) and resistive leakage current (I_R) components. Normal operating voltages causes ageing of ZnO discs whereas switching/ lightning over voltages may cause over loading of all or part of the ZnO blocks. The increase in resistive leakage current (I_R) due to above reasons may bring the arrester to thermal instability and may result in complete arrester breakdown. Hence, monitoring of leakage resistive



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current (I_R) is very important.

Harmonics are created in leakage current on application of fundamental frequency, due to non-linear voltage-current characteristic of surge arresters. Third harmonic is the largest harmonic component of the resistive current. Third harmonic in system voltage creates capacitive harmonic currents which affect the measured value. Error in the measured values may be considerable. As reported, 1% third harmonic in system voltage may introduce error up to 100% in the measured value.

b) Test Procedure

The measurement principle is based on the fact that harmonics are created in the leakage current by non-linear voltage-current (V-I) characteristics of surge arrester. Resistive current consists of fundamental, third harmonic, fifth harmonic and seventh harmonic components. The harmonic content depends on the magnitude of the resistive current and on the degree of Non-linearity of the voltage-current (V-I) characteristics. The harmonic content also varies with the voltage and temperature of the surge arrester.

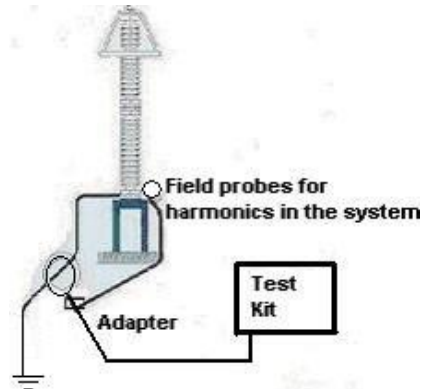
The third harmonic is the largest harmonic component of the resistive current and is most used for diagnostic measurements. No reference is needed for measurement of third harmonic resistive current since it is assumed that all harmonics arise from the non-linear resistive current. The conversion from harmonic to resistive current level, if required, relies on information supplied by surge arrester manufacturer or from measurements in the laboratory. Basic principle of this technique lies in the measurement of third harmonic resistive current duly compensated for third harmonics present in system voltage. A trend analysis of the 3rd harmonic resistive current is necessary to observe periodical changes due to normal/abnormal ageing. Sudden rise in THRC or very high value of third harmonic resistive current indicates degradation of ZnO discs and inspections/corrective actions are required to be taken in advance to prevent catastrophic failures.

c) Method of THRC Measurement

The total current (I_T) through the Surge Arrester having Capacitive component (I_C) and Resistive current (I_R) flows through the Leakage Current Monitor. The I_R component has 3rd, 5th harmonic and other harmonics present in it. However, when degradation of ZnO discs occur, the variation of 3rd Harmonic Resistive Current is more pronounced than other harmonics currents, total current (I_T) and total capacitive current (I_C).

The connection arrangement for the instrument is quite simple as shown in Figure. Connecting

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a clamp type CT probe to the ground wire allows the leakage current to be detected. Probe output passes through a shielded cable to the detector unit. A band pass filter is built into the initial stage to pass the third harmonic and to cut off the fifth and higher harmonics. After the third harmonic is selected, it is amplified and then displayed. Compensating antenna is also provided, output of which is fed to the test kit to compensate for the harmonics present in the system voltage. The harmonics present in the EHV system are compensated and filter circuits are provided for bypassing undesired currents and only 3rd harmonic resistive current is measured.

7.5.2 INSULATION RESISTANCE MEASUREMENT

Insulation resistance measurement for Surge Arresters also gives the condition of the arresters. Insulation resistance measurement to be carried out on individual stacks then carrying out the test for the complete arrester. As this is off line test, the arrester to be taken out of service for such measurement.

IR test proves good to find out any moisture entry through sealing inside the arresters. Keeping a stack with low IR value will stress the other stack also and failure is more prone on the other stacks also. Corrective action to be taken immediately.

7.6 THERMOVISION SCANNING

Thermal scanning plays a vital part in identifying defects in EHV switchyard. As it requires no shut down of switchyard equipment, it need to be widely used on condition monitoring. Any loose joint, improper isolator contacts and current transfer parts could be identified using thermal scanning.

Thermal scanning also helps in identifying defects in surge arresters. This may not be true always but if there is defect and more current is passing through the arrester, then the temperature inside arrester will increase and could be detected by thermal scanning. The below picture indicates a defective arrester from that of healthy one during thermal scanning. It is advisable to carry out this test during maximum voltage condition of the switchyard unlike thermal scanning of other current carrying components during maximum current flow condition. This test followed by IR measurement/ Tan δ and capacitance measurement shall indicate healthiness of surge arresters.

7.7 STATION EARTH RESISTANCE MEASUREMENT

7.7.1 MEASUREMENT OF STATION EARTH RESISTANCE BY FALL OF POTENTIAL

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METHOD

The measurement of the resistance of earth electrodes is done to see that they meet and remain within the required resistance limits. The resistance to earth of an earth electrode may change under different soil moisture condition and over periods of time. The establishment of satisfactory earthing systems (low total resistance to earth to perform satisfactorily) is an essential part of electrical supply wiring safety and installation economics.

a) Definitions

1) Ground/Earth

A conducting connection, whether intentional or accidental, by which an electric circuit or equipment is connected to the earth, or to some conducting body of relatively large extent that serves in place of the earth.

2) Grounding/Earthing Conductor

The conductor that is used to establish a ground and that connects equipment, device, wiring system or another conductor (usually the neutral conductor) with the grounding/earthing electrode(s).

3) Grounding/Earthing Electrode

A conductor used to establish a ground.

4) Grounding grid/Earthing mat

A system of grounding electrodes consisting of interconnected bare cables buried in the earth to provide a common ground for electrical devices and metallic structures.

5) Touch Potential

The potential difference between a grounded metallic structure and a point on the earth's surface separated by a distance equal to the normal maximum horizontal reach is approximately one meter.

6) Step Potential

The potential difference between two points on the earth's surface separated by a distance of one pace that will be assumed to be approximately one meter in the direction of maximum potential gradient.

7) Inductive Coupling

The association of two or more circuits with one another by means of inductance mutual to the circuits or the mutual inductance that associates the circuits.

b) Test Equipment

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Earth tester may be manual or automatic, analogue or digital of any standard make say Insulation tester, multi amp, etc.

Earthing test accessory kit containing:

Hammer 1.13 kg.	1 No.
Galvanized steel spikes 12 mm square	4 Nos. section 450 mm
long Spike extractors	2 Nos.
Cable or a cable winder complete with connector and clip (30 m)	
– 1 No Cable or a cable winder complete with connector and clip (50 m) – 1 No Leads, complete with connectors and clips (3 m) –	
02 Nos.	

c) Safety Precautions

It should be strongly impressed on all test personnel that a lethal potential (of the order of 1 kV to 10 kV) can exist between the station ground and a remote ground, if a power system fault involving the station ground (though the chances are remote) occurs while ground tests are being made and therefore use of insulating shoes, gloves and other protective devices are recommended whenever measurements are carried out at an energized Mat and handling the test leads.

d) Test Procedure

1) Method of Testing using the earth Testing Megger

The earth testing megger consists of basically a moving coil meter with two coils at right angles. A control coil is arranged to be in series with the current probes. The movement of the pointer is then dependent on the ratio of currents in the two coils and is thus dependent on resistance. In making measurement terminals P₁ and C₁ should be connected to main station earth E by separate copper wires (E is the Main earth mat electrode). A current probe, usually a 12 mm square section, 450 mm long galvanized steel spike is connected to a cable which is run out a Mat

terminal C₂. Another probe P (again 450 mm long, galvanized steel spike of 12 mm square section) is attached by another cable to terminal P₂, such that arrangement is obtained. When running wires to the electrodes they should be laid at least 2 m apart to minimize Inductive coupling.

In carrying out earth resistance checks it should be remembered that overhead lines with their aerial earth wires can provide a path for the earth megger current, so giving a false reading. The electrode should therefore be run out in a direction to avoid any incoming line routes and if possible, tests should be made in different direction.

At least 10 measurements are to be made at intervals from the C₂ electrode inwards as far as practicable. The resistance/distance curve so obtained should show a "plateau" region as shown in figure below. The resistance of this region is taken to be true earth resistance of the rod or plate. If a plateau region is not obtained, the test should be repeated with the C₂ spike further away typically 500

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m away from the Mat.

However, three measurements are recommended for simplicity with moving potential electrode P say by 6 m away from E and 6 m nearer to E and average can be taken.

1) Testing steps -Fall of Potential Method

P1 and C1 terminal is sometimes named as terminal X which is connected to earth electrode E. Connect C2 terminal to remote current spike C, at least 100 m away from any part of the earthing system. In addition, it should not be in line with any overhead transmission line, buried cable, metal water pipe or other earthed electrode. Connect P2 terminal to remote potential spike P, 61.8% of total distance d away from earth electrode E so that interference of electromagnetic field do not yield inaccurate readings. Make the first measurement- R1. Move the potential spike P nearer to electrode E by a selected amount (say 6 m). Make the second measurement- R2. Move the potential spike P further away from electrode E by 6 m (the other side of the center position). Make the third measurement- R3. Resistance of the earth electrode is the average (RA) of these three results (providing they agree within the specification limits of the electrode system). $RA = (R1+R2+R3) / 3$

e) Interpretation/Evaluation of Results

Any error in earth testing measurement is due to mainly low value of testing current used and maximum sensitivity of any instrument. On large electrode systems the instrument is working at the extreme end of its designed range and should the earthing system exhibit a high reactance, then the true impedance is not measured.

An ideal grounding should provide a near zero resistance to remote earth. In practice, the ground potential rise at the station site increases proportionally to the fault current, the higher the current, the lower value of a total system resistance thus must be obtained. For switchyard and power station, the ground resistance should be less than 1 Ω . The value measured should be within 20% of the previous value or if no previous history is available, it should be compared with the resistance of a similar rod or plate on the same site.

7.8 SECONDARY INJECTION TEST ON RELAYS & OTHER MEASURING DEVICES:

A. Principle:

Secondary injection tests are performed to verify the correct operation of the protection system with regard to everything that is connected to the secondary of the instrument transformers. The secondary injection test covers not only the testing of protective relays and their tripping settings, but also the checking of all associated circuits and devices involved in the proper performance of the secondary protection system. The purpose of secondary injection testing is to check that the protection scheme from the relay input terminals onwards is functioning correctly with the settings specified.

The secondary injection test involves disconnection of the device and/or circuits from its normal CT/VT connection.

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The primary injection test is always carried out after the secondary injection test, to ensure that problems are limited to the VT's and CT's involved, plus associated wiring, all other equipment in the protection scheme having been proven satisfactory from the secondary injection test.

The secondary injection test simulates the current and voltage transformer secondaries, by injecting single-phase or three-phase currents and voltages into relays, devices into the secondary circuits, performing all the corresponding checking and testing with normal and fault conditions.

B. Testing Instruments:

3 phase voltage & current injection unit including accessories

C. Safe practices & method:

Following Safe practices to be adhered prior to start or during secondary injection testing:

1. All the associated Circuit Breakers (Main & Tie both) & Isolators to be made OPEN before starting of testing.
2. Ensure the Breaker & Isolators open status at Busbar relays.
3. Proper PTW to be applied & JSA to be submitted duly signed by HOD (EMD).
4. Circuit breaker, Isolators & earth switch DC supply to be made OFF.
5. Testing work shall only be carried out by experienced area executive.
6. Before injection, ensure tightness checking of relay TBs.
7. CT & PT/CVT field wires to be disconnected or isolated at terminal blocks (TB).
8. Bay LBB initiation signal wire to be removed at relay panel.
9. For transmission lines, Zone-4 (Reverse zone) start signal wire to be removed at line relay panel. (To be used for Busbar trip under both CU fail)
10. Communication to be given to remote end substation before start of testing. Accordingly suitable action to be taken to avoid any spurious tripping at remote end due to testing at our end.
11. No work shall be carried out on Circuit breaker panel during relay testing.
12. Before starting of relay testing, first download the existing relay setting file from Relays & compare with the backup setting at Laptop/PC. If any deviation found then check Engg approved setting or consult Engg/COS.
13. Protection system DC supply for both Main & Tie bay remain ON during relay testing to ensure tripping behaviour as per as built scheme.
14. As built drawing of control & relay panels to be made available.
15. During secondary injection testing, check & ensure the correct measurement values, alarm, events, trip logs, DI/DO signals, trip relay operations etc as per approved scheme on all the integrated relays, meters, HMIs, LVS & other devices. If any deviation found, take appropriate action.
16. Injection must be carried out towards devices (relays, meters, transducer etc) instead of primary side instruments.
17. For transmission lines, Carrier & Direct trip transmit & receive signals to be checked alongwith remote end substation & ensure the correct operations

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as per scheme.

18. During Busbar relay testing, more precautions to be taken to avoid any maloperation of relay & subsequent outage of any charged element. Hence ensure logic integrated in CU & PUs are as per approved scheme.
19. During testing of Bay PU, corresponding Bus protection to be made OFF from CU. however during LBB testing, block both bus protections from CU. Ensure Bus out status at other CU for enabling of 1 out of 2 logic.
20. During testing of any CU, both bus protections to be made blocked by switch or function keys. Ensure Bus out status at other CU for enabling of 1 out of 2 logic.
21. Ensure the relays & other devices are GPS synchronized.
22. After completion of test, normalize all the CT & CVT terminations at TB & other control wires at relays & C/R panels. Remove all shorting wires, which was made for any simulation or bypass during the testing.
23. After testing, Download the relay setting files from relays & compare with the approved setting.
24. LBB initiation & Z4 start wires kept removed till the O/H works completion on Bay. It is to be normalized before charging of bay.
25. After normalization of bay Isolators ensure the Isolator CLOSE status & Breaker OPEN status at Busbar relays.
26. After charging of bay, ensure Breaker & Isolator CLOSE status & no alarm at CU & PUs.
27. Check & ensure the correct secondary current & Voltage readings at relay & panel TBs & primary readings at relay & SCADA.
28. Ensure no alarm is persisting in relays & SCADA after bay charging.

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8.0 SWITCHYARD EQUIPMENT MAJOR OVERHAULING

INTRODUCTION

The manufacturer prescribes the overhauling of the equipment at specified time interval in the lifetime of the equipment. The overhauling process considerably enhances the life of the equipment.

- CIRCUIT BREAKER**

As per COS guideline, Capital Overhauling of EHV Circuit Breaker to be done after 10 years of service through OEM. The parts which require overhaul in circuit breakers are:

- Operating drive
- Interrupting chambers

Operating drive require maintenance depending on the number of operation or specific time period like greasing and changing of gaskets etc., as per manufacturer's instructions. Experience reveals that in EHV circuit breakers, operating mechanism require more frequent maintenance than the main interrupting chambers provided the breaker is not subjected to fault tripping very frequently. The overhauling schedule of circuit breakers is as follows:

01.	Checking the arcing chamber, and overhauling all internal parts of the interrupter.
02.	Replacement of O" rings and gaskets.
03.	Checking the male, female contacts of the interrupter for erosion; checking the travel of the contacts, etc.
04.	Checking the working of all internal parts.
05.	Checking PIR and grading capacitor.
06.	Checking the working of operating mechanism.
07.	Checking the tripping and closing timings of all interrupters.
08.	Measurement of IR values of the breaker, both in open and close position.
09.	Checking the dynamic and static contact resistance.
10.	Checking of SF6 gas Dew point, purity & SO2 content after gas filling

The last three measurements are indicative of the working condition of the circuit breaker. Overhauling of interrupting chamber depends on number of operation and the amount of current interruption by the contacts. All the defective components and all gaskets are to be replaced. Based on the condition, sometimes overhauling of only operating mechanism can be taken up if interrupting chamber condition is indicated healthy in diagnostic tests.

- CURRENT TRANSFORMER AND POTENTIAL TRANSFORMER/CVT**

For monitoring the performance of the CTs, it is essential to watch the parameters like:

01.	IR values
02.	Tan δ values
03.	Capacitance

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04.	Oil level
05.	Oil test results
06.	DGA of oil
07.	Inspection with thermo-vision camera for hot clamps, if any.
08.	Physical checking of terminal joints

There are no specific set guidelines for overhauling / replacement of CTs. The CT being static equipment does not have any mechanical wear and tear. If the parameters as above are regularly monitored the CTs can give long service. However, the replacement criteria of CT & CVTs should be done on following condition.

- If the CT shows a higher trend of Tan δ & capacitance values, deterioration of oil quality and oil DGA indicates violation.
- If the deviation of secondary voltage of CVT is observed or CVT shows a higher trend of Tan δ & capacitance values, deterioration of oil quality and oil DGA indicates violation.
- As per COS guidelines, after 20 years of service the CT & CVTs are to be replaced irrespective of the condition of CT/CVT.
- MAJOR O/H CHECKLIST FOR HAND OPERATED ISOLATORS (AS PER BAY/BUS O/H SCHEDULE)

(ii)

Sl No.	Activity	Remarks (Done/Not done)
1	MAIN CONTACTS AND CONNECTIONS	
1.1	Clean off old contact lubricant from each moving contacts with solvent cleaner.	
1.2	Examine contacts for burning. Use Scotch bright Pad to remove burn marks. NOTE: Do not use emery cloth or harsh abrasive on plated surfaces.	
1.3	Apply a thin film of Petroleum jelly to each moving contact.	
1.4	Clean off old contact lubricant from each fixed contact finger.	
1.5	Examine contacts for burning (as 1.2)	
1.6	Apply thin film of Petroleum jelly to each contact.	
1.7	Close isolator smoothly by hand. Ensure that the moving contact enters the fixed contact centrally (equal movement of top and bottom contact fingers on each fixed contact) with 0.0015" feeler gauge. Ensure Isolator is dead from both end.	
1.8	Ensure that with the isolator closed there is even contact pressure maintained by each of the fixed contact fingers on the moving contact. (0.0015" Feeler gauge will not enter between fixed and	

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	moving contacts).	
1.9	Open isolator and check for any movement issues	
1.10	After opening of Isolator ensure that the primary connections to each of the fixed contacts are secure. Ensure that all nuts and bolts associated with contact heads and primary connections are tight.	
2	POST AND LINKAGES	
2.1	Clean insulator stacks unless greased.	
2.2	Examine surface of insulators for signs of burning or tracking	
2.3	Test for cracks by ringing with coin or small spanner.	
2.4	Ensure bolts at base of insulator stacks are tight.	
2.5	Ensure all bolts securing bearing housing to structure are tight.	

2.6	Ensure that all clevis pins at post level are secure and have split pins which have been opened.	
2.7	Lubricate all clevises.	
2.8	Lubricate bearings at base of each center Post with grease gun.	
2.9	Ensure that all 'U' bolts on drive rods are tight.	
3	MECHANISM BOX	
3.1	Remove mechanism box covers.	
3.2	Remove auxiliary switch covers.	
3.3	Clean and ensure wiring is tight on auxiliary switch and terminal blocks. Use Insulator brush for cleaning.	
3.4	Clean auxiliary switches with solvent then lightly smear with petroleum jelly.	
3.5	Clean bolt interlock mechanism. Verify that the connections are tight and the plunger is free.	
3.6	Lubricate moving parts and pivot points. Clean away any surplus oil. Do not oil or grease plunger.	
3.7	Clean out mechanism box with dust brush & cloth.	

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3.8	Where applicable unlock emergency bolt release controller and lubricate lock and bolt sleeve.	
3.9	Replace auxiliary switch cover and mechanism box covers.	
3.10	Operate isolator and ensure that locking off pin enters in both the open and closed positions. Ensure the open & close status at all relays and control circuits are as per scheme.	
4	RESISTANCE TESTS	
4.1	Close Isolator and measure resistance across each phase. Investigate and rectify any high readings. Include dropper connections to isolator posts on both ends. Digital Micro ohmmeter. Compare with earlier reading (consider 10 micro-ohm for each contact). Max allowed 300micro-ohm. Record data in given format.	
5	FINAL INSPECTION AND OPERATION	
5.1	Report on the general state of paint work, gaskets and galvanizing on the mechanism box and Operating rods. Replace gasket if required.	
5.2	Carry out a visual inspection of the isolator structure earthing.	
5.3	Operate the isolator to ensure that its movement is smooth and free. Check isolator close and open status at all end devices/relays. Record results. Ensure removal of AC and DC fuses after operation.	

RESISTANCE TESTS FORMAT:

Measure resistance across isolator including main connections.

	R PH	Y PH	B PH
Old value			
As found			
As left			

- Note: All checks shall be carried out in all phases of earth switches which are taken under O/H shutdown.

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- MAJOR O/H CHECKLIST FOR MOTORISED OPERATED ISOLATORS (AS PER BAY/BUS O/H SCHEDULE):

SI No.	Activity	Remarks (Done/Not done)
1	MAIN CONTACTS AND CONNECTIONS	
1.1	Clean off old contact lubricant from each moving contacts with solvent cleaner or spirit rag.	
1.2	Examine contacts for burning. Use Scotch bright Pad to remove burn marks. NOTE: Do not use emery cloth or harsh abrasive on plated surfaces.	
1.3	Apply a thin film of Petroleum jelly to each moving contact.	
1.4	Clean off old contact lubricant from each fixed contact finger.	
1.5	Examine contacts for burning (as 1.2)	
1.6	Apply thin film of Petroleum jelly to each contact.	
1.7	Close isolator smoothly by hand. Ensure that the moving contact enters the fixed contact centrally (equal movement of top and bottom contact fingers on each fixed contact) with 0.0015" feeler gauge. Ensure Isolator is dead from both end.	
1.8	Ensure that with the isolator closed there is even contact pressure maintained by each of the fixed contact fingers on the moving contact. (0.0015" Feeler gauge will not enter between fixed and moving contacts).	
1.9	Open isolator and check for any movement issues	
1.10	After opening of Isolator ensure that the primary connections to each of the fixed contacts are secure. Ensure that all nuts and bolts associated with contact heads and primary connections are tight.	
2	POST AND LINKAGES	
2.1	Clean insulator stacks unless greased.	
2.2	Examine surface of insulators for signs of burning or tracking	

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2.3	Test for cracks by ringing with coin or small spanner.	
2.4	Ensure bolts at base of insulator stacks are tight.	
2.5	Ensure all bolts securing bearing housing to structure are tight.	
2.6	Ensure that all clevis pins at post level are secure and have split pins which have been opened.	
2.7	Lubricate all clevises.	
2.8	Lubricate bearings at base of each center Post with grease gun	
2.9	Ensure that all 'U' bolts on drive rods are tight.	
3	MECHANISM BOX	
3.1	Open mechanism box doors. Lubricate hinges, catches.	
3.2	Remove auxiliary switch covers.	
3.3	Clean and ensure wiring is tight on auxiliary switch and terminal blocks. Use Insulated brush for cleaning.	
3.4	Clean auxiliary switches with solvent then lightly smear with petroleum jelly.	
3.5	Clean magnetic bolt interlock mechanism. Verify that the electrical connections are tight and the plunger has adequate clearance, moves freely and is not distorted.	
3.6	Lubricate moving parts and pivot points. Clean away any surplus oil. Do not oil or grease plunger	
3.7	Clean out mechanism box with dust brush & cloth	
3.8	Where applicable unlock emergency bolt release controller and lubricate lock and bolt sleeve	
3.9	Verify that the heaters are working. Replace and that are not. Also confirm integrity of working.	
3.10	Clean contacts on opening and closing contractors. Use a smooth file, if necessary, to restore contact surfaces. Leave contacts dry.	
3.11	Operate contactors by hand and ensure that they are free and that the contacts make correctly. Oil pivots sparingly and wipe away access.	

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3.12	Clean contacts on opening and closing relays with Spatula and Solvent. Operate by hand to ensure correct movement of armature and contact lift.	
3.13	Clean and lubricate contacts on overload relay. Ensure connections are tight.	
3.14	Lubricate sparingly control switch contacts. Ensure electrical connections are tight. Replace covers.	
3.15	Remove covers from control switches. Clean contacts and lubricate operating mechanism. Wipe off any excess oil.	
3.16	On hydraulic mechanism, check oil level on hydraulic reservoir. Top up if necessary.	
3.17	On direct drive type mechanism, check oil level in gear box and top up when necessary.	
3.18	Check condition of fuses, neutral links and fuse bases.	
4	ISOLATOR OPERATION	
4.1	Confirm that the operation of emergency bolt release is correct.	
4.2	Verify all motor supply/control fuses and links.	
4.3	Close and open isolator locally and confirm operation is smooth and that contacts are made correctly. Ensure the open & close status at all relays and control circuits are as per scheme	
4.4	Ensure that the mechanical (Castell type) interlock operates in either the open or closed position, as appropriate.	
4.5	Carry out torque test for operating and stall conditions. NOTE: The Engineer will measure the maximum normal and stall motor currents.	
4.6	Time the operation of the thermal overload relay by turning the controller to 'close' and then to 'open' when isolator is almost closed. Reset relay.	
4.7	If thermal overload relay fails to operate correctly, report to Supervisor. If adjustments are made, then the relay must be injected and the test repeated. Test to be done as per given format.	
4.8	Close isolator and remove motor supply/control fuses and links.	

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5	RESISTANCE TESTS	
5.1	Close Isolator and measure resistance across each phase. Investigate and rectify any high readings. Include dropper connections to isolator posts on both ends. Digital Microhmeter. Compare with earlier reading	
	(consider 10 micro-ohm for each contact). Max allowed 300micro-ohm. Record data in given format	
6	FINAL INSPECTION AND OPERATION	
6.1	Report on the general state of paint work, gaskets and galvanizing on the mechanism box and Operating rods. Replace gasket if required.	
6.2	Carry out a visual inspection of the isolator structure earthing.	
6.3	Operate the isolator to ensure that its movement is smooth and free. Check isolator close and open status at all end devices/relays. Record results. Ensure removal of AC and DC fuses after operation.	

RESISTANCE TEST FORMAT:

Measure resistance across isolator including main connections.

	R PH	Y PH	B PH
Old value			
As found			
As left			

THERMAL OVERLOAD RELAY CHECK FORMAT:

Operating time as found ----- seconds

If adjustments were made: ----- seconds

Operating time as left ----- seconds

(a) Note: All checks shall be carried out in all phases of earth switches which are taken under O/H shutdown.

Signature of Electrical Supervisor

Signature of Electrical Engineer

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- MAJOR O/H CHECKLIST FOR HAND OPERATED/MOTORISED EARTH SWITCHES (AS PER BAY/BUS O/H SCHEDULE)

SI No.	Activity	Remarks (Done/Not done)
1	MAIN CONTACTS AND CONNECTIONS	
1.1	Clean fixed & moving contacts of all phases with solvent cleaner.	
1.2	Examine contacts for burning. Use Scotch bright Pad to remove burn marks. NOTE: Do not use emery cloth or harsh abrasive on plated surfaces.	
1.3	Verify by hand that similar tension exists on each of the female spring contacts on each phase.	
1.4	Verify that the nuts and bolts securing the fixed contacts to the fixed contact heads are tight on all phases.	
1.5	Verify that the nuts and bolts securing the moving contacts to the moving contact arms are tight on all phases.	
1.6	Close the earth switch by hand to verify that on each phase: 9 The moving contacts enter the fixed contacts Centrally & aligned 10 The moving contacts enter the fixed contacts simultaneously 11 The moving contacts are making good contact with the fixed contacts on completion of the closing operation. Record all results.	
1.7	Lightly lubricate all fixed and moving contacts	
1.8	Verify the tightness of the primary electrical connections on each fixed contact	
1.9	Verify that the connections between the flexible contact arm and the flexible copper braids are sound and tight. Record results.	
1.10	Verify that the connections between the flexible Copper braids are sound and tight. Record results	
1.11	Carry out a visual inspection of all flexible copper Braids. Record	

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	results.	
1.12	Earth switch mechanical interlock with isolator to be checked for operation when isolator is not fully open.	
2	LINKAGES	
2.1	Grease all bearings through grease nipples	
2.2	Lubricate all connecting pins and clevises. Verify that all split pins are fitted and secure	
2.3	Verify that all pull rod clamp nuts and length adjusters are tight	
2.4	Verify that the nuts and bolts securing the balance weights are tight	
2.5	Verify that the bolts securing the bearing the bearing housing to the structure are tight	
3	INSULATORS	
3.1	Clean all insulators	
3.2	Carry out a visual inspection of all insulators to ensure that there are no signs of burning or tracking. Test each shed for cracks by ringing.	
3.3	Verify that the fixing bolts at the base of the Insulator stack are tight	
4	MECHANISM BOX	
4.1	Remove mechanism box covers	
4.2	Remove auxiliary switch cover and cable termination cover (if fitted).	
4.3	Clean all terminals with solvent	
4.4	Verify the condition of all terminal blocks and tightness of all electrical connections and lugs. Record results	
4.5	Clean auxiliary switches with solvent then lightly smear with petroleum jelly.	
4.6	Operate the earth switch and verify that the auxiliary switches make good contact in the closed position. Record results	

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4.7	Clean and replace the auxiliary switch cover and Cable termination cover	
4.8	Lubricate all moving parts, linkages and pivot points	
4.9	Verify that the interlock castell keys operate freely	
4.10	Verify that the nuts and bolts securing the mechanism box to the earth switch structure are sound and tight.	
4.11	Clean out mechanism box and place mechanism box cover. Change gasket if required	
4.12	Check the functioning of space heater with thermostat (if provided). Ensure no cable is nearby heater	
4.13	Verify electrical interlocks based on isolator status and 3-phase under- voltage by simulating conditions.	
4.14	Clean and lubricate contacts on overload relay. Ensure connections are tight	
4.15	Check condition of fuses, neutral links and fuse bases.	
5	FINAL INSPECTION AND OPERATION	
5.1	Report on the general state of paint work, gaskets and galvanizing on the mechanism box and Operating rods. Replace gasket if required.	
5.2	Carry out a visual inspection of the isolator structure earthing.	
	Report any defects to the area maintenance engineer for remedial action	
5.3	Operate the isolator to ensure that its movement is smooth and free. Check isolator close and open status at all end devices/relays. Record results. Ensure removal of AC and DC fuses after operation.	

- Note: All checks shall be carried out in all phases of earth switches which are taken under O/H shutdown.

Signature of Electrical Supervisor

Signature of Electrical Engineer

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9.0 ACCEPTANCE NORMS

INTRODUCTION: The acceptance norms/permissible limits for values obtained during maintenance tests which are considered safe have been indicated in Table – 1

TABLE -1 ACCEPTANCE NORMS SWITCHYARD EQUIPMENT

Sl. No.	Parameter	Values				Reference / Remark	
1.	CIRCUIT BREAKERS						
	(A) Dew Point of SF ₆ Gas	Dew Point values as per Table – 2 for SF ₆ Gas Dew Point for Various Breakers					
	(B) Dew point of operating air	(-) 45°C at Atmospheric Pressure				Manufacturers Recommendations.	
	(C) CB Operating Timings, ms	765KV	400KV	220KV	132KV	IEC-62271-100 (2001)	
	Closing time (Max), milli sec	110	150	150	150		
	Trip time (Max), milli sec	25	25	25	25		
	Pole discrepancy at rated operating pressure for OPEN between Ph- Ph(max) in ms	4	5	5	5		
	Pole discrepancy at rated operating pressure for OPEN between Break to Break of same phase (max) open in ms	3	3.33	3.33	3.33		
	CO time (minimum)	As per GTP of manufacturer					
	Trip delay time for DCRM test (CO operation)-minimum deviation from standard timings Close/Open, in ms	300	300	300	300		
(D) Travel of operating rod	ABB – 200mm CGL – 230 mm BHEL – 130 mm GE/ALSTOM –184 mm SIEMENS –155 mm				Manufacturers Recommendations		

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	(E) PIR time	400 kV	Manufacturers Recommendations
	• BHEL make 12 - 16 ms • CGL make 8 – 12 ms • ABB make 8 – 10 ms – GE make 8 – 12 ms (R) ALSTOM make 8 – 12 ms – TELK make 8 – 12 ms (S) SIEMENS make 8 - 12 ms		
	(F) PIR operating time prior to opening of main contacts (ABB, SIEMENS, CGL, NGEF make CB)	5 ms (Min) at rated pressure.	Manufacturers Recommendations
	(G) PIR and main contacts overlap time (BHEL, M&G, SIEMENS, ABB make CBs)	5 ms (Min) at rated pressure	Manufacturers Recommendations
	(H) Tan delta of grading capacitors	0.007	Since temp correction factor for Tan δ depends on make, type and also aging conditions, the correction factors for different types/makes are different. Hence, no std. temp error factors is applicable.
	(I) Rate of rise in Tan Delta	0.001 per year (max)	Manufacturer's GTP to be referred for value of one element.
	(J) Deviation in capacitance of grading capacitors	Corresponding to shorting of element	
	(K) Contact Resistance of CB (in Micro- Ohm)	400KV-150 220KV-100 132KV-100	
	(L) Contact Resistance of CB terminal connector	10 Micro-Ohm per connector	NGC, Recommendations. UK
	(M) Evacuation level before SF6 gas filling	5mbar (min)	Manufacturers Recommendations

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	(N) N ₂ leakage rate from N ₂ accumulator	3 bar per year (max)	
	(O) IR Value Phase – Earth	1000 MΩ (Min) by 5.0/10.0 kV Insulation Tester	NGC, UK Recommendation
	(P) IR Value across open contacts	1000 MΩ (Min) by 5.0/10.0 kV Insulation Tester.	
	(Q) IR Value of Control Cables	50 MΩ (Min) by 0.5 kV Insulation Tester	
	(R)PRESSURE SWITCH SETTINGS. (i) SF ₆ gas pressure switches (ii) Operating air pressure switches (iii) Operating oil pressure switches	Within ±0.1 Bar of set value Within ±0.1 Bar of set value Within ±1.0 Bar of set value	Manufacturers Recommendations
	(S)BDV of oil used for MOCB • At the time of filling • During O&M	40kV at 2.5 MM GAP (Min) 20kV at 2.5 MM GAP (Min)	Manufacturers Recommendations
Sl. No.	Parameter	Values	Reference / Remark
2.	CURRENT TRANSFORMERS		
	(A) IR Value by Insulation Tester		
	Primary – Earth	1000 MΩ (Min) by 5.0/10.0kV	NGC, Recommendation
	Secondary – Earth	50 MΩ (Min) by 0.5kV	UK
	Control Cables	50 MΩ (Min) by 0.5kV	UK

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	(B) Tan delta value	0.007	Since temperature correction factor for Tan-Delta depends on make, type and also aging conditions, the correction factors for different types/makes are different. Hence, no standard temp. error factors can be applied.
	(C) Rate of rise in Tan Delta	0.001 per year (max)	
	(D) Rate of change in Capacitance	1% (max) per year	Cumulative change not exceeding -5% and 10% from the Factory results.
	(E) Monitoring of Tan delta	Up to 0.007 (rise @ 0.001)	-Yearly monitoring -Half yearly monitoring with Diagnostic of Oil and CT -replace the CT IEC 60599 (for limiting values)
	(F) Tan delta tip up test (Difference of tan delta measured at 2kV and 10kV)	0.001(max)	Replace the CT
	(G) Terminal connector contact resistance	10 Micro-Ohm per connector	NGC, UK Recommendations.
	(H) CT ratio errors	According to accuracy class on name plate	IS-2705
3.	CAPACITIVE VOLTAGE TRANSFORMERS		

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	(A) Tan δ	0.007		Since temperature correction factor for Tan δ depends on make, type and also ageing conditions, the correction factors for different types /makes are different. Hence, no standard temperature error factors can be applied.
	(B) Rate of rise in Tan Delta	0.001 per year (max)		
	(C) Change in Tan δ from pre commissioning value	Measurement Value	Frequency	
		up to +0.002	3-Yearly	
		+0.002 to +0.003	Yearly	
		Above +0.003	Replace	

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	(D) Contact resistance of terminal connector.	10 $\mu\Omega$ per connector	NGC, UK Recommendations
	(E) Change in Capacitance from pre commissioning value	Measurement Value	Frequency
		up to $\pm 2\%$ of	3-Yearly
		$\pm 2\%$ to $\pm 3\%$	Yearly
		Above $\pm 6\%$	Replace
	(F) IR Value:		NGC. UK Recommendations
	Primary – Earth	1000 M Ω (Min) by 5.0/10.0 kV IR tester	
	Secondary – Earth	50 M Ω (Min) by 0.5kV IR Tester	
	Control Cable	50 M Ω (Min) by 0.5kV IR Tester	
	(G) Drift in secondary Voltage (to be measured in 0.2/0.5 accuracy class multimeter)	Condition	Measurement Frequency
		up to ± 0.5 V	Fortnightly
		± 0.5 to ± 0.8 V	Fortnightly
		± 0.8 to ± 1.2 V	Weekly
		± 1.2 to ± 2.0 V	Daily
		above ± 2.0 V	Daily
		-0.8 to -4.0 volts less than -4.0 volts	Replace
	(H) EMU Tank Oil Parameters		IS-1866
	BDV, kV (at 2.5 mm Gap)	30 (Minimum)	
	Moisture Content, ppm	35	
	Moisture Content at 90°C, Ω -cm	0.1×10^{12}	
	Acidity	0.5 mg KOH gm (Maximum)	
	IFT at 27°C, Newton/Meter	0.018 (Minimum)	
	Tan δ at 90°C	1.0 (Maximum)	
	Flash Point, °C	125 (Minimum)	
	(I) CVT Voltage Ratio Errors	AS per accuracy class	IEC-186
4.	ISOLATORS		
	(A) Contact Resistance, $\mu\Omega$	300 (Maximum) across both terminal assembly	

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	(B) Contact Resistance of terminal	10 (Maximum)	
	connector, $\mu\Omega$ /Connector		
	(C) IR Value		
	Phase to Earth	1000 M Ω (Minimum) by 5/10 kV Insulation Tester	NGC, UK Recommendation
	Across Open Contact	1000 M Ω (Minimum) by 5/10 kV Insulation Tester	
	Control Cables	50 M Ω (Minimum) by 0. 5 kV Insulation Tester	

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Sl. No.	Parameter	Values	Reference / Remark
5.	SURGE ARESTER (Gapless type)		
	Third Harmonic Resistive Current (THRC) – for all make (ALSTOM / OBLUM / CGL/ELPRO/LAMCO)	1.5 times of commissioning value In case commissioning values are not available, following may be referred:	
	For new LAs	up to 30 μ A	
	For LAs in service	up to 150 μ A – Normal	
		150 to 350 μ A – to be tested for insulation test & if value found low - to be removed from service Beyond 350 μ A, the LA is to be removed from service	
	IR Value/stack	1000 M Ω (Minimum)	NGC, UK recommendations
6.	MISCELLANEOUS		
	Station Earth Resistance	1 Ω (Maximum)	IS
	Thermal Scanning		IEEE/C 37.010.1979
	Temp up to 15°C (above ambient)	Normal	
	Temp. 15-50°C (above ambient)	Alert	
	Temp. above 50 °C (above ambient)	To be immediately attended.	
	Terminal Connectors	10 $\mu\Omega$ per connector	NGC, UK Recommendations
	IR Values		IS 900
	• All LT Motors	50 M Ω (Minimum) by 0.5 kV IR Tester	
	• Control Cables	50 M Ω (Minimum) by 0.5 kV IR Tester	
	• LT Transformers	100 M Ω (Minimum) by 0.5 kV IR Tester	

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	LT Switchgears	100 MΩ (Minimum) by 0.5 kV IR Tester	
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Table – 2 SF₆ Gas Dew Point for Various Breakers

Sl. No	Make of CB	Dew point at rated pressure	Dew point at atmospheric pressure	Remarks
01	BHEL	- 15 °C	- 36 °C	At the time of commissioning
		- 7 °C	- 30 °C	During O & M
		- 5 °C	- 27 °C	Critical
02	CGL	- 15 °C	- 35 °C	At the time of commissioning
		- 10 °C	- 31 °C	During O and M
03	ABB	- 15 °C	- 35 °C	At the time of commissioning
		- 5 °C	- 26 °C	During O and M
04	GE/ALSTOM	- 15 °C	- 36 °C	At the time of commissioning
		- 7 °C	- 30 °C	During O and M
		- 5 °C	- 27 °C	Critical
05	SIEMENS	- 15 °C	- 35 °C	At the time of commissioning
		- 10 °C	- 30 °C	During O and M
		- 5 °C	- 25 °C	Critical

Table – 3: DGA Evaluation Standards for Current Transformer

- IEC 60599 maximum admissible values for sealed instrument transformers without any action to be taken on the transformer (Values in microliters per liter)

H ₂	CO	CO ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂
300	300	900	30	50	10	2

- (CIGRE WG 12.16) Acceptance Criteria for oil Impregnated paper Insulated instrument Transformers

	Gas Concentration, ppm	H ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	CO	CO ₂
New Equipment	Acceptable Limit	<10	<5	<2	<5	none	<100	<200
In-Service Equipment	Do Nothing	<300	<30	<50	<10	<2	<300	<900
	Remove	>1000	>200	>500	>100	>10	No data	No data

- As CTs and PTs / CVTs are sealed units, it is not recommended to open these units at site. These units may need to be sent to the OEM for repairs/refurbishment.

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10.0 SWITCHYARD OPERATIONS UNDER EMERGENCY CONDITIONS

10.1 SAFE PRACTICES DURING HEALTHY CONDITION:

Following observations/ activities shall be recorded twice in every shift (start and end of shift) and separate daily logging by maintenance department:

- i. Executive-in-charge shall take round of complete switchyard and inform/record any abnormality observed in switchyard.
- ii. Healthiness of paging system in switchyard.
- iii. Print out of Disturbance recorder and Event logger.
- iv. Any alarm appearing on Disturbance recorder and event logger.
- v. Time synchronization status of Disturbance recorder and Event logger with GPS.
- vi. Switchyard DCDB P-E, N-E & P-N DC voltages.
- vii. Any alarm appearing on Alarm facia/protection relays.
- viii. Healthiness of all Alarm facia in switchyard control room.
- ix. Time synchronization of ABT metering system with GPS.
- x. Disturbance recorder & Event logger print outs after every fault.
- xi. Downloading of Fault event on Disturbance recorder & Event logger PC hard disk immediately after fault incidence and clearing of Buffer memory of Disturbance recorder & Event logger after downloading.
- xii. Weekly clearing of Disturbance recorder and event logger buffer memory and logging of same.

10.2 SAFE PRACTICES DURING EMERGENCY CONDITION:

Following activities/observations shall be done during emergency condition at switchyard:

- i. On observation of fault in any switchyard equipment, faulty equipment to be isolated first manually.
- ii. In case faulty equipment can't be isolated, the Bus bar to which faulty equipment is
- iii. connected shall be completely isolated manually from remaining Bus bars.
- iv. Unit shutdown, if required, to prevent large scale damage, shall be taken.
- v. DCDB P-E, N-E and P-N DC voltages to be corrected immediately for healthiness of DC system.
- vi. Faulty equipment to be isolated completely and all related protection system to be tested and healthiness ensured before charging of equipment.
- vii. Reason for non-operation/ mal-operation, if any, of protection system to be ensured before taking protection system into service and if required help of OEM may be taken to establish reason for non-operation/ mal-operation of protection system.
- viii. Complete protection/ interlock system to be tested and healthiness to be ensured before taking equipment into service.

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11.0 SWITCHYARD BUS & LINE CHARGING INSTRUCTIONS:

11.1 BUS CHARGING INSTRUCTIONS (FIRST CHARGING/AFTER SHUTDOWN)

- Ensure healthiness by:
 - Physical inspection of the bay by Operation & Area Maintenance Engineer
 - Measure insulation resistance of Bus for healthiness, if possible.
- Ensure that all PTW's have been canceled, and system has been normalized and physically verified.
 - All Earth switches of bus should be open and any local/temporary earthing provided has been removed.
 - Operation and Maintenance jointly to check for removal of man and material from the site.
- All outgoing/incoming breakers/Bus isolators of the bus should be open.
 - CT circuit testing by primary and secondary injection of current (3phase and 1- phase) from CT MB 's end is carried out. (for new bays and where, major job or augmentation in protection or equipment is done)
 - Ensure Bus CVT (wherever provided) circuit has been verified by voltage measurements through primary/secondary injections.
 - Complete protection/interlock system is tested and healthiness to be ensured before CHARGING. It must be ensured settings and Logics in the relay (BUS BAR PROTECTION) are as per approved settings from PE-ELECT/COS (for new bays and where, major job or augmentation in protection or equipment is done.). Operation and maintenance engineer to sign the isolation/normalization slip.
 - After operation of earth switch and isolators the AC supply shall be kept off (except cases where isolator auto operates on interlocks). Also wherever DC supply is not used for status or interlocks in isolator & earth switches after desired operation, same shall also be kept off.
 - Dead bus is to be charged using a non-critical line feeder in radial mode. (tie breaker open) after taking permission from LDC.

11.2 LINE CHARGING INSTRUCTIONS (FIRST CHARGING/AFTER SHUTDOWN)

- Ensure healthiness by
 - i. Physical inspection of the bay by Operation & Area Maintenance Engineers.
 - ii. Getting a clearance/charging message from remote end.
 - iii. Measure insulation resistance online for healthiness, if possible.
- Ensure that all PTWs have been canceled, and system has been normalized and physically verified (both by operation and maintenance).

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- CT circuit testing by primary and secondary injection of current (3-phase and 1- phase) from CT MB end is carried out. (for new bays and where, major job or augmentation in protection or equipment is done)
- Ensure Line CVT (wherever provided) circuit has been verified by voltage measurements through primary/secondary injections.
- Normalize/Connect any disconnecting link/control cable disconnected during testing.
- Complete protection/ interlock system is tested and healthiness to be ensured before taking the LINE into service. It must be ensured the settings and Logics in the Relay (MAIN and LBB) are as per the approved settings from PE-Electrical/COS (for new bays and where, major job or augmentation in protection or equipment are done). Operation and maintenance engineer to sign the isolation/normalization slip.
- Ensure protection systems, LBB are put in service & breaker trip circuit 1&2 are healthy. Ensure Isolator status and CTs provided in busbar protection has been verified for correct inputs. The spill currents shall be recorded before and after the line charging.
 - During first charging-
 - i. Enable SOTF for all Zones
 - ii. Reverse Zone (Zone-4) is to be set at 100% of line length with no delay (Possible in most of the Numerical relays).
- After operation of earth switch and isolators the AC supply shall be kept off (except cases where isolator auto operates on interlocks). Also, wherever DC supply is not used for status or interlocks in isolator & earth switches after desired operation, same shall also be kept off.
- First (1st) charging of line is to be preferably carried out by a single feeder / line fed from remote end of the sub-station so that in case of non-operation of primary protection, the faulted line is cleared by remote end on back up protections instead of generating units. Issue is to be taken up with respective Load Dispatch Center for their acceptance.
- Charging of line after shutdown/line fault is to be preferably carried out from the remote end substation (PGCIL/State board). If both end of the lines are fed through NTPC generating stations, charging of line may be done from the station who has taken the line shutdown or more distant from the fault location.

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12.0 BRIEF OF SWITCHYARD RELATED FAULTS WHICH MAY LEAD TO UNIT TRIP:

Sr.no	Fault description	Recommendations/Maintenance practices
1.	Hot spots at GT bay Isolator & CT connectors	- Monthly thermography to be done. Record & attend the hot spots as opportunity basis. Attend immediately if temp diff is >50degC. - Proper alignment to be checked during Isolator closing. - CRM to be done during bay testing. - Proper cleaning & greasing to be done on Isolator main contacts. - Proper tightness to be done during bay O/H. - Use of GI bolts to be ensured & cleaning to be done at terminal contacts.
2.	Oil leakage from GT side CT & CVTs	- Daily walk down monitoring of bay equipment to be ensured. - In case of minor seepage, kept under close monitoring & attend on opportunity else Immediately replace. - Proper alignment of IPS tube at terminal contacts to be ensured.
3.	GT LA failure	- LCM measurement & record keeping on Quarterly basis. During monsoon & Winter, measure on monthly basis. - On increasing trend, LCM to be done on daily/weekly basis. Replace immediately, If $I_r(3^{rd}) > 350$ microamp. For new LA $I_r(3^{rd})$ should be less than 30microAmp. - Cleaning & IR testing during bay O/H or any opportunity S/D. - Daily walk down monitoring of bay equipment to be ensured & record total leakage current at counter. - Replacement to be done after 15 yrs of service.
4.	GT bay CT & CVT failure	- All routine test to be done annually. Replace if any deviation from acceptance limit found as per OGN. - Replacement to be done after 20 yrs of service.
5.	Snapping of A-row conductor	- A-row tightness to be checked on Unit O/H.

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6.	Failure of Unit feeder & Main Bus string Insulators	- Insulator cleaning to be ensured on bay O/H. - Insulator PID test to be carried out after 10 yrs of service. Replace the faulty insulator, if found defective. Hot line washing to be done as per COS guidelines.
7.	Failure of control cable for breaker trip circuit & CT circuit due to low IR	- Replacement of control cables after 15 years of service with armoured type new cable. - IR to be taken during Unit OH @ 1000V DC & compare with previous results. If major deviation found, then replace the cable. - Measurement of CT loop resistance from GRP panel - on bay O/H.
8.	Failure of control cable due to fire at Swyd cable trench	- Complete Cleaning of cable trenches to be ensured. - Power & control cables to be laid separately. - Water logging should not be inside the cable trench & availability of dewatering pumps to be ensured during - rainy season.
9.	Oil/Air/SF6 leakage from Breaker assembly & breaker lockout	- Complete OH of breaker after 10 years of commissioning. - Replacement of breakers after 25 years of service. - Daily walk down monitoring & record keeping of bay equipment to be ensured.
10.	Failure of Circuit breaker	- Complete OH of breaker after 10 years of commissioning. - Replacement of breakers after 25 years of service. - Daily walk down monitoring & record keeping of bay equipment to be ensured. - Breaker routine testing to be done as per guidelines & - acceptable limit on annual basis.
11.	Malfunctioning of Pole discrepancy relay	- Testing of PD relay to be done during Unit OH. - Tightness of wire termination ensured.

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12.	Malfunctioning of bay Protection relay	- Testing of relay to be done during Unit OH. - Tightness of wire termination on relay & panel TBs to be ensured. - CT & PT link connection to be normalised as per latest - as built scheme.
13.	During Changeover of main breaker to transfer breaker through N/T switch	- SOP to be available for Main breaker to transfer bay changeover & vice versa at control room. - Special precaution to be taken for dead machine protection, where breaker status is used. - Before closing of Transfer breaker, switch to be positioned at T (For N/T switch) & at I (for NIT switch) & it is to be normalised after opening of Transfer breaker.
14.	Earth fault in DC tripping circuit lead to spurious opening of BKR or Isolator	- Daily monitoring & checklist for Swyd DC system. - DC E/F alarm should be hooked up to control room. - Immediate rectification of DC E/F issue. - Isolators & E/S fuses to be removed & kept at KB during running condition.
15.	Fire/ash water spray jet on Main Bus bar conductors lead to Bus bar tripping	- Fire hydrant pipelines should not laid around Switchyard live conductor area. If available, canopy must be provided. - Canopy to be provided over ash water pipelines near - swyd fencing.
16.	Blocking of Bus bar protection due to BU communication failure or CU/BU relay failure	- Daily monitoring of all relay & control panels & checking of any alarm appeared. - Bus bar protection logic to be implemented in relays - as per IOM Ref: OS/ED/Elec/2019/01, dt: 30.05.2019.

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17.	Mal operation of Isolators/earth switches due to Human error during Unit on bar	- Any bay operation to be supervised in presence of Operation & maintenance executives. - Local operation of bay equipment to be avoided. - Large size SLD to be displayed in Swyd control room. - Bay Isolation/normalisation protocol to be prepared by Oprn & Maint executives before any operation.
18.	Accidental opening of Isolators due to storm/cyclone	- Daily monitoring of bay equipment & checking of Isolator alignments & proper making of Male & female contact.
19.	Water/moisture ingress in Swyd equipment Panel/MBs	- Space heater service availability to be ensured in all Swyd panels & MBs. - Daily monitoring of Swyd equipment & checking of proper closing of all panels/MBs. - Faulty gaskets to be replaced in MBs. - All panel holes to be covered by aluminium tape.
20.	Unit/Bus tripping due to fall of metallic wires on live conductors	- Bird nest to be removed from all Swyd gantries. - All the metallic wires & sheets present inside the Swyd to be removed.
21.	Looseness of CT wire at CT MB or Relay panels	- CT, CVT & all tripping related terminations & links tightness to be checked during Unit O/H.
22.	Operation of Unit Intertrip relay due to bypassing of Tie Breaker Open status on Busbar protection.	- Busbar protection logic to be configured as per Engg/OS guidelines in all PUs & CUs. - Complete scheme checking & simulation to be checked during testing & commissioning. - Tie Breaker open status to be hooked up in series with Busbar protection for Intertrip.
23.	Maloperation of Busbar relay during relay testing of any bay	- Before starting any testing on relay & bay equipment, LBB to Busbar trip wire to be removed from relay panel. - Before testing online relay, Zone-4 pickup B/O wire (used as an Input for Busbar relay) to be removed at line relay panel.

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13.0 SWITCHYARD STANDARD OPERATING PROCEDURES (SOPs):

13.1 GENERAL PROCEDURE FOR OPENING OF ISOLATORS:

- i. Collect portable Walkie-talkie set / Cell phone, Bay key from Switchyard control room.
- ii. After reaching the required bay, ensure the concerned breaker is in OFF condition.
- iii. Establish communication with switchyard control room.
- iv. For remote operation of the Isolator unlock & open the isolator marshalling box and switch on the DC supply & 3 Φ AC supply and put the Local-Remote switch in Remote position. Communicate the same to SWYD CR.
- v. Now open the said Isolator from SWYD Control panel/SCADA. Check for complete opening of all the arms of the 3-poles at local.
- vi. Switch off the DC supply & 3 Φ AC supply and put the Local-Remote switch in OFF position inside the isolator-marshalling box. Close & lock the Isolator marshalling box.
- vii. Put the Danger Tag on Isolator control switch at switchyard control panel.

Note: -

- The operator should keep safe distance from the point of operation of the isolator as much as possible.
- Isolator opening command is to be executed only from switchyard control room, i.e. Remote.
If Isolator fails to open by remote command, the opening of Isolator from local marshalling box to be carried out only in presence & supervision of Electrical Maintenance.

13.2 GENERAL PROCEDURE FOR CLOSING OF ISOLATORS

- i. Collect the Walkie-talkie set / Cell phone, Torch-Light & the corresponding bay key from the Switchyard control room.
- ii. Remove the Danger Tag from Isolator control switch at switchyard control panel.
- iii. Send the operator to the local (the bay, which is to be normalized).
- iv. The operator deputed to the local should physically check and ensure the following things:
 - a. The corresponding breakers are in OFF condition

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- b. All the Earth Switches of the corresponding bay are in OPEN condition.
- c. Establish communication with the Control Engineer at the switchyard control room and apprise him with the situation.
- d. Wait for further instruction from the Switchyard Control Engineer
- e. Keep safe distance from the point of operation as much as possible.
- v. After getting instruction from the control engineer for closing any isolator, the operator deputed to local should do the following:
 - a. Unlock & open the marshalling box of the isolator, which is to be CLOSED.
 - b. Switch ON the DC supply of the Isolator
 - c. Switch ON the 3- Φ AC supply of the Isolator
 - d. Put the Local-Remote selector switch of the Isolator in Remote position.
 - e. Keep safe distance from the point of operation as much as possible
 - f. Give the Clearance/feed back to the Control Engineer about the readiness of the Isolator.
- vi. After getting clearance/feedback from the operator at local, CLOSE the isolator from the Bay control panel/SCADA in the switchyard control room.
- vii. After the command is given to a particular isolator, ask the operator to check for the closing of all the three phases/poles and alignment of the Isolator.
- viii. After everything found in order, ask the operator to do the following:
 - a. Switch OFF the DC supply of the Isolator
 - b. Switch OFF the 3- Φ AC supply of the Isolator
 - c. Put the Local-Remote selector switch of the Isolator in Local Position
 - d. Close & lock the marshalling box of the isolator, which is closed.
 - e. Give the feed back to the control engineer about the points mentioned above.
 - f. Wait for further instruction from the Switchyard Control Engineer.

Note: -

- g. From the safety point of view, the operator should keep away from the Isolator, while the same is in operation.
- h. The closing command to the isolator is to be given from the switchyard control room, i.e., REMOTE. If the Isolator fails to close from REMOTE, then the same is to be informed to Electrical maintenance. In no case the operator should CLOSE the Isolator from LOCAL.

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13.3 GENERAL PROCEDURE FOR CLOSING OF EARTH SWITCHES

- i. Collect portable Walkie-talkie set / Cell phone, Bay key from Switchyard control room.
- ii. After reaching the concerned bay, establish communication with switchyard control room.
- i. Check if the CB & the concerned isolators of the said bay / busbar / line are in open condition.
- ii. Earth-switch can be operated only Locally. For operation of the Earth-switch unlock & open the isolator marshalling box and switch on the DC supply & 3Φ AC supply and put the Local- Remote switch in Local position. Communicate the same to SWYD CR. Close the Earth-switch Locally by pressing the RED push button.
- iii. Ensure from local that the Earth-switch is completely closed for all three poles & communicate the confirmatory message to switchyard control room.
- iv. Switch off the DC supply & 3Φ AC supply and put the Local-Remote switch in OFF position inside the Earth-switch marshalling box. Close & lock the Earth-switch marshalling box.
- v. Put Danger Tag on the same at switchyard control panel.

Note: -

- Any Earth Switch closing operation in switchyard is to be carried out only in consultation of Electrical Maintenance.

13.4 GENERAL PROCEDURE FOR OPENING OF EARTH SWITCHES

- i. Collect portable Walkie-talkie set / Cell phone, lock & key from Switchyard control room.
- ii. After reaching the concerned bay, establish communication with switchyard control room.
- iii. Check if all man & material has been removed from the concerned bay and the CB & the concerned isolators of the said bay / busbar / line are in open condition.
- iv. Earth-switch can be operated only Locally. For operation of the Earth-switch unlock & open the isolator marshalling box and switch on the DC supply & 3Φ AC supply and put the Local- Remote switch in Local position. Communicate the same to SWYD CR. Open the Earth-switch Locally by pressing the green push button.
- v. Ensure that the Earth Switch has completely opened. Communicate the same to the SWYD CR.

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13.5 GENERAL PROCEDURE FOR OPENING OF CIRCUIT BREAKERS

- i. DC supply is kept ON & the Local/Remote switch is kept in Remote position always for the Breaker (unless specifically made otherwise during maintenance activity).
- ii. Trip the breaker from the SWYD control panel/SCADA.
- iii. Put the Danger Tag at the Breaker control knob.
- iv. Switch off DC control supply of the Breaker from local marshalling box (Only if specifically requested for).

13.6 GENERAL PROCEDURE FOR CLOSING OF CIRCUIT BREAKERS

- i. Remove the Danger Tag from the Breaker position indicator.
- ii. Switch on DC control supply & keep the Local/Remote switch in Remote position of the Breaker from local marshalling box.
- iii. Check whether the Master Trip Relay 86 for the Breaker has got operated & reset the same if required.
- iv. If SCADA control is not available at Switchyard CR, then bring the Synchronizing Trolley close to the concerned control panel. Put the Synchronizing Trolley cable plug at the synchronizing socket for the Breaker to be normalized by properly pushing & rotating the cable plug anticlockwise (Check for Breaker No. written at the synchronizing socket for correct insertion).
- v. Put the Synchronoscope ON/OFF switch on the Synchronizing Trolley to ON position and if voltage is available on either side of the Breaker put the SYNC/BYPASS switch in SYNC mode or to BYPASS mode if voltage is available only on one side of the Breaker.
- vi. Check whether the 'Synchronization In-Limit' green lamp is glowing in the Synchronizing Trolley.
- vii. If the 'Synchronization In-Limit' lamp glows, then give close command to the breaker from the control panel.
- viii. Put off the Synchronoscope switch on the synchronizing trolley.
- ix. Take out the Synchronizing Trolley cable plug from the synchronizing socket by pushing & moving it clockwise.
- x. Keep the Synchronizing Trolley at its designated position. (If SCADA control is available then close the breaker through SCADA.)

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13.7 FOR TRANSMISSION LINE ISOLATION:

Line description

Date:.....

Sl No.	Activity	Responsibility	Yes/No
1	Ensure RLDC code received for isolating/tripping the line.	Operation	
2	Ensure same RLDC code received at other end substation.	Operation	
3	By taking other end substation operation engineer online over telephone, give trip command at one end of the line and ensure other end breaker tripped at same instant.	Operation	
4	Ensure line voltages and currents are zero after tripping the line.	Oprn & EMD	
5	Open line isolators and bus isolators. Ensure Breaker & isolator open status in Bus bar PU relays.	Oprn & EMD	
6	If closing of earth switch is required (either by EMD or PGCIL):	Operation	
	>>Ensure permit is applied.	Operation	
	>>Ensure other end line isolator also opened.	Operation	
7	After switching of Isolator & earth switch remove fuses & kept at JB. JB locked with safety tag.	Operation	
8	Remove trip circuit (TC1, TC2, LKT1, LKT2) fuses and Z4 start O/P for Busbar relay & LBB links at relay panel.	EMD	
9	Take NBFC (No back feeding clearance) charging code from other end and record in code register.	Operation	
10	After ensuring NBFC from other end, issue permit.	Operation	

* Details of Equipment (Isolator, Breaker, Earth Switch) to be isolated is to be noted:

Signature of Operation Representatives

Signature of EMD representative

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13.8 FOR TRANSMISSION LINE NORMALIZATION:

Line description

Date:.....

SI No.	Activity	Responsibility	Yes/No
1	Ensure all permits on the line are cancelled.	Operation	
2	Ensure all PTW at other end substation also cancelled.	Operation	
3	Ensure NBFC on the line is cancelled.	Operation	
4	Open line earth switches LE2, LE1, BE & AE.	Operation	
5	Physically ensure all local earthing rods are removed.	Oprn & EMD	
6	Ensure other end line earth switch is also opened.	Operation	
7	Ensure CT TTB is normalized.	EMD	
8	Ensure code received for charging of the line.	Operation	
9	Normalize trip circuit (TC1, TC2, LKT1, LKT2) fuses & Z4 start O/P wire and LBB links and reset all trip relays.	EMD	
10	Close line, reactor (if applicable) and bus isolators and ensure Breaker & isolator status in Bus bar PU relays.	Operation	
11	Ensure other end isolators are also closed.	Operation	
12	After switching of Isolator & earth switch remove fuses & kept at JB.	Operation	
13	Put synchronizing trolley in service position and switch it on.	Operation	
14	If dead line is to be charged, Put switch in Manual mode. Close CB and ensure all three phase voltages and give clearance for synchronization at other end.	Operation	
15	Preferably line to be charged from remote end. If the line is being charged from other end, put check synchronization relay in Auto mode, ensure voltages in all phases after line charging from other end. Ensure synchronization permissive is available and close the breaker. Remove synchronizing trolley after switching off.	Operation	
16	Ensure no alarm/relay appeared in line protection, control and Bus bar protection panel appeared after line taken in service.	Oprn & EMD	
17	Ensure currents in all three phases.	Oprn & EMD	
18	Physically inspect local bay equipment for any abnormality.	Oprn & EMD	
19	Ensure no change in PLCC counters reading.	Oprn & EMD	

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a) Details of Equipment (Isolator, Breaker, Earth Switch) to be normalized is to be noted:

Signature of Operation Representatives

Signature of EMD representatives

13.9 FOR GENERATING UNIT ISOLATION:

Description.....

Date:

S.No	Activity	Responsibility	Yes/No
1	Ensure from corresponding UCB, whether isolations can be done on generator bay for issuing PTW.	Operation	
2	Open line isolator and bus isolator. Ensure Breaker & isolator open status in Bus bar PU relays.	Oprn & EMD	
3	If closing of earth switch is required:	Operation	
	>> Ensure permit is applied.	Operation	
4	After UAT/UT breaker rack out, close the line earth switch (GE2).	Operation	
5	Remove trip circuit (TC1, TC2, LKT1, LKT2) fuses and LBB links at relay panel.	EMD	
6	After switching of Isolator & earth switch remove fuses & kept at JB. JB locked with safety tag.	Operation	

* Details of Equipment (Isolator, Breaker, Earth Switch) to be isolated is to be noted:

Signature of Operation Representatives

Signature of EMD representatives

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13.10 FOR GENERATING UNIT NORMALIZATION:

Description.....

Date.....

S. No	Activity	Responsibility	Yes/No
1	Ensure all permits on the generator bay are cancelled.	Operation	
2	Ensure from the corresponding UCB that all PTW on GT and UAT have also been cancelled.	Operation	
3	Normalize trip circuit (TC1, TC2) fuses, isolator fuses & LBB trip wire.	EMD	
4	Open line earth switch GE2, GE1, BE & AE.	Operation	
5	Ensure CT TTB is normalized.	EMD	
6	Physically ensure all local earthing rods are removed.	Oprn & EMD	
7	Close line and bus isolators and ensure Breaker & isolator status in Bus bar PU relays.	Oprn & EMD	
8	After switching of Isolator & earth switch remove fuses & kept at JB.	Operation	
9	Inform corresponding Unit control room that the bay is ready for synchronization.	Operation	
10	Ensure no alarm/relay appeared in bay protection, control and Bus bar protection panel appeared after unit synchronization.	Oprn & EMD	
11	Ensure currents in all three phases.	Oprn & EMD	
12	Physically inspect local bay equipment for any abnormality.	Oprn & EMD	

* Details of Equipment (Isolator, Breaker, Earth Switch) to be normalized is to be noted:

Signature of Operation Representatives

Signature of EMD representatives

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13.11 FOR MAIN BUS ISOLATION:

Description

Date.....

S. No	Activity	Responsibility	Yes/No
1	Ensure RLDC code received for isolating/tripping the line.	Operation	
2	Identify the feeders to make bus free & note the sequence of operation and accordingly sequence to be followed. Manual trip the Breaker one by one.	Oprn & EMD	
3	Open bus isolator from bus to be freed & keep the motor supply off. Ensure Breaker & isolator open status in Bus bar PU relays.	Oprn & EMD	
4	After ensuring bus is free, trip sectionalizer breaker.	Oprn & EMD	
5	Open Line isolators of sectionalizer (both end) and keep MCB switch in off position.	Operation	
6	Check voltage of all three phases of dead bus is zero at CVT panel.	Oprn & EMD	
7	Closing of Bus side earth switch as per maintenance person requirement & issue permit to Maintenance person.	Operation	
8	After switching of Isolator & earth switch remove fuses & kept at JB. JB locked with safety tag.	Operation	

* Details of Equipment (Isolator, Breaker, Earth Switch) to be isolated is to be noted:

Signature of Operation Representatives

Signature of EMD representatives

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LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

13.12 FOR MAIN BUS NORMALIZATION:

Description

Date:.....

S.No	Activity	Responsibility	Yes/No
1	Ensure all permits on the 400KV Bus are cancelled.	Operation	
2	Open Earth switch of bus-sectionalizer bay and open earth switch (BE) at any other bay where work is carried out.	Operation	
3	Physically ensure all local earthing rods are removed all through the bus (especially where work is carried out).	Oprn & EMD	
4	Ensure CT TTB is normalized. (if done).	EMD	
5	Ensure code received for charging of the Bus.	Operation	
6	Normalize trip circuit (TC1, TC2) fuses, isolator fuses and reset all trip relays.	EMD	
7	Close isolator of sectionalizer and ensure the Isolator status in PU relay. Check the Isolator contact physically.	Oprn & EMD	
8	Close the sectionalizer breaker with the help of synchronizing trolley. Check for Bus voltages.	Operation	
9	Ensure no alarm/relay appeared in Bus bar protection panel.	Oprn & EMD	
10	Close the Bus isolator of all the feeder & ensure the isolator actual status in Bus bar PU relay. Then Close the corresponding Breakers.	Oprn & EMD	
11	Ensure no alarm/relay appeared in line protection, control and Bus bar protection panel.	Oprn & EMD	
12	After switching of Isolator & earth switch remove fuses & kept at JB.	Operation	
13	Physically inspect local bay equipment for any abnormality.	Oprn & EMD	

* Details of Equipment (Isolator, Breaker, Earth Switch) to be normalized is to be noted:

Signature of Operation Representatives

Signature of EMD representatives

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TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

13.13 FOR MAIN TO TRANSFER BREAKER CHANGEOVER OF UNIT:

Line description

Date:.....

SI No.	Activity	Responsibility	Yes/No
1	Ensure all permits on the TBC are cancelled.	Operation	
2	Ensure Isolators & Earth switches of transfer bay are Open	Operation	
3	Physically ensure all local earthing rods are removed from Transfer BUS	Operation	
4	Check physically whole TBC bay for any local/ system earthing.	Operation	
5	Ensure CT TTB of TBC is normalized & tightness checked at local & relay end by EMD	EMD	
6	Ensure TBC breaker local/remote switch in breaker MB in remote.	Operation	
7	Ensure code received for charging the bay. Code No..... Time.....	Operation	
8	Ensure that 89C isolator of all remaining Bays are in Open Condition. Isolator contacts/ its state are to be verified at local	Operation	
9	In control Panel, Unlock the NIT Switch & toggle the NIT switch of BCU to Inter position.	Oprn & EMD	
10	Ensure Normal inter & Inter-transfer lamp is picked up in BCU panel of GT Bay.	Oprn & EMD	
11	Ensure NIX & ITX relays are picked up.	Oprn & EMD	
12	Ensure sl. 8 & 10 in SCADA for line bay	Oprn & EMD	
13	Charge Transfer Bus by Closing line 89C Isolator from SCADA. Visually ensure isolator proper contacts in all 3 phases.	Operation	
14	Ensure no alarm/relay appeared in bay & Busbar relay & control panel.	Oprn & EMD	
15	Check for Differential & bias currents in CUA & CUB	EMD	
16	After Charging Transfer Bus, monitor the Voltage of transfer Bus & do thermography of complete transfer Bus/support string insulators. Report any abnormality	EMD	

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LMI NO		LMI/OGN/OPS/ELEC/42, Rev:01	
17	In Bus Bar Panel, Check Bus Bar CU-A&B (Central Unit) for Zone-3 creation in CU Bus bar topology from front facia.	EMD	
18	In Bus Bar Panel, Check line Bay Bus Bar PU-A&B for Zone-3 creation in PU Bus bar topology from front facia. Line 89C close status LED should be available in both PU-A & PU-B.	EMD	
19	Close TBC 89C isolator from SCADA & ensure 89C status in Transfer Bay PU-A & PU B	Oprn & EMD	
20	After Compliance of Sl. 9 to Sl.19, Close the Bus isolator of TBC 89A or 89B from SCADA. Selection of 89A or 89B shall be based on isolator status of GT Bay. i.e. Position of Bus 1 & Bus 2 isolator of TBC Bay shall be same as Bus 1 & Bus 2 isolator of GT Bay.	Operation	
21	After Closing Bus isolator of TBC Bay, Bay thermography shall be done of Transfer Bay, if any abnormality, report. Bus isolator close status must be ensured in PU-A & PU-B. It should be same in both PU of Transfer Bay	EMD	
22	Ensure BCU state in SCADA, TBC breaker local/remote switch in breaker BMK are in remote.	EMD	
23	In UCB, open ECD panel & select 400 KV Transfer Breaker by pressing Release button & Transfer Breaker select Button simultaneously. Blue indication will glow	Operation	
24	Toggle to CHECK option in the ECD panel. ECD panel incoming & running voltage & frequency should match each other, synchroscope should read 12 o'clock & only "In synch" LED must glow, if not then check the SKE relay logic & voltages in ECP panel.	Operation	
25	Close the Transfer Breaker by pressing release button & ON button (RED). Ensure ON feedback Red led. Then press RESET button in ECD. Breaker Selection Blue LED should turn off.	Operation	
26	Ensure the close status of circuit breaker of Transfer Bay in PU-A & PU-B of bus bar panel, SCADA & Relay panel.	Oprn & EMD	
27	Transfer the NIT switch to Transfer position.	Operation	

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TITLE		Guidelines on AIS maintenance practices	
LMI NO		LMI/OGN/OPS/ELEC/42, Rev:01	
28	Ensure Transfer & Inter-transfer lamps are picked up in BCU panel of Bay.	Oprn & EMD	
29	Ensure TX & ITX relays in Panel are picked up & perform thermography of TBC Bay.	EMD	
30	In Unit control Room, open ECD panel & select 400KV Main Breaker by pressing Release button & Main Breaker select Button simultaneously. Blue indication will glow	Operation	
31	Toggle to CHECK option in the ECD panel. ECD panel incoming & running voltage & frequency should match each other, synchroscope should read 12 o'clock & only "In synch" LED must glow, if not then check the SKE relay logic & voltages in ECP panel.	Operation	
32	Open Main Bay Circuit breaker from remote by pressing release button & OFF button (GREEN). Ensure OFF feedback Green led. Then press RESET button in ECD. Breaker Selection Blue LED should turn off.	Operation	
33	Confirm the Open status of main circuit Breaker in Bay PU-A & PU-B of bus bar panel, SCADA & GRP & Perform sl.no.15	Oprn & EMD	
34	Confirm the Closed status of TBC 89C isolator in Transfer Bay PU-A & PU-B of bus bar panel BB4C	Oprn & EMD	
35	Open Main Bay 89A&B bus isolators from remote.	Operation	
36	Confirm the close status of main bay 89C isolator in PU-A & PU-B of bus-bar panel & SCADA.	Oprn & EMD	
37	Perform sl.no.15 & TBC Bay thermography	EMD	

Signature of Operation Representatives

Signature of EMD representative

TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

13.14 FOR TRANSFER TO MAIN BREAKER CHANGE OVER OF UNIT:

Line description

Date:.....

Sl No.	Activity	Responsibility	Yes/No
1	Ensure all permits on the Main Bay are cancelled.	Operation	
2	Ensure isolators & earth switches of main bay are Open	Operation	
3	Physically ensure all local earthing rods are removed from main bay.	Operation	
4	Ensure R,Y,B Circuit Breaker of Main bay are in OFF condition	Operation	
5	Ensure CT TTB of Main Bay is normalized	EMD	
6	Ensure breaker local / remote switch in breaker MB is in remote.	Opn & EMD	
7	Ensure code received for charging Code No..... Time.....	Operation	
8	In control Panel, Unlock the NIT Switch & Toggle the NIT switch of BCU to Inter-Transfer position.	Operation	
9	Ensure Normal Inter & Inter-transfer lamp is picked up in BCU panel of main Bay.	Opn & EMD	
10	Ensure NIX & ITX relays in Panel are picked up.	Opn & EMD	
11	Ensure sl. 2 & 10 in SCADA for main Bay.	Opn & EMD	
12	Close main 89G isolator from SCADA & visually ensure isolator proper contacts in all 3 phases	Operation	
14	Close the Bus isolator 89A or 89B from SCADA. Selection of 89A or 89B shall be based on bus isolator status of TBC Bay. i.e. Position of Bus 1 & Bus 2 isolator of main bay shall be same as Bus 1 & Bus 2 isolator of TBC Bay. Confirm Bus isolator 89A or 89B LED status in main bay PU-A & PU-B.	Operation	
15	After Closing Bus isolator of main bay, Bay thermography shall be done, if any abnormality, report. 89A or 89B Bus isolator close status must be ensured in PU-A & PU-B. It should be same in both PU of Main Bay.	EMD	

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16	Check for Zone -3 topology creation in the following: a) In Bus Bar Panel Check main Bay- PU-A, PU-B b) In Bus Bar Panel Check TBC Bay-PU-A, PU-B	EMD	
17	Ensure no alarm/fault in panel Main & TBC bay relay/control & Busbar panel.	Oprn & EMD	
18	Check for Differential & bias currents in CU-A, CU-B, UNIT PU-A & PU-B, TBC PU-A & PU-B	EMD	
19	In Unit control Room, open ECD panel & select 400KV Main Breaker by pressing Release button & Main Breaker select Button simultaneously. Blue indication will glow	Operation	
20	Toggle to CHECK option in the ECD panel. ECD panel incoming & running voltage & frequency should match each other, synchroscope should read 12 o'clock & only "In synch" LED must glow, if not then check the SKE relay logic & voltages in ECP panel.	Operation	
21	Close main bay Circuit breaker from remote by pressing release button & ON button (RED). Ensure ON feedback RED led. Then press RESET button in ECD. Breaker Selection Blue LED should turn off.	Operation	
22	Confirm the close status of main circuit Breaker in PU-A & PU-B of bus bar panel & SCADA & Perform sl.no.18	EMD	
23	Perform sl. 19 with TBC breaker & Open TBC Circuit breaker from remote by pressing release button & OFF button (GREEN). Ensure OFF feedback Green led. Then press RESET button in ECD. Breaker Selection Blue LED should turn off.	Operation	
24	Confirm the Open status of TBC breaker in Transfer Bay PU-A & PU-B of bus bar panel & perform sl.no.18	Oprn & EMD	
25	Open main bay 89C isolator & ensure open status in Bay PU-A & PU-B of bus-bar panel & SCADA.	Oprn & EMD	
26	Open TBC 89C isolator & ensure open status in Bay PU-A & PU-B of bus-bar panel & SCADA.	Oprn & EMD	

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TITLE		Guidelines on AIS maintenance practices	
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27	Open TBC 89A or B bus isolator from remote & ensure open status in Bay PU-A & PU-B of bus-bar panel & SCADA.	Oprn & EMD	
28	Perform sl.no. 8 & toggle to Normal. Normal Inter & Normal lamps are picked up in BCU panel of main Bay & ITX, TX relays are dropped.	Oprn & EMD	
29	Perform sl.no.18 & Bay thermography. Report if any abnormality	EMD	

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13.15 FOR MAIN TO TRANSFER BREAKER CHANGEOVER OF LINE:

Line description

Date:.....

SI No.	Activity	Responsibility	Yes/No
1	Ensure all permits on the TBC are cancelled.	Operation	
2	Ensure Isolators & Earth switches are Open	Operation	
3	Physically ensure all local earthing rods are removed from Transfer BUS	Operation	
4	Check physically whole TBC bay for any local/ system earthing.	Operation	
5	Ensure CT TTB of TBC is normalized & tightness checked at local & relay end by EMD	EMD	
6	Ensure TBC breaker local/remote switch in breaker BMK is in remote.	Oprn & EMD	
7	Ensure code received for charging the line. Code No..... Time.....	Operation	
8	Ensure that 89C isolator of all remaining Bays are in Open Condition. Isolator contacts/ its state is to be verified at local	Operation	
9	In line bay control Panel, Unlock the NIT Switch & Toggle the NIT switch of BCU to Inter position.	Oprn & EMD	
10	Ensure Normal inter & Inter-transfer lamp is picked up in BCU panel of main Bay.	Oprn & EMD	
11	Ensure NIX & ITX relays in Panel are picked up.	Oprn & EMD	
13	Charge Transfer Bus by Closing main bay 89C Isolator from SCADA. Visually ensure isolator proper contacts in all 3 phases.	Operation	
14	Ensure no alarm/relay appeared in relay & Busbar panel.	Oprn & EMD	
15	Check for Differential & bias currents in CUA & CUB	EMD	
16	After Charging Transfer Bus, monitor the Voltage of transfer Bus & do thermography of complete transfer Bus/support string insulators. Report any abnormality	EMD	

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LMI NO		LMI/OGN/OPS/ELEC/42, Rev:01	
17	In Bus Bar Panel, Check Bus Bar CU-A&B (Central Unit) for Zone-3 creation in CU Bus bar topology from front facia.	EMD	
18	In Bus Bar Panel, Check main Bay Bus Bar PU-A&B for Zone-3 creation in PU Bus bar topology from front facia. 89C close status LED should be available in both PU-A & PU-B.	EMD	
19	Close TBC 89C isolator from SCADA & ensure 89C status in Transfer Bay PU-A & PU B	Oprn &EMD	

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20	After Compliance of Sl. 9 to Sl.19, Close the TBC bay Bus isolator 89A or 89B from SCADA. Selection of 89A or 89B shall be based on isolator status of main Bay. i.e. Position of Bus 1 & Bus 2 isolator of TBC Bay shall be same as Bus 1 & Bus 2 isolator of main Bay.	Operation	
21	After Closing Bus isolator of TBC Bay, Bay thermography shall be done of Transfer Bay, if any abnormality, report. Bus isolator close status must be ensured in PU-A & PU-B. It should be same in both PU of Transfer Bay	EMD	
22	Ensure TBC breaker local/remote switch in breaker BMK are in remote.	Oprn &EMD	
23	Close the EHV CB of TBC bay.	Operation	
24	Ensure the close status of TBC Bay circuit breaker in Transfer Bay PU-A & PU-B of bus bar panel & SCADA.	Oprn &EMD	
25	Transfer the NIT switch in to Transfer position for line Bay.	Operation	
26	Ensure Transfer & Inter-transfer lamps are picked up in BCU panel of main Bay	Oprn &EMD	
27	Ensure TX & ITX relays in Panel are picked up & perform thermography of TBC Bay	Oprn &EMD	
28	Open main Circuit breaker from remote	Operation	
29	Confirm the Open status of main circuit Breaker in PU-A & PU-B of bus bar panel & SCADA & Perform sl.no.15	Oprn &EMD	
30	Confirm the Closed status of TBC 89C isolator in Transfer Bay PU-A & PU-B of bus bar panel.	Oprn &EMD	
31	Open main bay 89A or B isolator from remote.	Operation	
32	Confirm the Open status of main bay 89A&B isolator in Bay PU-A & PU-B of bus-bar panel & SCADA.	Oprn &EMD	
33	Perform sl.no.15 & TBC Bay thermography	EMD	

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13.16 FOR TRANSFER TO MAIN BREAKER CHANGEOVER OF LINE:

Line description

Date:.....

Sl No.	Activity	Responsibility	Yes/No
1	Ensure all permits on the Main Bay are cancelled.	Operation	
2	Ensure Isolators & Earth switches of main bay are Open	Operation	
3	Physically ensure all local earthing rods are removed from main Bay	Operation	
4	Ensure R,Y,B Circuit Breaker of Main bay are in OFF condition	Operation	
5	Ensure CT TTB of Main Bay is normalized	EMD	
6	Ensure main breaker local / remote switch in breaker MB is in remote.	Oprn &EMD	
7	Ensure code received for charging of main bay Code No..... Time.....	Operation	
8	In Panel of main Bay, Unlock the NIT Switch & Toggle the NIT switch of line BCU to Inter-Transfer position.	Operation	
9	Ensure Normal Inter & Inter-transfer lamps are picked up in BCU panel of main Bay.	Oprn & EMD	
10	Ensure NIX & ITX relays in Panel are picked up.	Oprn & EMD	
12	Close line 89L isolator from SCADA & visually ensure isolator proper contacts in all 3 phases	Operation	
13	Confirm 89L Isolator status in SCADA	Oprn &EMD	
14	Close the Bus isolator of main bay 89A or 89B from SCADA. Selection of 89A or 89B shall be based on bus isolator status of TBC Bay. i.e. Position of Bus 1 & Bus 2 isolator of main Bay shall be same as Bus 1 & Bus 2 isolator of TBC Bay. Confirm main bay Bus isolator 89A or 89B LED status in Bay PU-A & PU-B.	Operation	

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15	After Closing Bus isolator of main Bay, Bay thermography shall be done, if any abnormality, report. Main bay 89A or 89B Bus isolator close status must be ensured in PU-A & PU-B. It should be same in both PU of main Bay.	EMD	
16	Check the status of Zone 1/2/3 topology in the following: 14.0 In Bus Bar Panel Check main Bay PU-A, PU-B 15.0 In Bus Bar Panel Check TBC Bay PU-A, PU-B 16.0 In Bus Bar Panel Check CU-A, CU-B	EMD	
17	Ensure no alarm/fault in bay relay/control panel, Busbar panel & SCADA.	Oprn & EMD	
18	Close the EHV CB of main bay from SCADA.	Operation	
19	Ensure the close status of main Bay circuit breaker in Bay PU-A & PU-B of bus bar panel BB4B & SCADA	Oprn & EMD	
20	Ensure no alarm/fault in bay relay/control panel, Busbar panel & SCADA.	Oprn & EMD	
21	Check for Differential & bias currents in CU-A, CU-B, LINE PU-A & PU-B, TBC PU-A & PU-B	EMD	
22	Open TBC Circuit breaker from SCADA	Operation	
23	Confirm the Open status of TBC circuit Breaker in Bay PU-A & PU-B of bus bar panel & SCADA	Oprn & EMD	
24	Open main 89C isolator from SCADA & confirm 89C OFF status LED in main bay PU-A & PU-B of bus bar panel.	Oprn & EMD	
25	Open TBC 89C isolator from SCADA & confirm TBC 89C OFF status LED in Transfer Bay PU-A & PU-B of bus bar panel.	Oprn & EMD	
26	Open TBC 89A or B isolator from SCADA & confirm 89A or B OFF status LED in Transfer Bay PU-A & PU-B of bus bar panel.	Oprn & EMD	
27	Toggle Switch to Normal position. Ensure NIX & NX relays in Panel is picked up & ITX, TX relays are dropped	Oprn & EMD	
28	Perform sl.no.21 & Bay thermography. Report if any abnormality	EMD	

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Signature of Operation Representatives

Signature of EMD representatives

Note: All SOPs (Bay isolation/normalization & Main to transfer changeover & vice versa) are prepared here as per NTPC Switchyard general scheme, however Stations are advised to prepare & adhere the SOPs for every available bay during Isolation & normalization according to his existing scheme & methodology.

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14.0 LIST OF MINIMUM MANDATORY SPARES (FOR 765/400/220/132/66KV SWYD):

Equipment details	Min Quantity
CIRCUIT BREAKERS	
One complete pole (phase) of circuit breaker including CSD/Closing resistor, grading capacitor (as applicable), pole column, interrupter, Unit operating mechanism, marshalling box excluding supporting structure, terminal connectors and common operating Mechanism	3 No. of each type and Rating
Set of gaskets, rings & seals	1 Set of each type
Molecular filter for Circuit breaker	1 Set of each type
Tripping coils with resistors and closing coil with resistor	6 Nos of each type
Density / pressure monitoring system for circuit breaker	1 Set of each type
Relays, power contactors, switch-fuses for electrical control circuit (one set consisting of one no. each of all types for one circuit breaker)	1 Set of each type
Fixed and moving arcing contact assemblies including insulating Nozzles etc. for 1 interrupter (each type & rating of Breaker)	2 Set of each type
Terminal Pads & connector	3 Nos. of each type
Set of Valves for pneumatically operated circuit breaker	1 set of each type
Pressure switch set for pneumatically operated circuit breaker	1 set of each type
Corona rings	2 set of each type
Grading capacitor if applicable	3 nos of each type
SF6 gas	15% of total used quantity in switchyard
Spring charging motors (if applicable)	3 nos of each type
DISCONNECTORS	
One complete pole of each type of HCB isolator with 1 E/S along with operating mechanisms, insulators etc. without support structure	3 no. of each type & rating

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One complete pole of Pantograph isolator along with operating mechanisms, etc. (without Structure) If applicable	3 no. of each type & rating
Relay, power contactors, switch fuses, timers, key interlock pushbutton switch for electrical control circuit	1 Set of each type
Limit switch & aux. switch	6 Nos. of each type
Terminal Pads and connectors	3 Nos. of each type
Rotary Bearing for Isolator	3 Nos. of each type
Motor with gear assembly	1no of each type
Coronal shield rings	3nos of each type
Copper contacts fingers for male & female contacts	2sets of each type
Support insulators	1set of each type
Current Transformer complete in all respects including terminal connectors etc. (120% Rating)	3 nos
Surge Arrester complete in all respects with surge counter, terminal connector etc.	3 nos
Surge counter/Monitor	5 Nos
Capacitor Voltage Transformer complete in all respects including terminal connectors etc.	3 no of each type
Bus Post Insulator Assembly (Complete)	3 Nos of each type
Clamps and connectors	10% of total population (Minimum 3 Nos. of each type)
String Insulators and associated hardware's	3 nos of each type
Disc Insulators / Long rod insulators	5% of the total number of discs of each voltage class installed at the switchyard
SAS INCLUDING PROTECTION SYSTEM	
Bay Control unit (complete with all components)	2 No of each type & configuration
Numerical Relays comprising various bay protection units, busbar & islanding scheme	1 No. of each type & configuration

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Switchgear control unit	1 No. of each type & configuration
Operator workstation (OWS)/ Engineering Workstation (EWS) along with software, monitor, mouse, keyboard, printer etc.	1 No.
Complete Network Controller / Server along with software	1 No.
LAN switches for LAN	1 No. of each type
Firewall	1 No. of each type
Fuses	10 No. of each type and rating
Terminal Blocks	5 nos. of each type, make, model and rating
Relays other than numerical relays	10% of each type of total population (min 1 no.)
GPS equipment	1 set comprises hardware Modules of each type
Media converters (applicable as per scheme) for SAS application	1 set
Interface cables containing standard length of each type of cable and its connector for each type of peripheral	2 Sets
MCBs	1 No. of each type, make and model used in the system
LIU of OFC including necessary connectors	1 no. of each type
ABT Energy Meter – 0.2s	1 No. of each type
Cards/Modules of line DR (if stand-alone)	1 No. of each type
Complete programming station along with software, monitor etc.	1 No.
Gateways	1 No. of each type
Modem/Firewall	1 no. Of each type
Transducers	1 no. Of each type
Auxiliary CT/PT	

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15.0 LIST OF T&P ITEMS FOR AIS:

Sr.no.	Item description	Min required Quantity
i.	SF6 Gas handling (Evacuation/filling/Vacuum) plant	1 set
ii.	SF6 gas purity checking & SO2 measurement kit	1 set
iii.	SF6 gas dew point monitor	1 set
iv.	Consumables spares for SF6 gas handling plant	1 set
v.	SF6 gas leak detector	2 set
vi.	Infrared thermography camera	1 set
vii.	Contact resistance measurement kit	1 set
viii.	Breaker DCRM test kit	1 set
ix.	CT analyzer test kit	1 set
x.	Breaker timing test kit	1 set
xi.	LA leakage current analyzer (LCM)	1 set
xii.	Tan delta test kit	1 set
xiii.	Relay secondary injection test kit	1 set
xiv.	IR test kit (From 500V to 5KV) digital	1 set
xv.	Fully automatic battery discharge unit up to 1000 AH DC system, with selectable current discharge current	1 set
xvi.	Primary injection kit suitable for testing of EHV Current transformers (Current range: 0-1000 amp.)	1 set
xvii.	Precision grade Multimeter (Digital)	03 nos
xviii.	Precision grade AC Clamp meter- from 0mA to 5A	03 nos
xix.	Precision grade DC Clamp meter- DC (up to 1000A)	01 no
xx.	Precision grade AC Clamp meter- AC (up to 400A)	01 no
xxi.	Portable earth resistance measurement kit	1 set

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xxii.	Fully Automatic portable HT cable Fault Locator (cable test kit)	1 set
xxiii.	Analog Phase sequence indicator	02 nos
xxiv.	Crimping tools suitable for sizes: 0.5 sq.mm to 300 sq. mm)	01 no
xxv.	High sensitivity (able to detect high resistive faults), Precision grade Fully automatic DC earth fault locator in live DC circuit.	1 set
xxvi.	Fully automatic digital Level oscillator & level measurement meter for PLCC system.	1 set
xxvii.	Battery operated trolleys for handling of motors	01 no
xviii.	Articulating boom lifter suitable for work up to height of A-row conductor	01 no
xxix.	Discharge earthing rod suitable for EHV system	06 nos
xxx.	High Voltage detector up to system max voltage rating	01 no

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16.0 IMPLEMENTATION & RESPONSIBILITY & RECORD KEEPING.

Switchyard Electrical Equipment Maintenance: By EMD

Normalization and Isolation: By Operation in presence of EMD representative.

Fire System: Operation, Offsite and C&I department

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17.0 REVIEW

HOD(EMD) will review & take the approval of competent authority (HOP). The document will be reviewed on 2 yearly basis or earlier, if required

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18.0 CHECKLIST: ANNEXURE-A

MAINTENANCE CHECKLIST FORMATS

Sl. No.	CONTENTS
01.	DAILY CHECKLIST OF SWITCHYARD EQUIPMENT
02.	CIRCUIT BREAKERS
03.	CURRENT TRANSFORMERS
04.	CAPACITIVE VOLTAGE TRANSFORMERS
05.	ISOLATORS AND EARTH SWITCHES
06.	LIGHTENING ARRESTORS
07.	BUS BARS & BPIs
08.	WAVE TRAPS
09.	ILLUMINATION SYSTEM
10.	THERMOVISION SCANNING
11.	FIRE FIGHTING SYSTEM
12.	PLCC PANELS
13.	DR & EVENT LOGGER
14.	HOT LINE WASHING OF INSULATORS

TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

18.1 DAILY MONITORING CHECKLIST OF SWITCHYARD EQUIPMENT

CIRCUIT BREAKERS:

BAY	SF6 GAS PRESSURE			HYDRAULIC/PNEUMATIC PRESSURE			OIL/AIR LEAKAGE OBSERVED			ANY OTHER ABNORMALITY
	R PH	Y PH	B PH	R PH	Y PH	B PH	R PH	Y PH	B PH	

CURRENT TRANSFORMERS:

BAY	OIL LEVEL			OIL LEAKAGE OBSERVED			ANY OTHER ABNORMALITY
	R PH	Y PH	B PH	R PH	Y PH	B PH	

CAPACITIVE VOLTAGE TRANSFORMERS:

BAY	OIL LEVEL			OIL LEAKAGE OBSERVED			ANY OTHER ABNORMALITY
	R PH	Y PH	B PH	R PH	Y PH	B PH	

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LIGHTNING ARRESTERS:

BAY	COUNTER READING			LEAKAGE CURRENT			ANY OTHER ABNORMALITY
	R PH	Y PH	B PH	R PH	Y PH	B PH	

PLCC:

LINE BAY	CHANNEL-1						CHANNEL-2						ANY ALARM
	CARR-1		CARR-2		DT		CARR-1		CARR-2		DT		
	TX	RX	TX	RX	TX	RX	TX	RX	TX	RX	TX	RX	

SWYD 220V PILOT CELL:

CELL NO	BATTERY BANK-1		BATTERY BANK-2		CHARGER-1		CHARGER-2		ANY ALARM IN CHARGER	ANY LEAKAGE FROM BATTERY
	VOLT	SPEC GRAVITY	VOLT	SPEC GRAVITY	VOLTAGE (P-E, N-E & TOTAL)	CURRENT	VOLTAGE (P-E, N-E & TOTAL)	CURRENT		

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SWYD 48V PILOT CELL:

CELL NO	BATTERY BANK-1		BATTERY BANK-2		CHARGER-1		CHARGER-2		ANY ALARM IN CHARGER	ANY LEAKAGE FROM BATTERY
	VOLT	SPEC GRAVITY	VOLT	SPEC GRAVITY	VOLTAGE	CURRENT	VOLTAGE	CURRENT		

SWYD OTHER AREAS:

BAY NAME	ISOLATOR CONTACT ALIGNMENT			ANY HOT SPOT	BAY JB/M B CLOSE	C/R PANEL ALARM	BUSBAR PANEL ALARM	SCADA ALARM	CONTROL ROOM TEMP	ACDB ROOM	DCDB ROOM	CABLE GALLERY	DEWATERING P/P AVAILABILITY
	R PH	Y PH	B PH										

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18.2 CIRCUIT BREAKERS:**CIRCUIT BREAKER – HALF YEARLY MAINTENANCE RECORD**

Date of Commissioning		Make		Rating		Sl. No.		Bay Location	
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A. HALF YEARLY MAINTENANCE – WITHOUT SHUT

DOWN ACTIVITY PTW NO..... DATE.....

SF6 CIRCUIT BREAKER (PNEUMATIC OPERATING MECHANISM) WITHOUT SHUT
DOWN ACTIVITY

MAINTENANCE OF AIR DRYER- As per manufacturer's recommendations

Signature of Maintenance Engineer

Signature of HOD-
Electrical

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TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

Date of Commissioning		Make		Rating		Sl. No.		Bay Location	
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B. YEARLY MAINTENANCE - S/D ACTIVITY**(i) SF6 CIRCUIT BREAKER****a) FUNCTIONAL CHECKING OF AUTO-STARTING OF AIR COMPRESSOR****b) PRESSURE DROP DURING DUTY CYCLE OPERATION - O-0.3 S-CO**

PTW NO.....

DATE.....

	OIL PRESSURE		AIR PRESSURE	
	Pre commissioning / Previous Results.	Actual	Pre commissioning / Previous Results.	Actual
Before Operation				
After Operation				

c) CHECKING OF HYDRAULIC PUMP OPERATION

	START (Pressure)	STOP(Pressure)
Pre-Commissioning		
Actual		

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LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

(ii) AIR BLAST CIRCUIT BREAKER - W-S/D VALUE

Temp Ambient.

Humidity (DEW POINT) Measurement of operating air

Permissible Limit: -45 ° C at atmospheric Pressure

(iii) Minimum Oil Circuit Breaker (MOCB)- W S/D

BDV of Oil KV (Average of Six Readings)

(iv) BREAKER OPERATION CHECKS - S/D ACTIVITY

(d) CB timing including PIR checks

Details of Kit used: (Make & Sl. No.) .. /

PHASE	BREAK	CLOSE	TRIP		CLOSE TRIP		REMARKS & OBSERVATION
			TC-1	TC-II	TC-I	TC-II	
R - PHASE MAIN CONTACT R - PHASE PIR PIR & MAIN CONTACTS OVERLAP TIME/ PIR OPENING TIME PRIOR TO MAIN CONTACTS R-PHASE AUXILIARY CONTACT	BREAK 1						
R - PHASE MAIN CONTACT R - PHASE PIR PIR & MAIN CONTACTS OVERLAP TIME/ PIR OPENING TIME PRIOR TO MAIN CONTACTS	BREAK 2						
Y - PHASE MAIN CONTACT Y - PHASE PIR PIR & MAIN CONTACTS OVERLAP TIME/ PIR OPENING TIME PRIOR TO MAIN CONTACTS Y-PHASE MAIN CONTACT	BREAK 1						

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LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

PHASE	BREAK	CLOSE	TRIP(O)		CLOSE TRIP (C-O)		REMARKS & OBSERVATION
			TC-1	TC-II	TC-I	TC-II	
Y - PHASE MAIN CONTACT Y - PHASE PIR PIR & MAIN CONTACTS OVERLAP TIME/ PIR OPENING TIME PRIOR TO MAIN CONTACTS	BREAK 2						
B - PHASE MAIN CONTACT B - PHASE PIR PIR & MAIN CONTACTS OVERLAP TIME/ PIR OPENING TIME PRIOR TO MAIN CONTACTS B-PHASE AUXILIARY CONTACT	BREAK 1						
B - PHASE MAIN CONTACT B - PHASE PIR PIR & MAIN CONTACTS OVERLAP TIME/ PIR OPENING TIME PRIOR TO MAIN CONTACTS	BREAK 2						

NOTE – C-O Time is to be measured with simultaneous Close-Trip Commands

(d).Dynamic Contact resistance and contact travel measurement

(DCRM)

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TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

CB POLES	BREAK	CONTACT WIPE	ARCING CONTACT LENGTH	OBSERVATION & REMARKS
R phase	BREAK 1			
	BREAK 2			
Y phase	BREAK 1			
	BREAK 2			
B phase	BREAK 1			
	BREAK 2			

NOTE – Dynamic contact resistance signature for CO operation for all the breaks of CB to be recorded and compared with the earlier signatures. Minimum CO time should be 300ms.

(e).Measurement of SF6 Gas parameters (For SF6 CB)

DATE.....

CB POLE	DEW POINT				SO2 ppm		PURITY	
	PRE-COMMISSIONING		ACTUAL		PRE-COMMISSIONING	ACTUAL	PRE-COMMISSIONING	ACTUAL
	TEMP, °C	PRESSURE, KG/cm ²	TEMP, °C	PRESSURE, KG/cm ²				
R phase								
Y phase								
B phase								

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TITLE	Guidelines on AIS maintenance practices
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TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

(f).Checking Pressure Setting of switches

CB Pole	SF6 Pressures Switch Setting		SF6 Pressure Switches Actual		Ambient Temp
	ALARM	LOCKOUT	ALARM	LOCKOUT	
R-Phase					
Y-Phase					
B-Phase					

Pressure Switch setting should be within +/- 0.1 Bar of set value.

(g).Checking of Pole Discrepancy Relay

TRIP "R" phase	Measure Tripping Time of Y & B pole	
	SET VALUE, Seconds	Operating Value, Secs
Pre commissioning		
Actual		

Close any one pole and observe the tripping of same pole

CLOSE „R" pole	Measure Tripping Time of R pole	
	SET VALUE, Seconds	Operating Value, Secs
Pre commissioning		
Actual		

(h).Duty Cycle operation including rapid re-closing

O-0.3 S-CO	Timings, mSec			
CB Pole	BREAK 1		BREAK 2	
	PRE-COMMISSIONING.	ACTUAL	PRE-COMMISSIONING.	ACTUAL
R-Phase				
Y-Phase				
B-Phase				

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TITLE	Guidelines on AIS maintenance practices
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(i).Measurement of Static Contact Resistance: Maximum Contact Resistance Acceptable for all Breaks: 150 Micro Ohm

CB POLES	CONTACT RESISTANCE IN MICRO OHMS			
	Interrupter 1 (CT SIDE)	Interrupter 2	Interrupter 3	Interrupter 4
R phase				
Y phase				
B phase				

(j).Capacitance & Tan δ Measurement of Grading Capacitor

Details of Kit used: (Make & Sl. No.)...../

Ambient Temp. Deg C

DESCRIPTION	INTERUPTER 1			INTERUPTER 2		
	R-PH	Y-PH	B-PH	R-PH	Y -PH	B-PH
CAPACITANCE						
PRE-COMM VALUE						
MEASURED VALUE						
TAN δ						
PRE-COMM VALUE						
MEASURED VALUE						

Note: Normally Capacitance and Tan measurement is to be carried out in UST mode but if interference is more then, same may be done in GST_g mode with one side grading capacitor grounded and other side connected to guard.

(k).CHECK OF INTERLOCKS

- (i) Local Closing Interlock -
- (ii) Closing of Line Isolator - (In case Pressure reduces) in

ABCB.

(l).CHECK OF OPERATIONAL LOCKOUTS

Sl. No	Test-Description	Pressure Values		
		A/R Lockout	Closing L/O	Operational Lockout

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		Set value	Actual	Set Value	Actual	Set Value	Actual
1.	Driving mechanism						
	(Hydraulic/Pneumatic)						

(m).Cleaning of Breaker Pole

(i) Support Insulators	DONE / NOT DONE
(ii) PIR and Grading Capacitor	DONE / NOT DONE
(iii) Interrupter Chamber	DONE / NOT DONE

(n).Checking of Close-Trip Coil

CB POLE	CURRENT DRAWN BY THE COIL IN AMPERES		
	Closing Coil	Trip Coil-1	Trip Coil-2
R-Phase			
Y Phase			
B Phase			

(o).Healthiness of Operation Counter

R phase	OK/NOT OK	REMEDIAL ACTION TAKEN
Y phase	OK/NOT OK	REMEDIAL ACTION TAKEN
B phase	OK/NOT OK	REMEDIAL ACTION TAKEN

- (i) Ch
- (i) Checking of tightness of all termination
- (ii) Checking of Door Sealing gaskets and replacement thereof, if necessary
- (iii) General cleaning of MB and Repainting of Metallic Surfaces, if required
- (iv) Check functioning of space heater/ illumination

Signature of Maintenance Engineer

Signature of HOD-
Electrical

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TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

18.3 CURRENT TRANSFORMERS:**CURRENT TRANSFORMER – YEARLY MAINTENANCE RECORD**

Date of Commissioning		Make		Rating / Type		Sl. No.		Bay Location	
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(i) Marshalling Box

- (a) Cleaning of MB.....
- (b) Checking the tightness of all electrical connections including earthing of MB
- (c) Cleaning and tightness of CT secondary terminals and checking healthiness of sec terminal bushing
- (d) Checking of Space Heater. Sealing & Illumination.....

(ii) Thermo scanning of CT, Top dome & Body Kit used: Load:

	Ambient Temp.	Scanned Temp. R Phase	Scanned Temp. Y Phase	Scanned Temp. B Phase	Remarks
CT Tank					
Top Dome					
CT Body Temp					

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TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

(iii) Measurement of Capacitance & Tan δ

Make of the Kit..... Permissible Value of Tan δ is 0.007 at 20° C

PHASE	Sl.No.	Voltage Applied, kV	Capacitance(C1/C2), pF		Tan δ , %		Remarks
			Pre-commg Values	Measured Value	Pre-commg Values	Measured Value	
R-Phase		2.0					Measurement to be taken after cleaning the porcelain surfaces
		10.0					
Y-Phase		2.0					• Results to be compared with Factory test /Pre
		10.0					
B-Phase		2.0					commissioning values. Evaluate Tan delta tip up
		10.0					

(iv) IR Measurement (DAR)

Signature of Maintenance Engineer

Signature of HOD-
Electrical

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LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

CURRENT TRANSFORMER - 2 YEARLY MAINTENANCE RECORD

Date of Commissioning		Make		Rating/Type		Sl. No.		Bay Location	
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(i) Nitrogen pressure checking

Caution: Nitrogen filling if required to be done as per manufacturer's procedure. Care to be exercised as not to over pressurize the CT.

Signature of Maintenance Engineer

Signature of HOD-
Electrical

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TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

18.4 CAPACITIVE VOLTAGE TRANSFORMERS:**CAPACITOR VOLTAGE TRANSFORMER - MAINTENANCE RECORD**

Date of Commissioning		Make		Rating/Type		Sl. No.		Bay Location	
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A. FORTNIGHTLY CHECKS (WITHOUT SHUT DOWN)

- (i) Measurement of the Secondary Voltage of all the core of each CVT at the switchyard control room (at the first point) is to be monitored.

CORE No.	SECONDARY VOLTAGE, VOLTS					
	R – N	Y – N	B – N	R – Y	Y – B	B – R
CORE-1						
CORE-2						
CORE-3						

a. NOTE –**QUARTERLY MAINTENANCE (WITHOUT SHUT DOWN)**

- (ii) Measurement of voltage at switchyard MB (in volts)

CORE No.	SECONDARY VOLTAGE, VOLTS					
	R – N	Y – N	B – N	R – Y	Y – B	B – R
CORE-1						
CORE-2						
CORE-3						

Signature of Maintenance Engineer

Signature of HOD-
Electrical**CAPACITOR VOLTAGE TRANSFORMER - MAINTENANCE RECORD**

NTPC**SIMHDARI**

TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

Date of Commissioning		Make		Rating/Type		Sl. No.		Bay Location	
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18.5 ISOLATORS AND EARTH SWITCHES:**YEARLY MAINTENANCE (Shut Down Activity)****(i) OPERATING MECHANISM**

- i. Maintenance of Linkages including transmission gears -
- ii. Maintenance of Stopper bolts -
- iii. Cleaning of Aux. switch contacts & Greasing with Silicon Grease -
- iv. Checking of Electrical/Mechanical Interlock with E/S & CB -
- v. Lubrication of operating Mechanism hinges, Lock Joints - on Levers, Bearings.
- vi. Checking & Tightening of all the mounting bolts

(ii) MAIN CONTACTS

- i. Cleaning and Lubrication of Main Contacts
- ii. Alignment
- iii. Tightening of Bolts & Nuts, Pins Etc.
- iv. Cleaning of Support Insulators and check for cracks in insulators, if any
- v. Checking of interlocks

(iii) MARSHALLING BOXES OF ISOLATORS AND EARTH SWITCHES

- i. Checking of space heater & illumination
- ii. Checking of healthiness of Rubber Gaskets
- iii. Visual Check of auxiliary contacts
- iv. Cleaning and tightness of all terminations

(iv) EARTH SWITCH

- i. Checking and Alignment of Earthing Blades
- ii. Cleaning of Contacts
- iii. Operation of Earth Switch
- iv. Checking of Aluminum/ Copper flexible conductor
- v. Checking of earth connections of structure & MOM box
- vi. Contact resistance measurement for isolators and earth switches

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TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

vii. Main Contact Resistance Measurement- (In $\mu \Omega$)

(v) Format for Records:

Date of Commissioning		Make	
Rating/Type		Bay No.	Location
PTW No.		Date	

- **Note:** Complete format & checklist for annual overhauling & maintenance work in Isolators & Earth switches are given at point 8.4, 8.5 & 8.6 under Major O/H work heading in this OGN.

TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

18.6 BUSBAR AND BUS POST INSULATOR - MAINTENANCE RECORD
YEARLY MAINTENANCE OF BUS BAR & BPI

Sl. No.	ACTIVITY	SCHEDULED DATE/ ACTUAL DATE	MEASURED VALUE	REMARKS
1.	MEASUREMENT OF STATION EARTH RESISTANCE			
2.	THERMOVISION SCANNING OF ALL CONDUCTOR JOINTS AND TERMINAL CONNECTORS/ CLAMPS			AS PER ENCLOSED FORMAT GIVEN IN THIS DOCUMENT SEPARATELY
3.	CLEANING OF INSULATORS			
4.	CHECKING OF INSULATORS FOR CRACKS			

Signature of Maintenance Engineer

Signature of HOD-
Electrical

18.7 YEARLY MAINTENANCE FORMAT/RECORD OF WAVE TRAP

Date of Commissioning		Make	
Rating/Type		Bay No.	
PTW No.		Date	

A. MAINTENANCE ACTIVITY (WITH SHUT DOWN)

Sl. No.	ACTIVITY	SCHEDULED DATE	ACTUAL EXECUTION DATE	REMARKS
01.	BOLT TIGHTNESS AND CLEANLINESS			
02.	GENERAL INSPECTION/ CLEANING OF TUNING UNIT			
03.	IR of LA			

Signature of Maintenance Engineer

Signature of HOD-
Electrical

TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

18.8 YEARLY MAINTENANCE FORMAT/RECORD FOR SWITCHYARD ILLUMINATION SYSTEM

Date of maintenance		Location	
PTW No.		Mtce. Done by	

Sl. No.	Job Description	Observation & Remarks	Date	Signature
01.	Check healthiness of light fittings in all circuits in the station bldg. PH, and DGS bldg. Repair, replace as required			
02.	Check if all switchyard fittings are in working condition (nos. as per list). Repair, replace as required.			
03.	Check Lighting panel, receptacle panel tightening of terminals.			
04.	Check OUTPUT SUPPLY after fuse in receptacle panel			

Signature of Maintenance Engineer

Signature of HOD-
Electrical

TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

18.9 THERMAL SCANNING OF SWITCHYARD - MAINTENANCE RECORD (MONTHLY)

Name of Substation					
BAY NO.		BAY NAME			
LWC No.		DATE OF SCANNING		AMB. TEMP, °C	
Sl. No.	POINT OF MEASUREMENT	TEMPERATURE, °C			REMARKS
		R-PH	Y-PH	B-PH	
(I) MAIN BAY EQUIPMENT					
01.	CONNECTOR / CLAMP OF WAVE TRAP				
	A) TOWARDS LINE SIDE				
	B) TOWARDS BUS SIDE				
02.	CVT				
	A) TOP CLAMP				
	B) FIRST JOINT				
	C) SECOND JOINT				
	D) ELECTRO MAGNETIC UNIT				
03.	CONNECTOR / CLAMP OF S / R ISOLATOR (89 R)				
	A) TOWARDS LINE SIDE				
	B) TOWARDS S / R SIDE				
	C) ISOLATOR CONTACT				
04.	CONNECTOR / CLAMP OF S / R BUSHING				
05.	CONNECTOR / CLAMP OF S / R ISOLATOR (89 L)				
	A) TOWARDS LINE SIDE				
	B) TOWARDS BUS SIDE				
	C) ISOLATOR CONTACT				
06.	CONNECTOR / CLAMP OF ISOLATOR- BUS SIDE (89)				
	A) TOWARDS BREAKER SIDE				
	B) TOWARDS BUS SIDE				
	C) ISOLATOR CONTACT				
07.	CONNECTOR / CLAMP OF ISOLATOR – CT SIDE (89)				
	A) TOWARDS BREAKER SIDE				
	B) TOWARDS BUS SIDE				
	C) ISOLATOR CONTACT				
08.	CONNECTOR / CLAMP OF CIRCUIT BREAKER (52)				
	A) TOWARDS LINE SIDE				
	B) TOWARDS BUS SIDE				
09.	CONNECTOR / CLAMP OF C T				
	A) TOWARDS BREAKER SIDE				
	B) TOWARDS LINE SIDE				

TITLE	Guidelines on AIS maintenance practices
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THERMAL SCANNING OF SWITCHYARD - MAINTENANCE RECORD

Name of Substation					
BAY NO.		BAY NAME			
LWC No.		DATE OF SCANNING		AMB. TEMP, °C	
Sl. No.	POINT OF MEASUREMENT	TEMPERATURE, °C			REMARKS
		R-PH	Y-PH	B-PH	
(II) TIE BAY EQUIPMENT					
01.	CONNECTOR / CLAMP OF WAVE TRAP				
	A) TOWARDS LINE SIDE				
	B) TOWARDS BUS SIDE				
02.	CONNECTOR / CLAMP OF CIRCUIT BREAKER (52)				
	A) TOWARDS LINE SIDE				
	B) TOWARDS BUS SIDE				
03.	CONNECTOR / CLAMP OF C T- A				
	A) TOWARDS BREAKER SIDE				
	B) TOWARDS LINE SIDE				
04.	CONNECTOR / CLAMP OF C T - B				
	A) TOWARDS BREAKER SIDE				
	B) TOWARDS LINE SIDE				
05.	CONNECTOR / CLAMP OF ISOLATOR- (89 A)				
	A) TOWARDS BREAKER SIDE				
	B) TOWARDS BUS SIDE				
	C) ISOLATOR CONTACT				
06.	CONNECTOR / CLAMP OF ISOLATOR-(89 B)				
	A) TOWARDS BREAKER SIDE				
	B) TOWARDS BUS SIDE				
	C) ISOLATOR CONTACT				
(III) ALL JUMPER / DROPPER JOINTS RELATED TO					
01.	A) MAIN BAY				
	B) TIE BAY				
	C) STRUNG BAY				
	D) RIGID BUS (IPS TUBE)				

PERMISSIBLE LIMITS OF TEMPERATURE

TEMPERATURE UP TO 15 °C ABOVE AMBIENT - NORMAL

TEMPERATURE BETWEEN 15 °C TO 50 °C ABOVE AMBIENT – ALERT/UNDER

OBSERVATION TEMPERATURE BEYOND 50 °C ABOVE AMBIENT – TO BE

ATTENDED IMMEDIATELY

Note: This is a reference format. The format to be modified as per actual site layout. Thermal scanning is to be done once in every month.

Signature

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TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

18.10 MAINTENANCE FORMAT FOR SWITCHYARD FIRE FIGHTING SYSTEM

SUB-STATION:		DATE OF MAINTENANCE	
PTW NO:		MAINTENANCE DONE BY	

Sl. No.	EQUIPMENT	JOB DESCRIPTION		REMARKS & OBSERVATION
1	Operate Deluge System for oil filled equipment, where ever installed	Operate Electrically or manual (Releasing locking pressure at deluge valve) for followings:		
	EQUIPMENT	OPERATION METHOD		
1	Check that all FIRE HOUSE cabinets are complete with hose reels and nozzles. Seal the cabinet with lead seal after inspection.	HOSE CABINET NO -1	NOS OF HOSES	
		HOSE CABINET NO -2	NOS OF HOSES	
		HOSE CABINET NO -3	NOS OF HOSES	
		HOSE CABINET NO -4	NOS OF HOSES	
2	HYDRANT SYSTEM	-Checking of pressure of hydrant syst at the remotest end, auto starting of pumps, diesel engine etc		
3	ELECTRICAL PANELS	Cleaning		
		Tightening of terminals		
4	GENERAL	Painting of pipes, air lines, marshalling box		

Signature:

Name:

Designation

Date:

Signature:

Name:

Designation

Date:

Signature:

Name:

Designation

Date:

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TITLE	Guidelines on AIS maintenance practices
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YEARLY MAINTENANCE FORMAT FOR FIRE EXTINGUISHERS

SUB-STATION:		DATE OF MAINTENANCE	
PTW NO:		MAINTENANCE DONE BY	

01. Check for fully charged Cartridge & change if necessary.
02. Check for quality of charge & refill if required.
03. Check if ready for operation.
04. Check for cleanliness.
05. Seal using lead seal.

Sl. No.	TYPE	LOCATION	QUANTITY	REMARKS & OBSERVATION	DATE	SIGNATURE
1						
2						
3						
4						
5						
6						
7						
8						

Signature of Maintenance Engineer

Signature of HOD-
Electrical

NTPC**SIMHDARI**

TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

YEARLY MAINTENANCE FORMAT FOR FIRE ALARM SYSTEM

SUB-STATION:		DATE OF MAINTENANCE	
PTW NO:		MAINTENANCE DONE BY	

01. Check for operation of fire alarm system installed at various location by incense stick or some smoke device
02. Check for alarm in the control panel & Swyd Kiosk.
03. Check the condition of battery
04. Check for cleanliness.

Sl. NO	TYPE	LOCATION	QUANTITY	REMARKS & OBSERVATION	DATE	SIGNATURE
1						
2						
3						
4						
5						
6						
7						
8						

Signature of Maintenance Engineer

Signature of HOD-
Electrical

NTPC**SIMHDARI**

TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

18.11 PLCC EQUIPMENT MAINTENANCE RECORD

Name of Line		Make/Model		PTW NO:	
Date of Commissioning		Cabinet serial No.		MTCE DONE BY	

(A) YEARLY MAINTENANCE - S/D Activity

(a) General Cleaning of Cabinets. (b) Checking of healthiness of Ventilation Fans in Cabinet. (c) Level Measurements

Sl. No.	MAINTENANCE ACTIVITY	TEST POINTS (T.P.) WHERE MESUREMENTS TO BE DONE	SPEECH		PROTECTION-I		PROTECTION- II	
			Tx	Rx	Tx	Rx	Tx	Rx
01	POWER SUPPLY MEASUREMENTS							
02	INPUT VOLTAGES							
03	STABILISED DC VOLTAGES							
04	TRANSMITTER CHECKS							
05	FM OSCILLATOR Frequency measurement							
06	AM OSCILLATOR- Time measurement							
07	OUTPUT LEVEL MEASUREMENT							
08	RECEIVER CHECKS							
09	Receiver level FM							
10	Receiver level AM							
11	ALARM CHECKS							
12	Check Alarm Contacts with Buzzer/ Ohm meter after inserting test plug" IN POSITION"							
13	TRANSMISSION OF PROTN. CODE CODE I, CODE-II, CODE-III							
15.	RECEIPT OF PROTN. CODE CODE I, CODE II, CODE III							
16.	LOOP TEST/REFLEX TEST							

Signature of Maintenance Engineer

Signature of HOD-
Electrical

NTPC**SIMHDARI**

TITLE	Guidelines on AIS maintenance practices
LMI NO	LMI/OGN/OPS/ELEC/42, Rev:01

18.12 PREVENTIVE MAINTENANCE RECORD FOR DR & EVENT LOGGER**MONTHLY PREVENTIVE MAINTENANCE S RECORD -GENERAL**

Sl. No.	MAINTENANCE ACTIVITY	APR	MAY	JUNE	JULY	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR
	Testing of Disturbance Recorder for test prints by manual triggering (All Generator and Line DRs to be checked)												
	Testing of Event Logger (Min two events to be checked)												

Signature of Maintenance Engineer

Signature of HOD-
Electrical

18.13 CHECKLIST FOR HOTLINE WASHING OF INSULATOR

(Checking to be done prior to commencement of each hot line washing cycle)

S.No	Step/Activity	Check points	OK/Not OK	Responsibility
1	Permit system	Check for proper PPE i.e. Safety shoes, helmet etc. & Check for proper PTW.		NTPC
2	Availability of first aid box	To be made available		
3	Check equipment double earthing	Earth wire should be of minimum 6mm copper, multistrand wire. Tightness of connections		NTPC
4	Check the weather, wind direction & Speed	Speed < 35km/hr. Weather should be clear.		NTPC & Agency
5	Check calibration validity of both portable conductivity meter & online conductivity meter	Calibration certificate of online conductivity meter, leakage detector		NTPC & Agency
6	Check pressure gauge for working conditions.	Pressure gauge needle should be in stable position		NTPC & Agency
7	Check the pressure gauge	Pressure level between 800 to 1200psi		NTPC & Agency
8	Connection of Leakage current detector should be in series with earthing wire	Ensure leakage current detector is in working condition. Ensure leakage detector will not get wet due to washing water.		NTPC & Agency
9	Hooter alarm in case conductivity increase beyond 2 micro mho	healthiness of hooter		NTPC & Agency
10	Pump oil level be kept at indicator marking	Monitoring of change in oil colour. If oil colour turns turbid or oil has been used for 300 hours, change the oil.		NTPC & Agency
11	Water Flushing in hose pipe & Check for water leakage from nozzle	Water should not leak from nozzle. Flushing to be done		NTPC & Agency

12	Check for correct phase-sequence by judging from direction of motor at no-load.	Motor direction should be clock-wise while looking at the rotating shaft		NTPC & Agency
13	Electrical connections in starter panels should be checked for correct sequence and tightness.	Starter panel should be equipped with ELCB of 30mA, with appropriate ratings.		NTPC & Agency
14	Check for cleanliness of water tank. (FRP Tank)	Tank should be properly cleaned		NTPC & Agency
15	Check conductivity of DM water	Should be less than 5 micromho/cm		NTPC & Agency
16	Ensure locking arrangement of tank	Tank should be locked		NTPC & Agency

Sign of NTPC

Sign of agency

Daily monitoring of DM water quality

Sl No	Quantity	Time		PH	Conductivity (micro-mho) and sign	
		Chemistry Lab	Agency		By Chemistry at Lab	By Agency at pump
1						
2						
3						
4						
5						